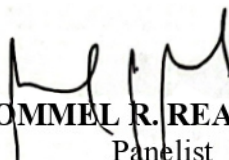
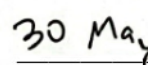


The thesis manuscript attached hereto, entitled “**CITY-TO-CITY (C2C) LEARNING THROUGH URBAN POLICY ANALYSIS – A TEXT MINING APPROACH**”, prepared and submitted by **KEISHA C. LOREJO** in partial fulfillment of the requirements for the degree of Bachelor of Science in Applied Mathematics is hereby accepted.

MAYNARD U. USARES, M.Sc.
Adviser

Date


ROMMEL R. REAL, Ph.D.
Panelist


30 May 2025

Date

VLADIMER B. KOBAYASHI, Ph.D.
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MA. MARGARITA R. LAVIDES, Ph.D.
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27 May 2025

Date

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SHERELYN A. EVANGELIO, M.Sc.
College Secretary

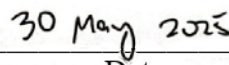
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KEISHA C. LOREJO

**SUBMITTED TO THE FACULTY OF
THE COLLEGE OF SCIENCE AND MATHEMATICS
UNIVERSITY OF THE PHILIPPINES MINDANAO
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF**

BACHELOR OF SCIENCE IN APPLIED MATHEMATICS

June 2025



UNIVERSITY OF THE PHILIPPINES MINDANAO

Bachelor of Science in Applied Mathematics

Keisha C. Lorejo

**City-to-City (C2C) Learning Through Urban Policy
Analysis – A Text Mining Approach**

Thesis Adviser

Maynard U. Usares, M.Sc.

Department of Math, Physics, and Computer Science

Date of Submission

June 2025

The research protocol involving human participants received ethics approval	NA
This manuscript can be made available to the public	No
This manuscript can be made available only after consultation with the author and thesis adviser	Yes
This manuscript can be made available to those bound by confidentiality agreement	Yes

Student's signature:

Signature of thesis adviser:

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BIOGRAPHICAL SKETCH

Keisha C. Lorejo, fondly known as “Keish” or “Shang” to her friends, is a vibrant and grounded individual known for her warm presence and fluid personality. Born and raised with an active lifestyle, Keisha’s early years were marked by her passion for volleyball, which not only honed her physical strength but also instilled in her a strong sense of discipline and teamwork.

Academically, Keisha's relationship with mathematics has been a journey of transformation. Although she initially struggled with the subject during her elementary years, her persistence and curiosity turned math from a challenge into a passion. This evolving interest eventually guided her towards pursuing a degree in Applied Mathematics, where she continues to embrace complex problems with enthusiasm and resilience.

Keisha sees herself as water—adaptive, reflective, and responsive. She believes in matching the energy people give her and often mirrors their best qualities, allowing her to build meaningful connections and cultivate a broad circle of friends. Her social nature has not only enriched her personal life but also played a significant role in her growth as an individual.

In earlier years, Keisha was driven by the fear of being forgotten or failing to leave a mark. Today, her outlook has matured. She now hopes to be remembered not for grand achievements, but for the kindness she extended and the authenticity she lived by. For her, being known as someone with a beautiful soul is the highest form of recognition—a legacy of compassion and sincerity.

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I would like to express my deepest gratitude to my mother and sister, whose unwavering love, encouragement, and belief in me have formed the bedrock of my academic journey. To my sister, Charisse, thank you for comforting me through countless sleepless nights. And to my Mama Angelica, thank you for your steadfast support and guidance during my most challenging times. I wouldn't be here without both your unwavering support and care.

To Omar and Jamae, my lifelong friends: though distance has separated us, your steadfast presence has been my lifeline through my college years.

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TABLE OF CONTENTS

	PAGE
INTRODUCTION	1
REVIEW OF LITERATURE	9
City-to-City (C2C) Learning	10
Urban Competitiveness	11
E-Governance	12
Text Mining of Urban Policies	14
Latent Dirichlet Allocation (LDA)	16
Structural Topic Modeling	17
MATERIALS AND METHODS	21
Data Collection	22
Text Mining: Developing of E-Governance Index	22
Text preprocessing	23
Text transformation	23
Classification of ordinances	23
Word2Vec model selection	26
Grouping of ordinances by category	27
E-governance index computation	29
Text Mining: Structural Topic Modeling (STM)	30
Topic modeling using LDA	30
Topic modeling using STM	31

Hyperparameter estimation	33
Posterior inference	37
Topics Generated	39
Text postprocessing	40
Visualization	42
Hardware and Software Specification	42
RESULTS AND DISCUSSION	43
Data Collection	43
Text Mining: Developing the E-governance Index from the Ordinances	45
Text preprocessing	45
Text transformation	47
Classification of ordinances	48
Text Mining: Structural Topic Modeling	69
Topic prevalence effect and correlation	74
Visualization	81
C2C Learning and its Practical Application	85
SUMMARY AND CONCLUSION	90
LITERATURE CITED	93
APPENDICES	97

LIST OF TABLES

TABLE	TITLE	PAGE
1	A summary of the studies mentioned in the literature review.	18
2	E-Governance Components Defined by the sub-indices of the UN's E-Government Development Index (EGDI).	28
3	Keyword list for each e-governance component.	28
4	Top 6 cities by total number of collected ordinances and their corresponding year range of publication.	43
5	Sample results showing the original ordinance text, city of origin, ordinance number, and the preprocessed ordinance.	45
6	Summary statistics of the TF-IDF scores.	47
7	Word2Vec models with their corresponding Spearman and Pearson correlations evaluated against the WordSm-353 dataset.	49
8	Summary statistics of the similarity scores between the ordinances and the online services category.	50
9	Summary statistics of the similarity scores between the ordinances and the telecommunication infrastructure category.	51
10	Summary statistics of the similarity scores between the ordinances and the human capital category.	52
11	The mode, along with the corresponding value 1% less than the mode, used to determine the similarity threshold.	54
12	Top 5 cities with the highest number of ordinances classified under the online services category.	54
13	Top 5 cities with the highest number of ordinances classified under the telecommunication infrastructure category.	54
14	Top 5 cities with the highest number of ordinances classified under the human capital category.	54

15	Top 5 cities with the highest number of ordinances not classified into any category.	56
16	Sample single and multiclass classification results using the similarity-based classification of ordinances.	57
17	The computed e-governance index for each city ranked from highest to lowest.	58
18	The CMCI score obtained from DTI for each city ranked from highest to lowest	59
19	Summary of diagnostic metrics for STM models with varying numbers of topics K.	66
20	The parameters chosen for the STM model.	68
21	The first 10 topics generated by the STM model with their corresponding labels and words.	69
22	Breakdown for each unique topic label.	71
23	Cluster of topics with their corresponding labels.	73
24	Regression table of the effect of the CMCI score on the topic proportion of the first 3 topics.	75
25	Regression table of the effect of the e-governance index on the topic proportion of the first 3 topics.	75
26	Topic 28 generated by the STM model with its corresponding labels and words.	76
27	Regression table of the effect of the CMCI score on the topic proportion of the Topic 28.	76
28	Regression table of the effect of the e-governance index on the topic proportion of the Topic 28.	76
29	Regression table of the effect of the city of origin on the topic proportion of the Topic 28.	85
30	The top 5 ordinances contributing to Topic 28, along with the corresponding city each ordinance originates from.	87

LIST OF FIGURES

FIGURE	TITLE	PAGE
1	Overview of the workflow for the methods of the study.	21
2	The generative process expressed in pseudocode of LDA (Lifted from Blei et. al., 2003)	30
3	The generative process expressed in pseudocode of STM (Lifted from Roberts et al., 2019).	32
4	The generative process expressed in pseudocode of CGS (Lifted from Darling, 2011).	38
5	Ordinance no. 987 from Pasig city's website with its corresponding category and date of publication.	44
6	A snapshot of the first page of Ordinance No. 987 from Pasay city.	44
7	A snapshot of the Google Sheets spreadsheet containing the collected information from the ordinances.	44
8	The distribution of TF-IDF scores across the initial vocabulary.	46
9	Histogram of the similarity scores between all the ordinances and the online services category.	50
10	Histogram of the similarity scores between all the ordinances and the telecommunication infrastructure category.	51
11	Histogram of the similarity scores between all the ordinances and the human capital category.	52
12	Scatter plot with regression line showing e-governance index vs. CMCI score.	61
13	Updated ordinance data in the Google spreadsheet, including columns for the e-governance index and CMCI score for each city.	64
14	A histogram plot of the number of terms based on how many documents they appear in.	64

15	Summary statistics of the number of documents each term appears in.	66
16	Plot of the effect of the e-governance index to the topic proportion of all 95 generated topics.	78
17	Plot of the effect of the CMCI score to the topic proportion of all 95 generated topics.	78
18	Homepage of the website containing information about what the website entails.	81
19	The “Model Exploration” with Topic 1 highlighted and the most relevant terms for Topic 1.	83
20	The section for how e-governance index affects the topic proportion of Topic 1.	83
21	The section for how CMCI score affects the topic proportion of Topic 1.	83
22	The section for how city of origin affects the topic proportion of Topic 1.	84
23	The most representative ordinances for Topic 1 labeled as Fisheries.	84

LIST OF APPENDIX TABLES

APPENDIX TABLE	TITLE	PAGE
1	95 topics generated by the STM model with their corresponding labels and words	98
2	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 1.	117
3	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 2.	117
4	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 3.	117
5	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 4.	117
6	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 5.	117
7	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 6.	118
8	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 7.	118
9	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 8.	118
10	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 9.	118
11	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 10.	118
12	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 11.	119
13	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 12.	119

14	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 13.	119
15	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 14.	119
16	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 15.	119
17	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 16.	120
18	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 17.	120
19	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 18.	120
20	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 19.	120
21	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 20.	120
22	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 21.	121
23	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 22.	121
24	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 23.	121
25	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 24.	121
26	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 25.	121
27	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 26.	122
28	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 27.	122

29	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 28.	122
30	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 29.	122
31	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 30.	122
32	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 31.	123
33	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 32.	123
34	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 33.	123
35	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 34.	123
36	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 35.	123
37	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 36.	124
38	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 37.	124
39	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 38.	124
40	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 39.	124
41	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 40.	124
42	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 41.	125
43	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 42.	125

44	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 43.	125
45	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 44.	125
46	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 45.	125
47	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 46.	126
48	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 47.	126
49	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 48.	126
50	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 49.	126
51	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 50.	126
52	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 51.	127
53	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 52.	127
54	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 53.	127
55	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 54.	127
56	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 55.	127
57	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 56.	128
58	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 57.	128

59	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 58.	128
60	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 59.	128
61	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 60.	128
62	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 61.	129
63	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 62.	129
64	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 63.	129
65	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 64.	129
66	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 65.	129
67	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 66.	130
68	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 67.	130
69	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 68.	130
70	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 69.	130
71	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 70.	130
72	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 71.	131
73	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 72.	131

74	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 73.	131
75	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 74.	131
76	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 75.	131
77	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 76.	132
78	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 77.	132
79	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 78.	132
80	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 79.	132
81	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 80.	132
82	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 81.	133
83	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 82.	133
84	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 83.	133
85	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 84.	133
86	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 85.	133
87	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 86.	134
88	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 87.	134

89	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 88.	134
90	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 89.	134
91	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 90.	134
92	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 91.	135
93	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 92.	135
94	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 93.	135
95	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 94.	135
96	Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 95.	135
97	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 1.	136
98	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 2.	136
99	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 3.	136
100	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 4.	136
101	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 5.	137
102	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 6.	137
103	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 7.	137

104	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 8.	137
105	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 9.	138
106	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 10.	138
107	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 11.	138
108	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 12.	138
109	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 13.	139
110	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 14.	139
111	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 15.	139
112	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 16.	139
113	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 17.	140
114	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 18.	140
115	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 19.	140
116	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 20.	140
117	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 21.	141
118	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 22.	141

119	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 23.	141
120	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 24.	141
121	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 25.	142
122	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 26.	142
123	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 27.	142
124	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 28.	142
125	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 29.	143
126	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 30.	143
127	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 31.	143
128	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 32.	143
129	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 33.	144
130	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 34.	144
131	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 35.	144
132	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 36.	144
133	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 37.	145

134	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 38.	145
135	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 39.	145
136	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 40.	145
137	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 41.	146
138	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 42.	146
139	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 43.	146
140	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 44.	146
141	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 45.	147
142	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 46.	147
143	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 47.	147
144	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 48.	147
145	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 49.	148
146	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 50.	148
147	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 51.	148
148	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 52.	148

149	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 53.	149
150	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 54.	149
151	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 55.	149
152	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 56.	149
153	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 57.	150
154	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 58.	150
155	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 59.	150
156	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 60.	150
157	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 61.	151
158	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 62.	151
159	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 63.	151
160	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 64.	151
161	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 65.	152
162	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 66.	152
163	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 67.	152

164	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 68.	152
165	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 69.	153
166	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 70.	153
167	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 71.	153
168	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 72.	153
169	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 73.	154
170	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 74.	154
171	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 75.	154
172	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 76.	154
173	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 77.	155
174	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 78.	155
175	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 79.	155
176	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 80.	155
177	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 81.	156
178	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 82.	156

179	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 83.	156
180	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 84.	156
181	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 85.	157
182	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 86.	157
183	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 87.	157
184	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 88.	157
185	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 89.	158
186	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 90.	158
187	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 91.	158
188	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 92.	158
189	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 93.	159
190	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 94.	159
191	Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 95.	159
192	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 1.	160
193	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 2.	160

194	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 3.	160
195	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 4.	160
196	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 5.	160
197	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 6.	161
198	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 7.	162
199	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 8.	162
200	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 9.	163
201	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 10.	163
202	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 11.	163
203	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 12.	163
204	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 13.	164
205	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 14.	164
206	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 15.	164
207	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 16.	165
208	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 17.	165

209	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 18.	165
210	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 19.	166
211	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 20.	166
212	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 21.	166
213	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 23.	167
214	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 24.	167
215	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 25.	167
216	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 26.	167
217	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 27.	167
218	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 28.	168
219	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 29.	168
220	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 30.	168
221	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 31.	168
222	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 32.	169
223	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 33.	169

224	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 34.	169
225	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 35.	170
226	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 36.	170
227	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 37.	170
228	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 38.	170
229	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 39.	171
230	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 40.	171
231	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 41.	171
232	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 42.	171
233	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 43.	171
234	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 44.	172
235	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 45.	172
236	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 46.	172
237	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 47.	172
238	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 48.	172

239	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 49.	173
240	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 50.	173
241	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 51.	173
242	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 52.	174
243	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 53.	174
244	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 54.	174
245	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 55.	175
246	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 56.	176
247	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 57.	176
248	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 58.	177
249	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 59.	177
250	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 60.	177
251	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 61.	177
252	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 62.	178
253	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 63.	178

254	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 64.	178
255	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 65.	178
256	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 66.	178
257	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 67.	179
258	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 68.	179
259	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 69.	179
260	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 70.	179
261	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 71.	179
262	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 72.	180
263	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 73.	180
264	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 74.	180
265	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 75.	180
266	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 76.	181
267	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 77.	181
268	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 78.	181

269	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 79.	182
270	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 80.	182
271	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 81.	182
272	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 82.	182
273	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 83.	183
274	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 84.	183
275	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 85.	183
276	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 86.	184
277	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 87.	184
278	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 88.	184
279	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 89.	184
280	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 90.	185
281	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 91.	185
282	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 92.	185
283	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 93.	185

284	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 94.	185
285	Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 95.	186
286	Rank of the Highly Urbanized Cities (HUCs) from the Department of Trade and Industry's (DTI) Cities and Municipalities Competitive Index (CMCI).	186
287	Cities by their total number of collected ordinances and their corresponding year range of publication.	187

LIST OF APPENDIX FIGURES

APPENDIX FIGURE	TITLE	PAGE
1	QR Code for Access of the Website Created.	188

ABSTRACT

**KEISA C. LOREJO, University of the Philippines Mindanao, June 2025,
CITY-TO-CITY (C2C) LEARNING THROUGH URBAN POLICY ANALYSIS – A
TEXT MINING APPROACH.**

Adviser: Maynard U. Usares, M.Sc.

Urban competitiveness is essential for attracting and retaining businesses and significantly influences economic growth and quality of life. In the Philippines, the Cities and Municipalities Competitiveness Index (CMCI) evaluates urban competitiveness across five key pillars; however, it lacks a measure of a city's capacity for data-driven, digital governance. As cities increasingly utilize information and communication technologies (ICT) through e-governance, integrating this dimension into the CMCI is crucial. This study aims to enhance and promote urban competitiveness by employing text mining to analyze urban policies through a policy analytics framework that facilitates city-to-city (C2C) learning and the sharing of best practices. Thousands of ordinances are now publicly accessible, but their volume hinders manual review, slowing C2C learning and data-driven policymaking. This gap directly impacts the country's progress toward Sustainable Development Goal (SDG) 16, which promotes inclusive institutions and transparent governance. A dataset of 13,853 ordinances from 22 Philippine cities was processed to create an e-governance index, mapped to UN E-Government Development Index (EGDI) dimensions and merged with CMCI scores. Structural Topic Modeling (STM) was used to examine how ordinance themes relate to CMCI scores, e-governance levels, and city origin. Outputs are presented through an interactive website. Findings show a clear positive correlation between the e-governance index and CMCI scores, indicating that cities with higher levels of digital governance tend to be more competitive. As cities mature, ordinance themes shift from basic compliance to more strategic, development-focused policies. The website supports actionable C2C learning by enabling local officials to explore which policy themes are most associated with higher CMCI scores, stronger e-governance signals, or both. By aligning these themes with their goals, policymakers can identify high-impact legislative strategies, benchmark performance, and adopt successful ordinances from peer cities. By treating ordinance texts as data, this study offers a new way for cities to self-assess, collaborate, and fast-track reforms—advancing transparent, accountable, and innovation-driven governance aligned with SDG 16.

Keywords: C2C Learning, Urban Competitiveness, E-Governance, Text Mining, SDG 16

INTRODUCTION

City-to-city (C2C) learning is a structured and flexible process where knowledge is shared among two or more cities to promote good governance and improve municipal service deliveries (Moodley, 2019). “Learning” in the context of C2C learning encompasses a wide array of multifaceted processes, such as policy transfer, policy learning, and policy mobility, that can help one city acquire knowledge from other cities. To facilitate C2C learning, it is crucial to identify the various phases involved in how learning unfolds in this context. C2C learning process involved sequential phases namely, exploration, acquisition, knowledge utilization, and internalization (Enseñado, 2024). The *exploration phase* involves an internal assessment concerning the existing knowledge, resources, strengths, and weaknesses within cities. This phase aims to identify information needs inside a city to serve as a starting point before seeking or sharing information with other cities. Subsequently, the *acquiring phase* involves determining information and, eventually, knowledge from other cities. This is followed by the *knowledge utilization* phase, which requires processing and interpreting the information acquired from the previous phase. Finally, the *internalization phase* involves the assessment of the quality and relevance of the data processed to determine its value. The relationship between the phases in the C2C learning process can be explained by understanding the mechanisms involved in how policies are being moved and changed (Enseñado, 2024). In line with this viewpoint, it underscores the importance of policies and how analyzing them across different cities for effective learning facilitation can be beneficial. In C2C learning, policies are used to transfer, share, and implement knowledge, experiences, and best practices

across different cities. Also, as policies are transferred, they can mutate by adapting to the other cities' local contexts, needs, and challenges (Farmer and and Perl, 2020).

In a systematic literature review by Enseñado (2024), out of the total articles reviewed, 41 were qualitative, 8 were quantitative, 4 used mixed methods, and 7 could not be classified. This exemplifies the limited literature on using quantitative approaches in facilitating C2C learning. This study aims to contribute to quantitative approaches to analyze urban policies a more rapid, robust, and reliable approach to facilitating C2C learning.

While C2C learning enables mutual growth, its long-term promise lies in its ability to foster urban competitiveness. Webster and Muller (2000) defines urban competitiveness as a reflection of a city's ability to produce and market goods and services that offer good value, both for export and local consumption. In the Philippine context, urban competitiveness is measured through the Development of Trade Industries' (DTI) Cities and Municipalities Competitiveness Index (CMCI). The CMCI framework adopted Porter's five competitive forces framework, which highlights factors such as critical distinctions among industries and their evolution. Moreover, this framework also assists businesses in identifying strategic positions for their endeavors (Porter, 1998). The CMCI is computed by adding the scores evaluated from the five primary pillars which include economic dynamism, government efficiency, infrastructure, resiliency, and innovation. The scores for each pillar are computed based on the actual data values and the completeness of the data submitted. The index is then used as a basis for a city or municipality's ranking (DTI, n.d.).

The multidimensional nature of urban competitiveness, as measured by the CMCI, presents a crucial opportunity for further refinement. While the CMCI provides valuable insights, it lacks an index that captures a city's commitment to data-driven governance, often referred to as e-governance (Villamejor-Mendoza, 2020). E-governance, leveraging ICTs for better governance and development, is crucial for modern cities (Kitchin, 2014). As highlighted by Paskaleva (2013), e-governance plays a crucial role in urban competitiveness by driving the necessary transformations for a city to become competitive. Adding an e-governance pillar (represented as an index) to the CMCI would assess digital governance practices, providing a more comprehensive view of competitiveness in the digital age (Acuto et al., 2019). This aligns with the global shift towards data-driven governance. This inclusion is particularly relevant in today's digital age, where many cities are embracing data-driven approaches to move beyond traditional urban development. By incorporating an e-governance index, the CMCI could provide a more comprehensive picture of a city's competitiveness. This index could assess several key aspects of a city's commitment to data-driven governance which can involve the availability and user-friendliness of online government services, effectiveness of citizen engagement platforms and the accessibility of government data for public scrutiny. Incorporating an e-governance index goes beyond simply measuring another aspect. It provides an indicator to a city's dedication to holistic development, aligning with the growing focus on data-driven comparisons in urban sustainability (Rosol et al., 2017). Furthermore, e-governance fosters collaboration, innovation, and knowledge sharing which are hallmarks of successful C2C learning (Pereira et al., 2018). E-governance leverages ICT tools to transform government services and interactions, to positively influence C2C learning by facilitating knowledge

sharing and improved government services (Villamejor-Mendoza, 2020). To contextualize the components of e-governance, a framework based on the United Nations' E-Government Development Index (EGDI) is considered. The EGDI assesses a country's digital governance readiness through three main dimensions: the Online Service Index (OSI), which evaluates the availability and quality of online services; the Telecommunication Infrastructure Index (TII), which reflects the status and reach of ICT infrastructure; and the Human Capital Index (HCI), which measures the capacity of citizens to effectively engage with digital services through education and skill levels. These three components collectively outline the essential pillars of effective e-governance.

In evaluating urban competitiveness and e-governance practices at the city level, local ordinances serve as the most relevant and direct form of policy data. While national laws and broader legislative frameworks offer high-level insights, ordinances reflect the actual priorities, actions, and innovations being implemented within each city. Because cities in the Philippines are autonomous in crafting local legislation, ordinances provide a rich, city-specific lens through which to assess how each city governs. This makes them particularly useful in a C2C learning context, where cities must understand the granular strategies other cities employ to improve service delivery and competitiveness. Analyzing ordinances allows for a grounded comparison of governance practices, facilitating more actionable knowledge exchange between local governments.

The vast number of ordinances presents a significant opportunity to advance C2C learning. Cities like Marikina have over 5,000 ordinances posted on their official ordinance websites, demonstrating the vast availability of online legislative data and the potential it holds for analysis and research. This data, which serves as the very foundation for

knowledge sharing and collaboration across cities, holds immense potential for effective C2C learning. However, extracting actionable insights from this vast dataset remains a critical challenge. Effective methods for sifting through and analyzing these policies are crucial to identify successful strategies and ensure C2C learning benefits citizens. Given the limitations of traditional methods in analyzing urban policies—such as their reliance on manual review, limited scalability, and difficulty in capturing patterns across large document collections—text mining presents a compelling alternative. Since most policy data, including local ordinances, exist in unstructured textual form, text mining offers a suitable and efficient approach for extracting meaningful insights, identifying recurring themes, and uncovering hidden patterns across large volumes of text. Text mining is considered a multidisciplinary field that utilizes processes, such as information retrieval and data mining, to explore knowledge in textual data by extracting information and unveiling patterns within the text (Fan et al., 2006). Thus, a promising avenue for exploration lies in applying text-mining techniques to urban policy analysis. Text mining in the context of urban policy analysis is a fairly new field. Thus, a significant potential lies in its further exploration. Example, Puri et al. (2018) applied text mining to analyze urban policies using ordinances and mapping them to respective Smart City Characteristics (SCCs) to determine the extent to which urban policies contribute towards the achievement of Smart Cities.

One popular and useful text mining technique is topic modeling. This approach identifies underlying themes within and across documents. A popular topic modeling technique is Latent Dirichlet Allocation (LDA), which analyzes documents as mixtures of topics, each represented by a probability distribution of words (Blei et al., 2003). However,

LDA assumes a single underlying process for all documents, which might not capture the influence of factors like author or time period on thematic content. To address this limitation, a new topic modeling technique was developed called Structural Topic Modeling (STM) (Roberts et al., 2019). Like LDA, STM models documents as topic mixtures. STM but in contrast with LDA, STM incorporates document-level metadata, such as author or publication date, to understand how these factors influence both the prevalence and content of topics within each document (Roberts et al., 2019). This allows for a more nuanced analysis, revealing how external variables shape the thematic landscape of the document collection.

Another critical aspect of C2C learning lies in ensuring citizens' awareness from the insights gleaned from urban policies. Bridging the gap between policy analysis and citizen awareness requires innovative approaches such as visualization of the topics generated from text mining. While text mining can identify core themes (or topics) within urban policies, these themes are often difficult to interpret, creating a barrier for citizens and policy makers to make sense of the themes. Hence, effective knowledge utilization hinges on clear communication. Visualizations play a critical role in this regard by transforming extracted policy insights into readily comprehensible and visually engaging formats. These visualizations can not only facilitate knowledge exchange and collaboration within C2C learning initiatives, but also foster citizen awareness. Informed citizenry is crucial for maximizing the potential of C2C learning, as their active engagement and participation are essential for effective knowledge exchange and policy implementation across diverse cities. Ultimately, these visualizations present key themes within urban policies, allowing citizens and policy-makers to gain a deeper understanding of the specific

ordinances that are relevant to the issues affecting them. Given this context, this study seeks to answer the following research question: How can text mining of local ordinances be used to enhance urban competitiveness and support city-to-city (C2C) learning among cities in the Philippines?

Objectives

Building upon the recognized significance of policy analysis in facilitating City-to-City (C2C) learning and the demonstrable capability of text mining to gain valuable information from textual data, this study aims to apply text-mining to promote the widespread adoption of best practices via C2C learning. To achieve this, the study specifically aims to:

1. develop an e-governance index by mapping ordinances to e-governance component/s;
2. integrate the e-governance index developed and urban competitive index into structural topic modeling to compare cities;
3. employ a visualization of the topic modeling results.

This study has the potential to significantly enhance City-to-City (C2C) learning within the domain of urban competitiveness. By leveraging text-mining techniques to analyze vast amounts of urban policy data, the study can provide a way to identify best practices that contribute to a city's success in a data-driven environment. Notably, the analysis will focus on uncovering effective strategies for integrating e-governance initiatives, a critical component of data-driven development. Once disseminated and actively utilized by various cities, these insights will foster a robust knowledge exchange ecosystem, empowering cities to learn from each other's triumphs and challenges,

ultimately leading to the widespread adoption of best practices that significantly elevate urban competitiveness. Ultimately, this will lead to the widespread adoption of best practices that significantly elevate urban competitiveness, thus contributing to the achievement of Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities.

Scope and Limitations

While aiming to contribute to the advancement of C2C learning by analyzing urban policies through text-mining techniques, this study acknowledges the inherent limitations associated with its scope. Currently, the CMCI ranking is limited to only highly urbanized cities (HUCs). Because of this, the scope of the study is restricted to only this specific category. Consequently, the analysis cannot account for the dynamics of urban policies in other city classifications within the Philippines. This limitation restricts C2C learning to only HUCs, potentially hindering a more comprehensive understanding of policy dynamics across all Philippine city types. An additional limitation may arise from the data availability. Data availability challenges can arise from the lack of publicly accessible websites or cities not consistently posting their official ordinances online. Without these crucial documents, the application of text-mining techniques would be hindered, and the analysis of urban policies would be limited to cities only having publicly accessible websites with their respective ordinances in them. To ensure a focused and manageable analysis, this study will primarily focus on ordinances as the data source for text mining. Ordinances represent a crucial component of urban policy, outlining specific regulations and procedures.

REVIEW OF LITERATURE

City-to City (C2C) Learning

City-to-city (C2C) learning is not a singular process but rather a multifaceted discipline encompassing a range of factors that influence knowledge exchange and policy improvement among cities. Enseñado (2024) was able to bridge the gap in the need to establish C2C learning and the conceptual framework involved in its processes by investigating its contexts, practices, and conditions. The study employed a systematic literature review to develop a comprehensive understanding of C2C learning, showing a growing interest in C2C learning over the past decade. Climate change, transport, and local governance were among the most common points of interest across the studies subjected to their review, which employed concepts of C2C learning. The growing interest in C2C learning and its utilization across studies involving climate change, transport, and local governance shows that it can effectively address issues in these respective fields.

Similarly, Haupt et al. (2020) employed a similar approach by examining studies in the context of local climate governance to examine how C2C learning can be employed across transnational municipal networks (TMNs). The study highlighted the potential of TMNs as platforms for knowledge exchange in C2C learning, which addresses the current gap in research involving TMNs, C2C learning, and policy learning.

To emphasize the diverse applications of C2C learning, the study of Ilgen et al. (2019) serves as a notable example of the utilization of C2C learning in building urban resilience in the context of water squares, which are multifunctional urban plazas designed to manage rainwater runoff, raise public awareness about water management, and improve the overall quality of public spaces. The investigation of C2C learning, specifically in

policy transfer, in the context of urban resilience, was conducted using an empirical, in-situ analysis in the cities of Mexico and Rotterdam. The study confirmed that a successful city-to-city knowledge exchange for water resilience existed between Mexico City and Rotterdam.

Further emphasizing the diverse applications of C2C learning, Ndebele-Murisa et al. (2020) provide an illustrative case of how C2C learning can be utilized to develop climate resilience across African cities. Through C2C learning, the study determined issues across urban ecosystems in Harare, Lusaka, Windhoek, and Durban. Among these issues was water insecurity, which posed a significant challenge for these African cities, influenced by factors such as limited infrastructure, polluted water sources, and dependence on unreliable electricity for water treatment.

These key studies in C2C learning highlight the vast potential of this field in addressing diverse urban challenges. This underscores the crucial need for further research and development in this field to contribute to developing more resilient and sustainable urban communities by fostering knowledge exchange across cities.

Urban Competitiveness

Urban competitiveness has evolved significantly from focusing on national economies and individual businesses to being a multidimensional perspective centered on cities. The study of Sgambati and Gargiulo (2022) provided a comprehensive review of the evolution of urban competitiveness studies over the last 30 years, contributing to the growing understanding of urban competitiveness and current studies in this field. Using a bibliometric analysis, the study highlighted core definitions and prevailing frameworks of

urban competitiveness. Additionally, the study identified emerging trends and areas for further research in the context of urban competitiveness.

The study by Yuan et al., (2017) illustrates the multifaceted nature of urban competitiveness by examining its indexes across Chinese cities and developing a novel approach for measuring it. The study employed a two-step process involving determining the urban competitiveness index and using a partial least squares structural equation modeling (PLS-SEM) to develop a measure of urban competitiveness based on the identified urban characteristics that influence a city's overall competitiveness and considering a city's internal characteristics and connections with other cities. Through the developed urban competitiveness measure, the study ranked cities across China, showing the top cities as Shanghai, Beijing, Shenzhen, Chongqing, and Tianjin. Another notable result from this study is that most cities fell into the lower three classes, highlighting a significant gap between the top and bottom-performing cities.

Further highlighting the nature of urban competitiveness and its importance in a city's development, the study of Sáez and Periañez (2015) presents another notable exploration in this domain. The study employed principal component analysis (PCA) to create an urban competitive index (UCI) for ranking cities. Through this approach, the study was able to benchmark competitiveness across European cities. The index developed showed that major Western European cities like London and Paris outperform their Eastern counterparts in attracting businesses and investment. Moreover, the UCI developed in the study was found to be highly correlated with its sub-indexes like government efficiency and technological advancement, indicating its robustness.

In the context of the Philippines, the examination of urban competitiveness across its cities is highlighted in the Development of Trade Industries' (DTI) Cities and Municipalities Competitiveness Index (CMCI) (DTI, n.d.). The development of this index provides an impactful tool for benchmarking and ranking cities and municipalities across the country. The CMCI identifies areas for improvement across five key domains: economic dynamism, government efficiency, infrastructure, resiliency, and innovation. Ultimately, utilizing this measure can provide a basis for collaboration across cities, which is vital in areas like policy-making and attracting investment and businesses within and across cities.

The studies highlighted in this section showcase urban competitiveness's crucial role as a multidimensional city development metric. From frameworks like the CMCI in the Philippines to UCIs used in Europe, these competitiveness indexes effectively capture a city's ability to attract businesses and investment and highlight key areas for improvement. However, a gap still remains in fully comprehending how competitiveness translates into city dynamics within and across cities that contribute to the growth and evolution of cities. While the Philippines' CMCI provides a valuable foundation, there is an opportunity to improve this framework to develop a more nuanced and insightful ranking system to develop and foster knowledge exchange across cities. This gap highlights a need for further exploration to strengthen existing frameworks in the context of urban competitiveness by delving deeper into the dynamics within and across Philippine cities.

E-Governance

The development of e-governance has been a point of exploration across various studies. Singh and Sahu (2018) provided an in-depth review of the development of e-

governance implementation across various studies. Through employing a classification approach in their extensive literature review, the study determined areas of successful e-government implementation that can be attributed to key factors such as supportive government policies and citizen literacy. Consequently, the study was also able to identify significant gaps in the context of e-governance, such as the lack of access to technology, citizen security concerns, and low trust in e-government services, showing a need for studies involving citizen awareness of e-governance in developing countries.

Another notable field of exploration in the context of e-governance from the global perspective is its adoption across various countries, from the national to the local level. The study of Manoharan et al. (2023) delves deeper into this field of exploration by assessing government websites across 100 cities considered the most wired according to the International Telecommunication Union (ITU) standard between 2015-2016 and 2018-2019. Through employing this assessment, the study was able to compare the e-government performance across these cities, ultimately determining an overall increase in the performance of e-governance based on the increase of e-government capacity scores. Furthermore, the study addressed the need for further research to explore the impact of factors such as privacy, security, and citizen engagement in the adoption of e-governance.

Within the Philippine context, the advancement of technology has led to efforts aimed at enhancing public policies and administration, which involves the adoption of e-governance. The studies of Siar (2007) and Urbina and Abe (2017) provide in-depth examinations of e-governance adoption in the Philippines. The study of Siar (2007) highlights the minimal adoption of e-governance across the Philippine city governments. The assessment of over 114 websites based on their content, usability, and responsiveness

showed issues in adopting e-governance in the Philippines, such as weak information and communication technology (ICT) infrastructures across local governments and unequal access to technology and the internet among its citizens. The results of this study underscore the need for further exploration in adopting e-governance in the Philippines, focusing on impactful areas of study such as identifying and analyzing successful e-governance initiatives and assessing the effect of improved e-governance on citizen participation, government efficiency, and overall development.

On the other hand, the study of Urbina and Abe (2017) framed their assessment of adopting e-governance in the Philippines from the citizens' perspective. Through administering a survey on a nationally representative sample of 1,200 adults in the Philippines and subjecting the results of this survey using logistic regression analysis, the study was able to identify a gap between the attitude of citizens towards e-governance and its actual adoption. Although citizens have a positive attitude toward initiatives to develop the adoption of e-governance in the Philippines, the study determined that some factors hinder citizens from using services in line with these initiatives. Factors such as the lack of access to the internet and the lack of awareness when it comes to the existence of these services were determined to hinder citizen's usage of e-government services. This study's results and implications highlight a need to develop e-governance adoption in the Philippines.

Text Mining of Urban Policies

Text mining in the context of urban policies is a recently developed area of study pioneered in the study of Puri et al. (2018) which utilized a text-mining approach to gain valuable insights from urban policies and their contribution to the development of Smart

Governance. The study initially employed text mining on ordinances and tweets. Then, the study developed a mapping algorithm to link tweets and ordinances to each other and one or more relevant Smart City Characteristics (SCCs). The mapping algorithm achieved good results, with ordinances being mapped to relevant SCCs with a higher 80% range precision. However, the mapping of tweets to relevant SCCs was slightly less accurate, with a precision falling in the lower 80% range. The study provided valuable contributions to the development of smart governance by facilitating a more comprehensive analysis of public opinion on ordinances, which can ultimately lead to the development of citizen-centric urban policies vital in smart governance.

Leveraging the ordinance-tweet mapping methodology presented in the work of Puri et al. (2018), several subsequent studies have explored its potential for further development. Varghese et al. (2020) developed an app to disseminate information about the results of the ordinance-tweet mapping approach. The development of the app in this study increases public awareness of Smart Cities' legislative activity, which aligns with the principles of Smart Governance by promoting public transparency and involvement in decision-making processes. A similar study also tackling public awareness of the result of the ordinance-tweet mapping approach can be found in the study of Du et al. (2020). The study developed a web-based platform portal called Large Scale Ordinance Mining Portal (LMSOP) to disseminate the results of the ordinance-tweet mining procedure. The development of this portal fosters transparency and potentially encourages public participation as citizens become more informed about local legislation and its connection to public sentiment, which furthers the initiatives in developing smart cities.

The works presented above in the context of text mining of urban policies show the remarkable potential of text mining for analyzing urban policies and fostering Smart Governance. The success of these studies suggests that text mining can be a powerful tool for analyzing the urban policies of various cities. This analysis of urban policies can serve as a powerful catalyst to foster the exchange of best practices across cities, ultimately contributing to a knowledge-sharing network that fosters C2C learning and enhances urban competitiveness.

Latent Dirichlet Allocation (LDA)

In text mining, a process called topic modeling is employed to gain valuable insights from textual data. In essence, topic modeling is an algorithm used in text mining to identify themes within and across unstructured sets of textual data (Blei et al., 2003). Among the topic modeling techniques developed, the most widely used is Latent Dirichlet Allocation (LDA) (Vayansky and Kumar, 2020). This widespread adoption of LDA stems from its superior performance compared to other topic modeling techniques. Numerous studies have validated this claim. For instance, the study of Bellaouar et al. (2021) showed that compared to Latent Semantic Analysis (LSA), LDA is demonstrably faster at processing data and produces more thematically coherent topics. Additionally, Anantharaman et al. (2019) highlighted that LDA outperforms LSA and Non-negative Matrix Factorization (NMF) when classifying large text documents. On the other hand, when extracting meaningful topics from short texts, LDA proved to be a strong contender for generating valuable, diverse, and meaningful topics as highlighted in the study of Albalawi et al. (2020). Although LDA is effective and efficient as a topic modeling technique in text mining, it still has limitations. In LDA, the model assumes that textual

data within the analyzed collection documents share a single underlying generation process which can make it less effective in capturing how other factors, like author or time of creation, influence the thematic content (Mishler et al., 2015).

Structural Topic Modeling (STM)

To address the limitation of an LDA model, a variant of LDA called Structural Topic Modeling (STM) was developed. In an STM model, documents are still modeled as mixtures of topics, with each topic defined by a word probability distribution, similar to LDA's core structure. The difference between an LDA model and an STM model can be seen in the STM's introduction of document-level metadata that influences both the topical prevalence and topical content. In an STM model, topical prevalence refers to the degree to which a document is associated with a particular topic. In contrast, topical content is the specific words used within each topic (Roberts et al., 2019). The STM model allows for various combinations of these metadata, exploring how factors like author, time period, or document source influence the thematic landscape of their document collection. This flexibility goes beyond LDA's single underlying generative process, offering an understanding of how external variables shape the topics discussed in different documents. The expanded capabilities of STM have increased its utility in a wide range of studies. For instance, among the first notable works in the field of STM was the study of (Roberts et al., 2014) which aimed to automatically analyze open-ended survey responses from the 2008 American National Election Survey (ANES). The STM model's employment in the study identified political-science-related topics in the surveys, such as "economy" and "war." The STM model also revealed election-specific topics like "Obama." The study highlighted STM's promising and efficient potential as an alternative for analyzing textual

survey data. Similarly, the study of Bai et al. (2021) also utilized STM to identify major research themes and trends across studies on maritime transport by analyzing over 3,000 articles. The analysis revealed key themes in the field, like container operations and port management, alongside a shift in research focus towards efficient, sustainable transportation practices, which can provide insights for practitioners and policymakers on future research agendas and funding priorities.

The studies mentioned above highlight how valuable topic modeling is in analyzing textual data. The capabilities of topic modeling techniques like LDA and STM have proven to be powerful tools for gaining valuable insights from the extracted topics from textual data.

Table 1. A summary of the studies mentioned in the literature review.

Authors, Year	Field of Study	Methodology	Key Finding
Enseñado (2024)	City-to-City Learning	Systematic literature review on C2C learning.	Growing interest in C2C learning, potential to address issues in climate change, transport, and local governance.
Haupt et al. (2020)	City-to-City Learning	Examined studies on local climate governance.	Potential of TMNs as platforms for knowledge exchange in C2C learning.
Ilgen et al. (2019)	City-to-City Learning	Empirical, in-situ analysis of C2C learning in water squares.	Confirmed successful city-to-city knowledge exchange for water resilience.
Ndebele-Murisa et al. (2020)	City-to-City Learning	Examined C2C learning for climate resilience in African cities.	Identified water insecurity as a major challenge for these cities.

Table 1 (cont.).

Singh and Sahu (2018)	E-governance	Classification approach in a literature review on e-governance.	Supportive government policies and citizen literacy are key factors for successful implementation.
Manoharan et al. (2023)	E-governance	Assessed government websites across 100 wired cities.	Overall increase in e-government performance based on e-government capacity scores.
Siar (2007)	E-governance	Examined e-governance adoption in Philippine city governments.	Weak ICT infrastructure and unequal access to technology hinder adoption.
Urbina and Abe (2017)	E-governance	Survey on citizen perspective on e-governance adoption in the Philippines.	Positive attitude towards initiatives but lack of access and awareness hinder citizen usage.
Bellaouar et al. (2021)	Latent Dirichlet Allocation (LDA)	Compared LDA to LSA for topic modeling.	LDA is demonstrably faster and produces more thematically coherent topics.
Anantharaman et al. (2019)	Latent Dirichlet Allocation (LDA)	Compared LDA to LSA and NMF for classifying large text documents.	LDA outperforms LSA and NMF.
Albalawi et al. (2020)	Latent Dirichlet Allocation (LDA)	Examined the use of LDA for short texts.	Effective in generating valuable, diverse, and meaningful topics.
Bai et al. (2021)	Structural Topic Modeling (STM)	Used STM to analyze research trends and themes of studies about maritime transports.	Identified topics like “port management” from studies aiding policymakers and practitioners.

Table 1 (cont.).

Roberts et al., (2019)	Structural Topic Modeling (STM)	Used STM to analyze open-ended survey responses from the 2008 American National Election Survey.	Identified topics like “war” and “economy” from the open-ended survey responses.
Puri et al. (2018)	Text Mining of Urban Policies	Text mining approach to analyze ordinances and tweets.	Developed a mapping algorithm to link them to Smart City Characteristics (SCCs).
Varghese et al. (2020)	Text Mining of Urban Policies	Developed an app to disseminate information on ordinance-tweet mapping.	Increased public awareness of Smart Cities' legislative activity.
Du et al. (2020)	Text Mining of Urban Policies	Developed a web-based platform for ordinance-tweet mining results.	Promoted transparency and public engagement.
Sgambati and Gargiulo (2022)	Urban Competitiveness	Bibliometric analysis on urban competitiveness.	Highlighted core definitions and frameworks, identified emerging trends.
Yuan et al. (2017)	Urban Competitiveness	Developed a novel approach for measuring urban competitiveness in China.	Identified a significant gap between top and bottom-performing cities.
Sáez & Periañez (2015)	Urban Competitiveness	Created an urban competitive index (UCI) for ranking European cities.	Major Western European cities outperformed Eastern ones in attracting business and investment.
Development of Trade Industries' (DTI) Cities and Municipalities Competitiveness Index (CMCI) (DTI, n.d.)	Urban Competitiveness	Framework for benchmarking and ranking Philippine cities.	Identified key areas: economy, governance, infrastructure, resilience, and innovation.

MATERIALS AND METHODS

The study employed three (3) main processes: data collection, text mining, and visualization. The text mining involves two main processes which are the e-governance index development and the structural topic modeling. First, ordinances were gathered from their respective websites. These ordinances were then processed to develop an e-governance index, which serves as a proxy measure of e-governance initiatives for each city. Next, the ordinances underwent a text mining technique called structural topic modeling (STM). This method identifies topics which are underlying themes within the ordinances, allowing for deeper insights into their content. The study then analyzed how various factors—such as the developed e-governance index, the CMCI score, and the city of origin—affect the topics generated by the STM. Finally, the results of the structural topic modeling were visualized through a website. This platform showcased the discovered topics, their relationships with the selected variables, and the most representative ordinances for each topic. The visualization of the methods is shown in the diagram below.

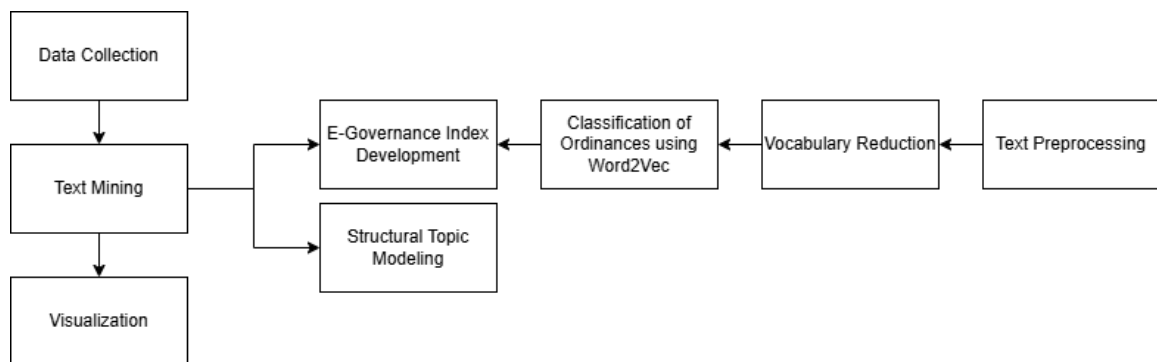


Figure 1. Overview of the workflow for the methods of the study.

Data Collection

Ordinances were collected from city websites classified as HUCs by the DTI. Web scraping techniques, including parsing and extracting text from HTML, as well as dynamically and automatically downloading files, were used to gather data from these websites. When extracting text data from the websites, key information such as the ordinance description and ordinance number was collected. Additionally, PDF files containing the full text of each ordinance were automatically downloaded and stored in a designated local folder. To extract text from the PDF files, Optical Character Recognition (OCR) was employed. This technique allowed for the conversion of ordinance documents into machine-readable text. OCR was particularly useful for ordinances stored as images, as it enabled text extraction even from scanned or non-editable PDFs. For PDFs that already contained selectable text, OCR automatically processed and converted them into plain text format. After collecting the ordinances, their data—including descriptions, ordinance numbers, full text, and city of origin—was compiled into an Excel spreadsheet. Each row in the spreadsheet represented a single ordinance, while each column contained specific ordinance details.

Text Mining: Developing of E-Governance Index

The development of the e-governance index involved four (4) key processes: text preprocessing, text transformation, ordinance classification, and e-governance index computation. First, the collected text data from the ordinances underwent text preprocessing techniques to clean the content. Next, text transformation was performed to convert the processed text into a structured format suitable for analysis and classification. After transformation, a similarity-based approach was applied to classify the ordinances

into three (3) predefined e-governance categories. Finally, the e-governance index was computed based on the classification results. This index provided a quantitative measure of e-governance implementation across the collected ordinances.

Text preprocessing

The collected ordinances were subjected to text preprocessing techniques including stop word removal, lowercase conversion, stemming, numerical character removal, duplicate word removal and tokenization.

Text transformation

The transformation of the preprocessed ordinances involved computing the Term Frequency-Inverse Document Frequency (TF-IDF) score for each term in its vocabulary. The TF-IDF scores were then plotted as a histogram to visualize their distribution, ultimately helping to determine an appropriate threshold. This threshold was then set based on the histogram to filter out less significant terms, thereby reducing the vocabulary size.

Classification of ordinances

Word2Vec is a technique that creates a vector representation of words developed by Mikolov et al. (2013). These vector representations capture the meaning and relationships of the words at play. Word2Vec uses text data as input and creates vectors for each word in said data. A vocabulary of the text data is first constructed and then the algorithms used in this technique learn the vector representation of the words from the text data. Distributional hypothesis is the key principle on which Word2Vec is generated on. This hypothesis states that words appearing in similar contexts tend to have similar meaning. Primarily, Word2Vec uses two training approaches to generate the vector

representation for each word. These methods include the Continuous Bag of Words (CBOW) and Skip-gram.

Using the CBOW approach allows the algorithm to predict a target word based on the words that come before and after it in a sentence. For an example sentence, “The sunflowers are yellow”, and the target word, “sunflowers”, the model will use the words “The”, “are”, and “yellow” to guess that “sunflowers” is the missing word. In this approach, the order of the surrounding words does not matter. The model will then convert each surrounding words into a vector. These vectors are then averaged together to create a combined vector representation of the context of the target word. This combined vector representation is then used to predict the target word. CBOW is an efficient model as it makes the model faster by removing unnecessary complexity and it can process large amounts of text data quickly while still learning useful relationships between words. The training complexity for this model architecture is computed using the equation below:

$$Q = N \times D + D + \log_2(V) \quad [1]$$

where N is the number of context words, D is the dimensionality of the word vectors and V is the size of the vocabulary.

The Skip-gram, unlike the CBOW model, uses a single target word to predict the words before and after the target word within a certain range, called the context window. If the context window is 5, then the algorithm will randomly choose 5 words before and after the target word to learn from. Using the same sentence “The sunflowers are yellow” with the target word “sunflowers”, the model will predict the words “The”, “are” and “yellow”. This approach allows the model to give more importance to words closer to the target word and less importance to distant words by sampling fewer distant words during

training. The training complexity for this model architecture is computed using the equation below:

$$Q = C \times (D + \log_2(V)) \quad [2]$$

where C is the context window, which is the maximum distance of the surrounding words considered, and the other variables are the same as those in the CBOW model architecture.

Different parameters and training process are involved in creating Word2Vec models. The above-mentioned are the two main model architecture that can be used in creating a model. On the other hand, Word2Vec has also configuration parameters to determine the model's behavior during training to allow the model to perform and learn effectively. In the context of Word2Vec, these configuration models include window size, vector size and learning rate. The window size of a Word2Vec model determines the number of words considered around the target word. On the other hand, the vector size determines the size or dimension of the word embeddings. Finally, the learning rate denoted by η which controls the adjustments of weights during training to minimize error in predicting the vector representation for each word. Another thing to be considered when creating a Word2Vec model is its optimization techniques. When creating a Word2Vec model, there are two options for optimization. One is by using hierarchical SoftMax for faster computation through its use of a binary tree representation for the vocabulary in computing the SoftMax layer. Another optimization technique that can be used in creating a Word2Vec model is negative sampling wherein the computation complexity is reduced by randomly sampling a small number of negative examples along with positive examples rather than using the entire vocabulary to update the weights. These parameters and

techniques determine the behavior and efficiency of Word2Vec models influencing the quality of the vector representation for each word.

Word2Vec model selection

The study employed an exploratory approach to determine the most appropriate Word2Vec model for the ordinances. It trained its own model and tested other pretrained models under Word2Vec. To classify the ordinances, the study built upon the methodology used introduced by Jatnika et al., (2019) wherein Word2Vec was used to compute the semantic similarity between English words. In the study, the Word2Vec model was created from a corpus of 320,000 articles from the English Wikipedia. A cosine similarity calculation was then used to determine how similar two words are in said corpus by computing the angle between two n-dimensional vectors which is defined by the equation below:

$$\text{similarity} = \cos \theta = \frac{\bar{x} \cdot \bar{y}}{||\bar{x}|| ||\bar{y}||} \quad [3]$$

Each word in the corpus obtained a cosine similarity in said study generated by the Word2Vec model. These cosine similarities for each word pair were then validated or tested for their accuracy by comparing them with data sets containing word pairs with human rated similarity scores from the WordSim-353 and SimLex-999 dataset. The comparison was conducted using the Pearson correlation, which measures the linear relationship between two datasets—in this case, the model's similarity scores and the human-rated scores. This method allowed the researchers to quantify how closely the model's output matched human judgment. This study compared the similarity scores generated by the Word2Vec model with human ratings from the WordSim-353 dataset and used the Pearson correlation to validate the model's accuracy. In addition to the Pearson

correlation, the study also applied the Spearman correlation to evaluate how well the model aligned with the ranked order of human judgment. This approach was based on Zhelezniak et al. (2019), which used the Spearman correlation as a measure of validity for model-generated similarity scores. Both Pearson and Spearman correlations were used in this study to identify the most appropriate model and to determine which model best aligned with human judgment. The usage of these correlations was employed in this study to determine which Word2Vec model aligned best with human judgment.

Grouping of ordinances by category

The United Nations established the e-Government Development Index (EGDI) to gauge a country's readiness for electronic government services (UN, 2022). This index traditionally relies on three sub-indices: the Online Service Index (OSI), the Telecommunication Infrastructure Index (TII), and the Human Capital Index (HCI). Currently, each sub-index is calculated by assigning points based on survey responses. Instead of awarding points, this study will explore a mapping technique that will link terms within the document-by-term matrix to the relevant e-government aspects addressed by each sub-index. These aspects encapsulate the key components of e-governance. This approach will use frequency with which ordinances are mapped to specific e-governance component/s. Table 2 outlines each element along with its corresponding definition:

Table 2. E-Governance Components Defined by the sub-indices of the UN's E-Government Development Index (EGDI).

E-Governance Component	Definition
Online Services	The scope and quality of online services. Refers to the comprehensiveness and effectiveness of government services offered through official online channels
Telecommunication and Infrastructure	Assesses the development status of telecommunication infrastructure. Measures how well-developed a region is in terms of its communication networks.
Human Capital	Focuses on a nation's inherent skills, knowledge, and abilities embodied in its population. Measures the potential of a country's workforce based on the level of education and health of its citizens

After the Word2Vec model was selected, the preprocessed ordinances will be classified into the categories mentioned above. A similarity-based matching process will be conducted by comparing the vector representation of each ordinance to a curated list of keywords corresponding to the following e-governance components. Table 3 details the curated keyword list for each e-governance component ensuring that it captures the definition of each component and that no two words are assigned to the same category. These keywords were obtained from the United Nations E-Government Survey 2022.

Table 3. Keyword list for each e-governance component.

E-Governance Component	Keywords
Online Services	"Online", "Digital", "Access", "Navigation", "Information", "Technology", "FAQs", "AI", "Chatbot", "W3C", "Data", "Portals", "Page", "Search", "Engine", "Site", "Framework", "Link", "Content", "Channels", "Feedback", "Browser"
Telecommunication and Infrastructure	"Telecommunication", "Mobile-cellular", "Subscription", "Cable", "Modem", "Wireless", "Broadband", "Telephone", "Analogue", "Postpaid", "Prepaid", "Computers", "Televisions", "Mobile", "Cellphone", "User", "Network", "Gadgets"
Human Capital	"Human", "Adult", "Literacy", "Enrolment", "School", "Child", "Education", "Enrolled", "Schooling", "Population", "Youth", "People"

For each ordinance, the cosine similarity between the ordinance's vector and each category's keyword list will be computed. This similarity score quantifies how closely an ordinance aligns with each e-governance category. To determine inclusion, a similarity threshold will be set based on the mode of the similarity score distribution for each category. The similarity threshold for each category was obtained by analyzing the distribution of similarity scores across all ordinances. Specifically, a histogram of the similarity scores was plotted for each category. The mode of the similarity score distribution was then identified, and a threshold range will be established by including scores within 1% of the mode. This ensures that the most representative ordinances for each category are included while allowing for slight variations in similarity. This approach allows for multi-class classification, meaning ordinances can be assigned to more than one category if they meet the similarity criteria.

E-governance index computation

The EGDI for each city was then calculated based on the number of ordinances classified under each category, which served as the score for each e-governance component. These scores will then undergo a min-max normalization to account for potential differences in the total number of ordinances analyzed per e-governance component. This ensures all e-governance components contribute equally to the final score. Finally, the EGDI is calculated as the average of these three normalized scores, which is defined as:

$$\text{EGDI} = \frac{1}{3} (\text{OS}_{\text{normalized}} + \text{TI}_{\text{normalized}} + \text{HC}_{\text{normalized}}) \quad [4]$$

The EGDI, along with the CMCI score for each city collected from DTI's websites, were added to the Excel spreadsheet containing all the ordinances collected for use in the next steps.

Text Mining: Structural Topic Modeling (STM)

Topic modeling using LDA

Latent Dirichlet Allocation (LDA) is a generative probabilistic model used in analyzing collections of documents. LDA operates under the assumption that each document can be represented as a probabilistic mixture of latent topics, where each topic reflects a distinct underlying theme. Each latent topic consists of words associated with that specific topic. The generative process starts with choosing N , which is the size of vocabulary, from a Poisson distribution. Next Θ , the topic proportion, which indicates how much each topic makes up of the overall document's content, is chosen from a Dirichlet distribution. Then, it randomly selects a topic z_n from the multinomial distribution defined by Θ , and based on this topic, it chooses the most likely word w_n from the multinomial distribution parameterized by β . This ensures that each word is generated according to the topic structure and associated word distributions (Blei et al., 2003). Figure 2 shows the generative process of an LDA model for a document in a corpus.

Algorithm 2: Generative Process of LDA

1. Choose $N \sim \text{Poisson}(\zeta)$
 2. Choose $\Theta \sim \text{Dir}(\alpha)$
 3. For each of the N words in w_n
 - Choose a topic $z_n \sim \text{Multinomial}(\Theta)$
 - Choose a word w_n from $P(w_n|z_n, \beta)$, a multinomial probability conditioned on the topic z_n
-

Figure 2. The generative process expressed in pseudocode of LDA (Lifted from Blei et. al., 2003)

Topic modeling using STM

Similar to LDA, structural topic modeling (STM) constructs latent topics from documents by assuming that documents are formed through a process based on these latent topics. The key difference between STM and LDA lies in the ability of STM to incorporate document-level metadata X_d into the analysis. This metadata, such as author or publication date, allows STM to explore how these external characteristics might influence the prevalence or content of topics within documents.

The STM model was applied to an Excel spreadsheet containing information on ordinances, including their descriptions, full text, city of origin, ordinance numbers, EGDI, and CMCI scores. To generate topics, the model used document-level metadata based on two key measures: the developed EGDI and the DTI's CMCI score. The generative process of an STM model starts with considering document-specific covariates or metadata in the vector X_d . This vector is then transformed using a coefficient matrix γ to influence the document's attention toward each topic represented by the vector $\overline{\theta}_d$. This attention is drawn from a logistic-normal distribution with mean $\mu = X_d\gamma$ and covariance matrix Σ . This ensures the topic proportions are positive and sum to one.

Next, the model builds a document-specific word distribution for each topic k . This distribution incorporates a baseline word usage vector m , common across all topics. It then adds topic-specific deviations $\kappa_k^{(t)}$ to capture how word usage differs for each topic. Additionally, the model can consider a content covariate y_d and its interaction with $\kappa_{y_d}^{(c)}$ and $\kappa_{y_d,k}^{(i)}$ to explore how content characteristics influence word choice within a specific topic. These components are combined and exponentiated to create the document-specific word distribution for topic k , denoted by $\beta_{d,k}$. In the absence of a content covariate, the

distribution simplifies to just the baseline and topic-specific deviations denoted by $\beta_{d,k} \propto \exp(m + \kappa_k^{(t)})$.

Finally, the STM generates words for each document position n within document d . It first assigns a topic $z_{d,n}$ by drawing from a multinomial distribution based on the document's overall topic attention $\overrightarrow{\theta_d}$. Given the assigned topic $z_{d,n}$, the model then selects the specific word $w_{d,n}$ from another multinomial distribution based on the document-specific word distribution for that topic $\beta_{d,k=z_{d,n}}$. The generative process of an STM model as outlined by Roberts et al. (2019) is shown below.

Algorithm 3: Generative Process of STM

1. Draw the topic proportion for a document using a logistic-normal model that considers document covariate X_d

$$\overrightarrow{\theta_d} | X_d, \gamma, \Sigma \sim \text{LogisticNormal}(\mu = X_d \gamma, \Sigma) \quad [5]$$

Where X_d is a $1 \times p$ vector

γ is a $p \times (K - 1)$ matrix coefficients

Σ is a $(K - 1) \times (K - 1)$ covariance matrix

2. For each topic k , form the document-specific word distribution given the document-level covariate y_d using the baseline word distribution m , topic-specific deviation $\kappa_k^{(t)}$, the covariate group deviation $\kappa_{y_d}^{(c)}$, and the interaction between the two $\kappa_{y_d,k}^{(i)}$

$$\beta_{d,k} \propto \exp(m + \kappa_k^{(t)} + \kappa_{y_d}^{(c)} + \kappa_{y_d,k}^{(i)}) \quad [6]$$

Where $m, \kappa_k^{(t)}, \kappa_{y_d}^{(c)}, \kappa_{y_d,k}^{(i)}$ are V -length vectors with each component corresponding to a particular word in the vocabulary.

3. For each word in the document $n \in \dots, N_d$
 - Draw a word topic assignment based on the document-specific distribution over topics.

$$z_{d,n} | \overrightarrow{\theta_d} \sim \text{Multinomial}(\overrightarrow{\theta_d}) \quad [7]$$

- Conditional on each topic chosen, draw an observed word from the topic

$$w_{d,n} | z_{d,n}, \beta_{d,k=z_{d,n}} \sim \text{Multinomial}(\beta_{d,k=z_{d,n}}) \quad [8]$$

Figure 3. The generative process expressed in pseudocode of STM (Lifted from Roberts et al., 2019).

Hyperparameter estimation

The STM model involves several key hyperparameters that influence the quality and interpretability of the topic estimation. One of the most crucial hyperparameters is the number of topics K , which defines how many distinct themes or topics will be extracted from the corpus. Another essential parameter is the initialization method (`init.type`), which determines how the STM model initializes its parameters. The model offers three methods: "Spectral," the default method recommended for its stability using moment-based estimation; "LDA," which uses collapsed Gibbs sampling; and "Random," which assigns random starting values to test model variability. Prevalence covariates allow the incorporation of external metadata to model how document-level factors influence topic proportions. In this study, the CMCI score, the developed e-governance index, and the city of origin were considered as this type of covariate. These were specified using R formula syntax, with the option to model non-linear relationships using splines. Content covariates, on the other hand, control how external metadata influences the language used within topics, which is useful for analyzing how different features change the discussion across categories.

The model's optimization process is regulated by several hyperparameters. Maximum EM iterations (`max.em.its`) sets the upper limit of Expectation-Maximization algorithm iterations, with a value of 100 used in this study to ensure sufficient optimization. Convergence tolerance defines the threshold for stopping optimization when the relative change in the variational objective falls below a specified value, which is set to $1e-5$ to ensure high precision. Regularization is handled through `sigma.prior` and `gamma.prior`. The `sigma.prior` controls the prior distribution on the topic covariance matrix, where a value of

0 applies maximum likelihood estimation and a value of 1 assumes a diagonal covariance matrix, affecting how strongly topics are allowed to correlate. The `gamma.prior` regulates the prevalence covariates using L1 (Lasso) regularization to encourage sparsity by shrinking small coefficients to zero or L2 (Ridge) regularization to prevent overfitting while preserving all variables. Similarly, `kappa.prior` regulates the content covariates and allows fine-tuning of the model's ability to capture nuanced language variations within topics by using different regularization techniques such as L1 for sparsity.

To assess the model's predictive performance, held-out likelihood estimation is used by splitting the dataset into training and validation subsets. This process evaluates how well the model generalizes to unseen data, ensuring the robustness of the topic identification. Additionally, document block updates (`ngroups`) enhance computational efficiency by dividing the corpus into groups and updating global parameters incrementally, which reduces memory consumption and speeds up convergence. The selection of these hyperparameters significantly affects the quality of the generated topic models.

In selecting the optimal STM model, the study systematically evaluated various hyperparameter combinations and assessed model quality through detailed diagnostic metrics. Specifically, the models were judged based on semantic coherence, exclusivity, residuals, and heldout likelihood, each of which provided unique insights into the model's quality. As part of the initial testing phase to determine an appropriate range of topics, the number of topics (K), the spectral initialization method was used. This spectral initialization is based on the algorithm developed by Mimno and Lee (2014) which automatically estimates the number of topics by identifying anchor words in a low-

dimensional representation of the word co-occurrence matrix. These anchor words serve as indicators of distinct topics, to approximate the number of latent themes present in the corpus without requiring a predefined K . This approach provides a starting point for selecting the optimal number of topics, which will be used for the informed selection of a narrower, more refined range of topic numbers to be further evaluated in the succeeding phase of model testing.

Semantic coherence was a critical metric used to measure the human interpretability and internal consistency of each topic. Specifically, the study employed the C_v coherence measure proposed by Röder et al. (2015), recognized for its strong correlation with human judgment. This metric quantifies coherence in four steps: First, it divides the top N words of a topic into word pairs. Second, it calculates the probabilities of word pairs using a sliding window approach, effectively capturing word proximity rather than simple co-occurrence. Third, it determines a confirmation score for each pair to reflect how strongly one set of words supports another, considering their similarity relative to all words in the topic. Finally, these confirmation scores are averaged across all pairs to produce a single coherence value. Formally, the coherence calculation involves the Normalized Pointwise Mutual Information (NPMI), represented by the following equations

$$\tilde{v}(W') = \left\{ \sum_{w_i \in W} , \text{NPMI}(w_i, w_j)^\gamma \right\}_{j=1, \dots, |W|} \quad [9]$$

where the NPMI is defined as:

$$\text{NPMI}(w_i, w_j)^\gamma = \left(\frac{\log \frac{P(w_i, w_j) + \epsilon}{P(w_i)P(w_j)}}{-\log(P(w_i, w_j) + \epsilon)} \right)^\gamma \quad [10]$$

The vector similarity contributing to the confirmation measure is given as:

$$\phi s_i(\vec{u}, \vec{w}) = \frac{\sum_{i=1}^{|W|} u_i \cdot w_i}{\|\vec{u}\|_2 \cdot \|\vec{w}\|_2} \quad [11]$$

Another key metric, exclusivity, measures how uniquely words characterize individual topics adapted from the works of Bischof and Airoldi (2012) and Roberts et al., (2019), utilizes the FREX metric, balancing word frequency and exclusivity. It involves two key components: exclusivity ($\text{Excl}(w)$) and frequency ($\text{Freq}(w)$). Exclusivity quantifies the proportion of the probability of word w within a given topic compared to its probabilities across all topics. Frequency represents how commonly word w occurs across the entire corpus. These two measures are standardized using the empirical cumulative distribution function (ECDF), ensuring comparability across different scales. A weighting parameter, denoted by frexw , balances the emphasis between exclusivity and frequency, typically set to 0.7 to prioritize exclusivity. Exclusivity is then computed as follows:

$$\text{Exclusivity}(w) = \left(\frac{\text{ECDF}(\text{Freq}(w))}{\text{ECDF}(\text{Excl}(w))} \right)^{\text{frexw}} \times \text{ECDF}(\text{Excl}(w)) \quad [12]$$

Additionally, residuals provided insight into the extent of unexplained variation left within the corpus after fitting the model, with lower residuals indicating improved model effectiveness. Heldout likelihood further validated model generalizability by quantifying predictive accuracy on previously unseen data. On the other hand, model convergence was rigorously monitored through the variational lower bound (lbound), achieved via iterative Expectation-Maximization (EM) optimization, with convergence indicating stability and robust estimation (Weston et al., 2023). Through the usage of these metrics, the most optimal STM model was selected.

Posterior inference

In topic modeling, posterior inference refers to reversing the generative process to discover the latent topics from the document instead of generating the document based on the topics. In an LDA model, the posterior inference amounts to finding a solution for the following equation:

$$p(\theta, \phi, \mathbf{z} | \mathbf{w}, \alpha, \beta) = \frac{p(\theta, \phi, \mathbf{z}, \mathbf{w} | \alpha, \beta)}{p(\mathbf{w} | \alpha, \beta)} \quad [13]$$

On the other hand, in an STM model, the posterior inference is given by this equation:

$$p(\eta, \mathbf{z}, \kappa', \gamma, \Sigma | \mathbf{W}, \mathbf{X}, \mathbf{Y}, k, m) \propto \left[\prod_{l=1}^D \text{Normal}(\eta_l | X_l, \gamma, \Sigma) \left(\prod_{n=1}^N \text{Multinomial}(z_{n,l} | \theta_l) \times \text{Multinomial}(w_n | \beta_{l,k=z_{d,n}}) \right) \right] \Pr(\kappa) \Pr(\tau) \quad [14]$$

However, these equations become intractable to compute. This is where approximate inference algorithms come into play. For an LDA model, a collapsed Gibbs sampling (CGS), a sampling algorithm, is used to approximate the posterior inference of the topic model. CGS is an efficient sampling algorithm because it relies solely on the latent topic assignments $\mathbf{z}$ to calculate the posterior distribution (Porteous et al., 2008). The CGS process involves iteratively refining word-topic assignments without explicitly estimating intricate topic-specific parameters. This is achieved using Markov Chain Monte Carlo (MCMC). During each iteration, the algorithm considers the current topic assignments of all words and then updates the topic assignment for a specific word. This process progressively refines topic assignments across the documents. Subsequently, after sufficient iterations, estimates of topic proportions per document and topic word distributions can be derived from the sampled assignments (Griffiths and Steyvers, 2004). The algorithm begins by randomly assigning topics $\mathbf{z}$ to each word. It then progresses through a set number of iterations, each involving revisiting every word in the corpus. For each word $w[i]$, its current topic assignment $z[i]$ is removed from the relevant counts

$n_{d,k}$, $n_{k,w}$ and n_k . Then, a new topic is probabilistically assigned based on a combination of the document-topic co-occurrence counts $n_{d,k}$, word-topic co-occurrence counts $n_{k,w}$, and total word counts per topic n_k , along with Dirichlet priors represented by hyperparameters α and β for topics and words, respectively. By incorporating background knowledge, these priors curb any one topic or word from dominating. Through iterative updates of topic assignments and counts, collapsed Gibbs sampling then uncovers the corpus's latent thematic distributions (Darling, 2011). The algorithm of CGS is defined as:

Algorithm 4: The process of a CGS algorithm

Input: words $\mathbf{w} \in$ document $\mathbf{d}$

Output: topic assignments $\mathbf{z}$ and counts $n_{d,k}$, $n_{k,w}$ and n_k

begin

 randomly initialize $\mathbf{z}$ and increment counters

foreach *iteration* **do**

for $i = 0 \rightarrow N - 1$ **do**

$word \leftarrow w[i]$

$topic \leftarrow z[i]$

$n_{d,topic} -= 1; n_{word,topic} -= 1; n_{topic} -= 1$

for $k = 0 \rightarrow K - 1$ **do**

$$p(z = k | \cdot) = (n_{d,k} + \alpha_k) \frac{n_{k,w} + \beta_w}{n_k + \beta \times W}$$

end

$topic \leftarrow \text{sample form } p(z | \cdot)$

$z[i] \leftarrow topic$

$n_{d,topic} += 1; n_{word,topic} += 1; n_{topic} += 1$

end

end

return $\mathbf{z}, n_{d,k}, n_{k,w}, n_k$

end

Figure 4. The generative process expressed in pseudocode of CGS (Lifted from Darling, 2011).

On the other hand, the posterior inference of an STM model is achieved through variational approximation. This approximation leverages a fast semi-collapsed variational expectation-maximization (EM) algorithm. Within the variational EM algorithm of an STM, the Expectation step (E-step) encounters logistic normal variables that are statistically inconvenient to handle directly. To expedite convergence, a Laplace approximation is employed.

Topics Generated

The words associated with each topic are determined using four ranking methods: highest probability, FREX (frequency and exclusivity), score, and lift. These methods identify the top words for each topic based on different criteria. The following formulas explain how each method is computed.

$$\text{HighestProb}(w, k) = P(w|z = k) \quad [15]$$

Highest Probability ranks words based on their likelihood of appearing within a given topic, as inferred directly from the topic-word distribution parameter β . Specifically, $P(w|z = k)$ represents the probability of word w given topic k .

$$\text{FREX}(w, k) = \left(\frac{w_{\text{freqrank}}^{-1} \cdot w_{\text{exclrank}}^{-1}}{\omega \cdot w_{\text{exclrank}}^{-1} + (1 - \omega) \cdot w_{\text{freqrank}}^{-1}} \right) \quad [16]$$

FREX highlights words that rank highly in both frequency and exclusivity. It is calculated by taking the harmonic mean of the inverse frequency rank, denoted as w_{freqrank}^{-1} , and the inverse exclusivity rank, denoted as w_{exclrank}^{-1} where both the frequency and exclusivity are estimated from the distribution of the topic given the word $P(z = k|w)$. The FREX weighting factor, denoted by ω , is typically set to a default value of 0.5

$$\text{Lift}(w, k) = \frac{P(w|z = k)}{P(w)} \quad [17]$$

Lift identifies words that are particularly distinctive or indicative of a specific topic. It is calculated by dividing the topic-word distribution $P(w|z = k)$ by the empirical probability distribution of the word $P(w)$.

Score measures how much more likely a word is to appear within a specific topic compared to its average likelihood across all topics.

$$\text{Score}(w, k) = P(w|z = k) \cdot \log \left(\frac{P(w|z = k)}{P(w)} \right) \quad [17]$$

The formula for score resembles a pointwise mutual information (PMI)-style formulation that rewards both high topic relevance and uniqueness.

Text postprocessing

For postprocessing, various methods can be used to validate and assess the generated topics from the topic model. In this study, both the topics generated from the topic modeling were evaluated using various metric. Besides semantic coherence and exclusivity, which were also used in selecting the best STM model, a regression analysis was performed to measure how the covariates: the developed e-governance index and the CMCI score, affect the proportion of different topics in documents. In this regression model for this analysis assumes that the proportion of topic k in document d , denoted as θ_k^d is influenced by document-level covariates X_n^d where $n = 1, \dots, p$ (Roberts et al., 2016). The regression model is a type of Generalized Linear Model (GLM) denoted as:

$$\theta_d^k = \beta_0 + \beta_1 X_1^d + \beta_2 X_2^d + \dots + \beta_p X_p^d + \epsilon^d \quad [14]$$

where the dependent variable θ_d^k represents topic proportions, which are constrained between 0 to 1. Since the topic proportions are only between 0 to 1, this study employed logistic regression instead of the standard linear regression through ordinary least squares (OLS) since it is unsuitable for this kind of range (Hosmer et al., 2013). Since topic prevalence represents fractions of a document's content, a typical linear model could produce invalid predictions outside this range. A logistic transformation is then used that maps any real number to the range (0,1). This is denoted as:

$$\text{logit}(\theta_k^d) = \log\left(\frac{\theta_k^d}{1 - \theta_k^d}\right) \quad [15]$$

For estimating the regression coefficients β_p , this study used Bayesian inference. The usage of Bayesian inference is appropriate since the topic proportions produced are uncertain. The uncertainty of these topic proportions is quantified through full posterior distributions instead of standard errors and confidence intervals typically used in frequentist inference methods. The goal of the Bayesian inference is to model the probability of the observed topic proportions given the document-level covariates. Given a logistic regression model for topic proportions θ_k^d , the posterior distribution β to achieve this goal is denoted as:

$$P(\beta|\theta_k^d, X^d) = \frac{P(\theta_k^d|X^d, \beta)P(\beta)}{P(\theta_k^d|X^d)} \quad [16]$$

Through this regression analysis, the study demonstrated how each covariate influenced topic prevalence, highlighting the varying levels of the e-governance index and CMCI scores across different types of ordinances. This analysis provides insights into how certain kinds of ordinances are associated with higher or lower e-governance and CMCI scores, offering implications for policy development and governance effectiveness.

Visualization

The visualization involved creating an interactive website that showcased the topic modeling results. The website featured various outputs, including plots for the regression analysis and an interactive visualization of topics, highlighting their differences. Multiple functionalities were implemented, presenting the results through informative plots and graphs. The website also illustrated the relationship between the e-governance index score, CMCI scores, and the topics. This functionality utilized graphical displays, such as plots, to highlight these relationships, demonstrating how these indexes influenced the thematic content of the ordinances. Finally, a section of the website presented the most representative documents or ordinances for each topic, showcasing which ordinances were most relevant to each topic.

Hardware and Software Specifications

The algorithms employed in this study was implemented using R, a powerful statistical programming language. R's extensive library of readily available packages streamlines the process by providing pre-built functions for text analysis and the specific algorithms relevant to this study. The analyses were run on an ASUS laptop with an 11th Gen Intel Core i5 processor and 20.0 GB of RAM to ensure efficient computation.

RESULTS AND DISCUSSION

Data Collection

A total of 13,853 ordinances were collected from 22 cities classified as highly urbanized (HUCs) by the DTI's CMCI. Most of these ordinances came from Makati, Muntinlupa, Quezon, Pasay, Pasig, and Mandaue. Table 2 presents the top six cities, along with the number of ordinances collected from each and the range of years during which these ordinances were published or approved. The collected ordinances are available either as brief descriptions or as full texts containing all relevant information about the ordinance. Figure 5 is an example of an ordinance retrieved from the Pasay City ordinance website. Figure 6 is a sample page of a full-text ordinance, which includes all relevant information pertaining to the ordinance. The attributes extracted from the ordinances were encoded into a Google spreadsheet, with each row representing a single ordinance and each column capturing specific ordinance attributes. This spreadsheet served as the foundational dataset for the subsequent analysis. Figure 7 is a sample of how the extracted information was organized in Google Sheets.

Table 4. Top 6 cities by total number of collected ordinances and their corresponding year range of publication.

City	Number of Ordinances	Publication Year Range
Makati	5522	1988-2022
Muntinlupa	1607	1997-2024
Quezon	845	2010-2024
Pasay	774	1992-2024
Pasig	735	2008-2024
Mandaue	726	2007-2018

987	AN ORDINANCE PROVIDING FOR THE ADMINISTRATIVE SUPPORT SERVICES TO THE OFFICE OF SENIOR CITIZENS (OSCA) AND PROVIDING FUNDS THEREFORE.	SENIOR CITIZENS AND PERSONS WITH DISABILITIES	01/13/1998	VIEW
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Figure 5. Ordinance no. 987 from Pasig city's website with its corresponding category and date of publication.

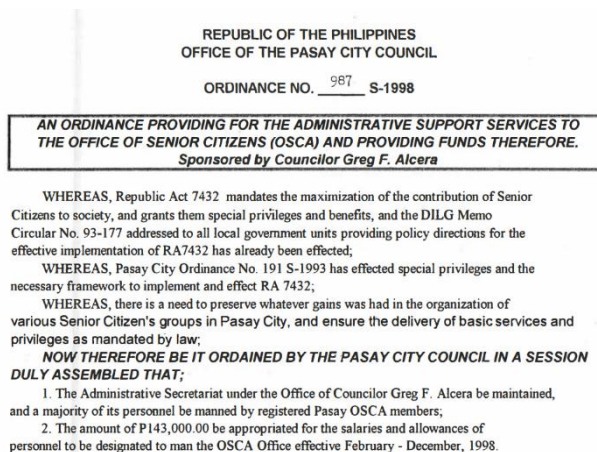


Figure 6. A snapshot of the first page of Ordinance No. 987 from Pasay city.

1	Ordinance Number	Ordinance Description	Full Text	City
2	AS OF 04-11-2024	ANCE ADOPTING THE	, equipment, mec	Pasay
3	989	SERIES OF 1997 WHIC	a Mite' a	Pasay
4	987	RVICES TO THE OFFIC	inistrative Secreta	Pasay

Figure 7. A snapshot of the Google Sheets spreadsheet containing the collected information from the ordinances.

Text Mining: Developing the E-governance Index from the Ordinances

Text preprocessing

The raw text data extracted from the “Ordinance Description” and “Full Text” columns underwent preprocessing to transform the unstructured data into a format for subsequent analysis. The text preprocessing yielded a list of lists of words with each list representing each ordinance. Below are the sample of the first 10 words yielded from the first 3 ordinances after preprocessing:

Table 5. Sample results showing the original ordinance text, city of origin, ordinance number, and the preprocessed ordinance.

Ordinance No.	City	Original Ordinance Description	After Preprocessing
Ordinance No. 6340 S. 2024	Pasay	An ordinance adopting the revised Pasay city revenue code.	['propos', 'amend', 'adopt', 'revis', 'revenu', 'code', 'republ', 'philippin', 'sangguniang']
Ordinance No. 989 S. 1998	Pasay	An ordinance providing for the mode and amount of payment for business plate number requirement of all business establishments in Pasay city as per Ordinance No. 977, Series of 1997 which requires all business establishments and occupational undertakings to register and secure a business permit registration plate before starting a business in the city.	['provid', 'mode', 'payment', 'busi', 'plate', 'number', 'requir', 'establish', 'seri']

Table 5 (cont.).

Ordinance No. 987 S. Pasay 1998	An ordinance providing for the administrative support services to the Office of Senior Citizens (OSCA) and providing funds therefore.	['provid', 'administr', 'support', 'servic', 'offic', 'senior', 'citizen', 'osca', 'fund']
---------------------------------	---	--

Each preprocessed ordinance contains an average of approximately 54 words.

However, some ordinances may end up with no usable words after preprocessing, particularly when a full-text version is unavailable. In such cases, only the ordinance description is used, which may be too brief to retain meaningful content. This limitation is often due to the structure of city ordinance websites. For instance, in cities like Las Piñas, full texts are not available online, and website restrictions prevent the downloading of ordinance files. As a result, only brief descriptions were collected for those entries. Conversely, when full-text versions were available, the complete ordinance documents were used as the primary data source.

Text transformation

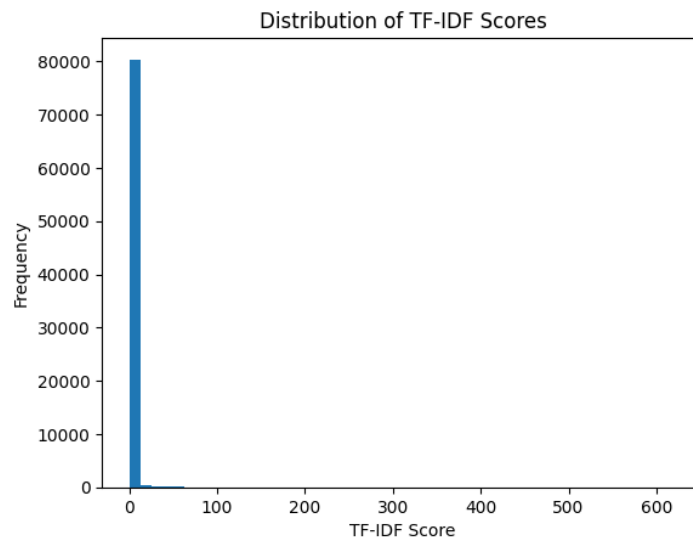


Figure 8. The distribution of TF-IDF scores across the initial vocabulary.

Table 6. Summary statistics of the TF-IDF scores.

Summary Statistic	Value
Min TF-IDF Score	0.0312272634770
Max TF-IDF Score	623.05596788443
Range	623.02474062095
Mean	0.9109076395240
2nd Quantile (Median)	0.0880503861859
3rd Quantile (75th percentile)	0.1739875473071
Standard Deviation	9.6919057794922

After computing the TF-IDF scores for each term from the initial vocabulary of the preprocessed ordinances, the distribution of these scores was plotted. Figure 8 shows the TF-IDF score distribution across the initial vocabulary. Also, to support a more in-depth analysis, summary statistics were computed, and the results are presented Table 6. The histogram revealed that a significant proportion of terms in the vocabulary had TF-IDF scores close to zero. A TF-IDF score near zero typically indicates that a term is either too common across many ordinances or not particularly relevant within the overall corpus. This observation is also reflected in the summary statistics. The distribution of TF-IDF scores is highly right-skewed, as indicated by the large difference between the mean (0.9109) and the median (0.0880), and the very high standard deviation (9.6919) relative to the mean. The maximum score of 623.055 compared to the minimum of 0.0312 further illustrates the wide dispersion in term importance across the corpus. To avoid an overly aggressive filtering, a threshold was set at the 25th percentile of the TF-IDF scores to remove low-value terms. By choosing the 25th percentile as a threshold, the filtering process excludes the lowest quartile of TF-IDF scores, which likely contains terms or documents with minimal contribution to the overall information structure. This approach resulted in the elimination of 20,415 words from the original vocabulary of 81,305 terms,

reducing it to 60,890 unique terms. The reduced vocabulary provided a more focused and informative representation of the ordinances for classification. The use of the 25th-percentile threshold is a well-grounded strategy, as it removes the bottom 25% of low-information terms—typically near-zero TF-IDF—while retaining the top 75% to avoid excessive pruning. This method also serves as a data-driven form of stop word removal, identifying uninformative terms based on their TF-IDF scores rather than relying on a static list. A similar filtering technique was employed by Lopez-Lira and Tang (2023) in their study evaluating ChatGPT’s reasoning capabilities in forecasting stock price movements.

Classification of ordinances

Following the vocabulary reduction, various Word2Vec models were developed and evaluated. A total of 130 custom Word2Vec models were trained on the dataset, alongside two pre-trained models, were evaluated. Among all the models evaluated, the pre-trained Word2Vec model “fasttext-wiki-news-subwords-300” was selected due to its superior performance, as demonstrated by its higher Spearman and Pearson correlation scores.

The Spearman and Pearson correlation scores of the pre-trained “fasttext-wiki-news-subwords-300” Word2Vec model are highlighted in bold. This model consists of 1 million word vectors trained on Wikipedia 2017, the UMBC webbase corpus, and the statmt.org news dataset, totaling approximately 16 billion tokens. Its performance surpassed that of both the pre-trained “word2vec-google-news-300” model—trained on a subset of the Google News dataset comprising around 100 billion words—and the two highest-performing Word2Vec models trained directly on the collected dataset, based on Spearman and Pearson correlation scores. The performance metrics are shown below:

Table 7. Word2Vec models with their corresponding Spearman and Pearson correlations evaluated against the WordSm-353 dataset.

1. Model	Spearman Correlation	Pearson Correlation
fasttext-wiki-news subwords-300	0.6361105762939401	0.62426605275059
word2vec-google-news- 300	0.6309060227466561	0.61940399464385
Trained model with a window size of 15 and a vector size of 290	0.4395250215380578	0.47150137714083
Trained model with a window size of 12 and a vector size of 250	0.4305793230755243	0.48052961097143

The selected Word2Vec model was then utilized to classify the ordinances based on their cosine similarity to predefined e-governance categories. To do this, both the ordinances and the categories were first represented as lists of preprocessed words. These words were then converted into vector representations using the Word2Vec model. For each ordinance, its similarity to a given category was calculated by comparing the words in the ordinance to the words defining the category. This involved measuring the cosine similarity between all possible word pairs—each word in the ordinance was compared with each word in the category. The resulting similarity scores from these pairwise comparisons were then averaged, producing a single similarity score that reflects how closely the ordinance aligns with that category. If an ordinance had no recognizable words, meaning the list representing that specific ordinance doesn't contain any word, a default similarity score of zero was assigned. This ensured that only valid comparisons contributed to the categorization process. Through this approach, each ordinance received a similarity score for each category, which was later used to determine its classification.

To determine the appropriate similarity threshold, the cosine similarity scores of all ordinances with respect to each e-governance category were visualized using histograms. The histogram plots below illustrate the distribution of similarity scores for each category. Following the histograms, summary tables present key statistics that describe the distribution of scores for each category

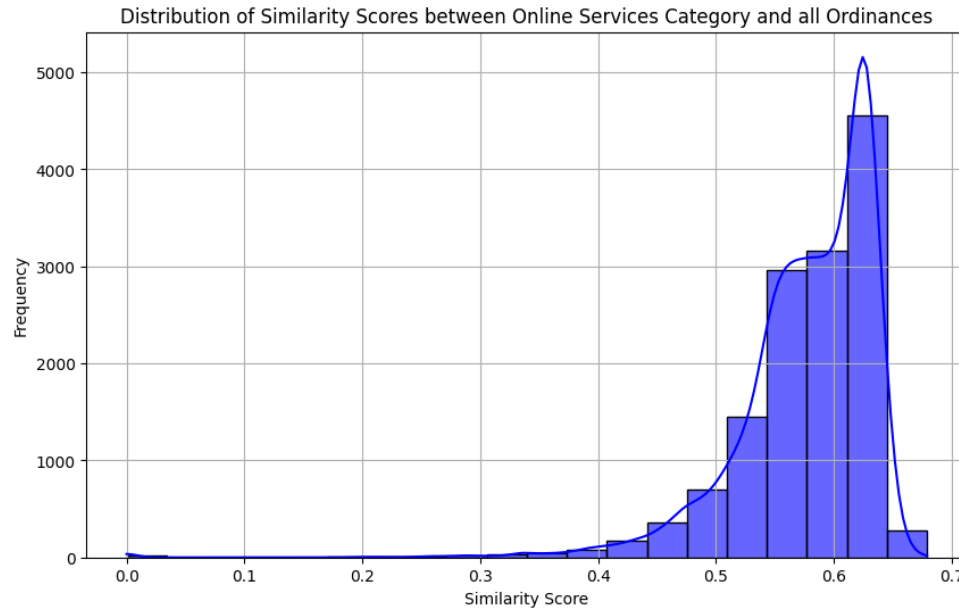


Figure 9. Histogram of the similarity scores between all the ordinances and the online services category.

Table 8. Summary statistics of the similarity scores between the ordinances and the online services category.

Summary Statistic	Value
Min Similarity Score	0.0000000000000000
Max Similarity Score	0.6791850328445435
Range	0.6791850328445435
Mean	0.5772445644939418
2nd Quantile (Median)	0.5887576341629028
3rd Quantile (75th percentile)	0.6225239038467407
Standard Deviation	0.0602894453149203

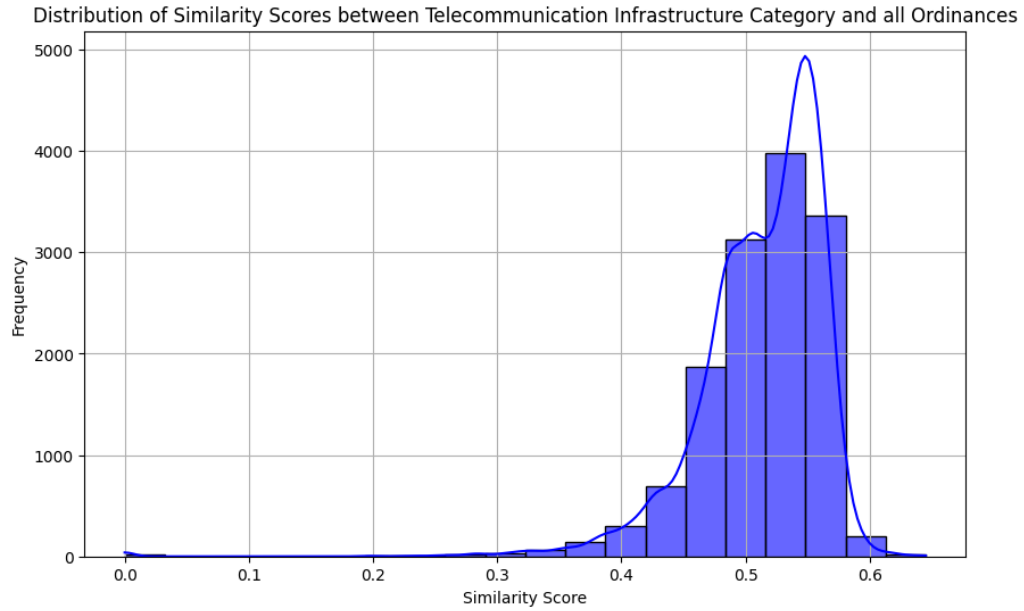


Figure 10. Histogram of the similarity scores between all the ordinances and the telecommunication infrastructure category.

Table 9. Summary statistics of the similarity scores between the ordinances and the telecommunication infrastructure category.

Summary Statistic	Value
Min Similarity Score	0.0000000000000000
Max Similarity Score	0.6452085375785828
Range	0.6452085375785828
Mean	0.5129398798147435
2nd Quantile (Median)	0.5225491523742676
3rd Quantile (75th percentile)	0.5491644144058228
Standard Deviation	0.0519512596135984

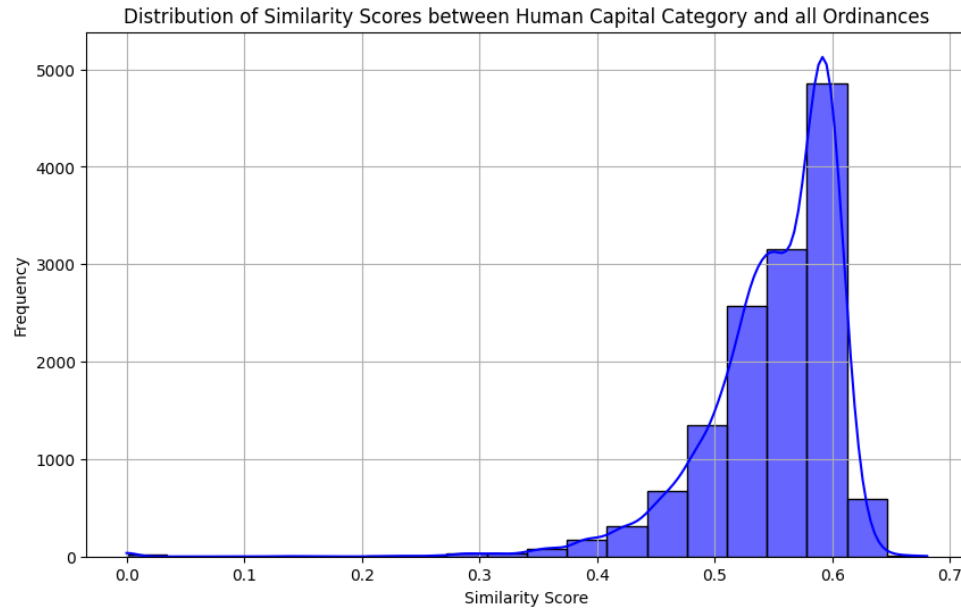


Figure 11. Histogram of the similarity scores between all the ordinances and the human capital category.

Table 10. Summary statistics of the similarity scores between the ordinances and the human capital category.

Summary Statistic	Value
Min Similarity Score	0.0000000000000000
Max Similarity Score	0.680492639541626
Range	0.680492639541626
Mean	0.549538346999631
2nd Quantile (Median)	0.562597692012787
3rd Quantile (75th percentile)	0.590995907783508
Standard Deviation	0.059041766835224

To determine the similarity threshold for each category, the mode of the similarity score distribution was used. Using the mode as the threshold implies that the most frequently occurring similarity scores are considered the most representative when comparing categories across all ordinances. To broaden the scope of the threshold, similarity scores below 1% of the mode were also included. This approach allows for the inclusion of scores that are close to the mode, as they are similarly representative of the

most common values resulting in a more inclusive classification process. Additionally, using a threshold enables multiclass classification, allowing ordinances to be categorized under more than one e-governance category. This is important, as an ordinance may address aspects that span multiple categories rather than being limited to just one. Through this, Table 11 shows the threshold scores obtained for each category. The Online Services category has a mode of 0.6248, resulting in a similarity threshold where an ordinance is classified under this category with a similarity score less than or equal to 0.6185. Similarly, the Telecommunication Infrastructure and Human Capital categories have modes of 0.5483 and 0.5919, respectively, with corresponding thresholds of less than or equal to 0.5483 and less than or equal to 0.5919. The Online Services category, having the highest mode among the three suggests that, overall, the ordinances are most similar in content to this category. This may indicate that many of the ordinances contain features aligned with the delivery of public services through official online channels, which is central to the definition of this category. In contrast, the Telecommunication Infrastructure category, having the lowest mode, implies that fewer ordinances closely align with this theme. This may suggest that the collected ordinances generally lack provisions or initiatives related to the development of telecommunication infrastructure, reflecting a potential gap in this area of e-governance.

Using this threshold, the ordinances were classified into the different e-governance categories. Tables 12, 13, and 14 present the number of ordinances classified under each category, ranked from highest to lowest, with only the top five cities shown for each category.

Table 11. The mode, along with the corresponding value 1% less than the mode, used to determine the similarity threshold.

Category	Mode	Low Threshold
Online Services	0.6247958410251606	0.618547882614909
Telecommunication Infrastructure	0.5483303787829997	0.542847074995170
Human Capital	0.5919400438054785	0.586020643367424

Table 12. Top 5 cities with the highest number of ordinances classified under the online services category.

City	Number of Ordinances Classified to Online Services category
Makati	2005
Mandaue	607
Pasay	543
Iloilo	183
Quezon	134

Table 13. Top 5 cities with the highest number of ordinances classified under the telecommunication infrastructure category.

City	Number of Ordinances Classified to Telecommunication Infrastructure category
Makati	1887
Pasay	697
Mandaue	604
Muntinlupa	231
Quezon	222

Table 14. Top 5 cities with the highest number of ordinances classified under the human capital category.

City	Number of Ordinances Classified to Human Capital Category
Makati	2097
Pasay	608
Mandaue	593
Quezon	195
Iloilo	184

Looking at the overall results, Makati, which has the highest number of ordinances, may have benefitted from this volume by ranking as the top city across all three e-governance categories. Focusing first on the Online Services category, one notable observation is that the top-performing cities tend to have well-developed government websites. These websites either organize ordinances effectively with complete information or, in the case of Mandaue, offer a comprehensive city web portal that includes various e-services such as a procurement system, a senior citizen system, and a document management system. Similarly, cities like Quezon and Pasay also feature a range of e-services on their official websites. This suggests that cities with ordinances strongly associated with the Online Services category tend to have robust and functional government web portals—effectively capturing the essence of this e-governance component.

Now, looking at the Telecommunication Infrastructure category, cities that ranked highly tend to have recently implemented significant improvements in their telecommunication systems. For example, a SunStar report detailed the final phase of Mandaue's citywide fiber optic network installation, completed on April 11, 2025. This initiative aimed to support high-resolution CCTVs with fully integrated systems across all 27 barangays (Virador, 2025). Another example is Quezon City's inclusion in the National Fiber Backbone (NFB) project, which seeks to improve internet connectivity and deliver more efficient and effective government services. The NFB is the country's first and only government-owned backbone network, spanning 1,245 kilometers and consisting of 28 nodes from Laoag, Ilocos Norte, down to Roces District in Quezon City (pco.gov.ph, 2024). These examples demonstrate how advanced telecommunications infrastructure

initiatives in these cities closely align with the core objectives of this e-governance component.

Finally, looking at the Human Capital category—since this component pertains to the education-related capabilities of a region—we can examine one of its indicators from the UN E-Government Survey: literacy rates. Cities like Makati, Pasay, and Quezon are all located in the National Capital Region (NCR), which recorded the highest literacy rate in the Philippines according to the 2020 Census of Population and Housing conducted by the Philippine Statistics Authority (PSA). (psa.gov.ph, 2020). The NCR achieved a literacy rate of 98.9%, indicating that these cities are part of a region with the most literate adult population.

These examples demonstrate how the classification of ordinances into the three e-governance categories (Online Services, Telecommunication Infrastructure, and Human Capital) aligns with observable real-world indicators—offering evidence that cities performing well in these categories also show strong performance in the actual factors those categories represent.

Conversely, the table below shows the top 5 cities with the highest number of ordinances that were not classified into any category.

Table 15. Top 5 cities with the highest number of ordinances not classified into any category.

City	Number of Ordinances Classified to Telecommunication Infrastructure category
Makati	2809
Muntinlupa	1285
Gensan	584
Quezon	553
Zamboanga	489

Table 16. Sample single and multiclass classification results using the similarity-based classification of ordinances.

Ordinance Number		Ordinance Description	Category	City
Ordinance 1406 S. 1999	No.	An ordinance establishing Pasay city tourism code and prescribing penalties for violations thereof.	Human Capital	Pasay
Ordinance No. 670: SP-2641, S-2017		An ordinance amending ordinance no. SP-2285, S-2014, entitled “An ordinance regulating the operation of electronic games (e-games), bingo games, and other games of chance, whether manual, mechanical, electronic internet broadband or otherwise, operating within the territorial jurisdiction of Quezon city and providing penalties for violations thereof.	Online Services and Telecommunication Infrastructure	Quezon

To assess the quality of the classification, both single and multiclass results were reviewed. Ordinances classified under specific categories showed thematic alignment with those categories. As the study used a unique cosine similarity-based approach, a qualitative evaluation was conducted by examining the content of each ordinance to determine whether its classification was contextually appropriate and logically consistent. Note that since the study employed Word2Vec, ordinances and categories were compared at the word level. As a result, there may be instances where ordinances are classified into categories

due to the presence of similar words, even if their overall content differs. For instance, an ordinance related to general infrastructure—such as road enhancement—might be classified under the telecommunication infrastructure category because of shared terminology. In such cases, the study still accepts these classifications as valid, given their conceptual relatedness.

Computing e-governance index

Using the formula used in the UN's 2022 E-Government survey in computing the e-governance index for each city, the following results were obtained for each city:

Table 17. The computed e-governance index for each city ranked from highest to lowest.

City	E-Governance Index
Cagayan de Oro	0.6666666667
Iligan	0.6666666667
Malabon	0.6666666667
Mandaue	0.5952380952
Manila	0.5701754386
Davao	0.5666666667
Quezon	0.5643939394
Muntinlupa	0.5386313466
Makati	0.5206349206
Pasay	0.4740259740
Zamboanga	0.4513888889
Valenzuela	0.4285714286
Iloilo	0.4000000000
Marikina	0.4000000000
Gensan	0.3988095238
Mandaluyong	0.3921568627
Pasig	0.3655913978
Caloocan	0.3611111111
Paranaque	0.3333333333
Las Piñas	0.3333333333
San Juan	0.3333333333
Bacolod	0.0000000000

Table 18. The CMCI score obtained from DTI for each city ranked from highest to lowest.

City	CMCI Score
Quezon	61.8475
Pasay	56.7826
Manila	56.2923
Iloilo	51.0439
Caloocan	49.5927
Muntinlupa	49.0816
Cagayan de Oro	48.6120
Makati	47.1135
Pasig	44.5875
Davao	44.5875
Valenzuela	44.2105
Gensan	42.1879
Iligan	42.0494
Mandaue	41.6661
Bacolod	41.1841
Mandaluyong	37.7354
Zamboanga	37.4007
Marikina	36.6775
Paranaque	36.3001
Las Piñas	35.1899
Malabon	34.8983
San Juan	34.1455

To provide a clearer comparison of how these two variables might influence each other, the CMCI score for each city is also considered. Table 18 presents the CMCI scores corresponding to each city ranked from highest to lowest. The analysis of the computed e-governance index was conducted in relation to each city's CMCI score, recognizing the significant connection between e-governance and urban competitiveness. E-governance and urban competitiveness go hand in hand, with e-governance often acting as a driver of urban competitiveness (Paskaleva, 2013). The implementation of ICTs through e-

governance allows local governments to remain adaptive and competitive in an increasingly digital world.

Among the top three cities with the highest CMCI scores, the following e-governance indices were recorded: Quezon City (0.5644), Pasay (0.4740), and Manila (0.6420). While Quezon and Manila achieved relatively high e-governance scores, Pasay City, despite its high CMCI score, obtained a noticeably lower e-governance index. While Pasay City shows a relatively low e-governance index, this may not necessarily indicate a lack of e-governance initiatives. For instance, Pasay City—despite being among the top cities contributing the most ordinances per e-governance category—also yielded a low e-governance index score. This discrepancy suggests that the issue may lie in the weighting used within the e-governance index scoring adopted from the UN’s 2022 E-Government Survey formula rather than in actual policy performance. In such cases, a low e-governance index despite a high CMCI score may not accurately reflect a city’s digital governance maturity but instead highlight potential limitations in how the index is constructed.

Conversely, the bottom three cities in terms of CMCI ranking—San Juan, Malabon, and Las Piñas—show a different trend. Their respective e-governance indices were 0.333 (San Juan), 0.6667 (Malabon), and 0.3333 (Las Piñas). While Las Piñas’ and San Juan’s score of 0.3333 aligns with its poor performance, Malabon present inconsistencies. Despite having low CMCI scores, Malabon achieved relatively high e-governance indices. Malabon had 106 ordinances, with just 1, 3, and 3 ordinances falling into each respective category. These low numbers would typically suggest a limited e-governance presence, yet both cities received moderately high indices. Again, this discrepancy points to a limitation in the e-governance formula. The method, which equally weights each category and bases

scoring on the number of ordinances classified per category, can lead to inflated indices for cities with very few but evenly distributed ordinances. As a result, cities with minimal actual engagement in e-governance-related legislation may appear to perform better than they do in practice.

Another perspective from which to interpret these results is through the lens of written urban policy. For instance, a low e-governance index in a city like Quezon—despite its high CMCI score—may indicate a lack of written ordinances explicitly related to e-governance initiatives. Conversely, cities like Malabon may have a high e-governance index but a low CMCI score, suggesting that while the city's overall competitiveness may be limited, it has documented a substantial number of ordinances tied to e-governance efforts. This implies that the presence or absence of ordinances plays a key role in shaping the e-governance index, regardless of whether the actual initiatives exist in practice. This is due to the e-governance index in this study being a proxy measure—derived from the number of ordinances categorized under specific e-governance components—rather than a direct assessment of on-the-ground implementation. As a result, cities with active but undocumented initiatives may score lower, while those with more written ordinances related to e-governance may score higher, even if actual implementation is limited.

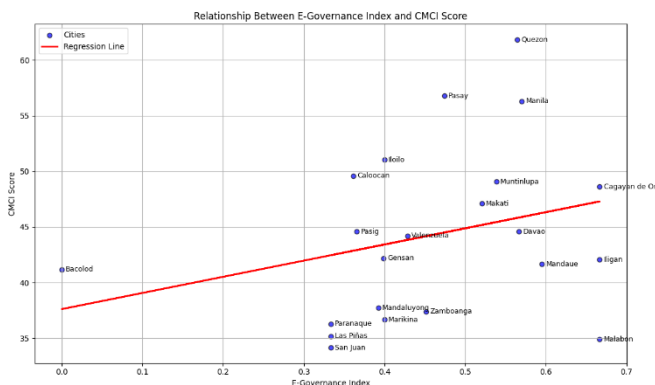


Figure 12. Scatter plot with regression line showing e-governance index vs. CMCI score.

Comparing each city's e-governance index and CMCI score individually may not fully capture the overall relationship between the two variables, especially given the limitations of the e-governance index formula. To gain a clearer and more accurate understanding of this relationship, a regression analysis was conducted to determine whether variations in the e-governance index are associated with changes in the CMCI score. The scatter plot in Figure 12, along with the regression line, illustrates the relationship between these two variables. The scatter plot illustrates a positive relationship between the e-governance index and the CMCI score. This suggests that cities with higher e-governance index values tend to have higher CMCI scores. While the data points appear somewhat dispersed, there is a noticeable upward trend indicating that increased e-governance is generally associated with stronger urban competitiveness. The slope and intercept of the regression line—14.5243 and 37.6157, respectively—support this relationship. Specifically, the CMCI score is expected to increase by approximately 14.52 for every unit increase in the e-governance index. Meanwhile, cities with no measurable e-governance initiatives (i.e., an index of 0) are projected to have a baseline CMCI score of around 37.62.

In summary, the regression plot validates the presence of a positive relationship between the e-governance index and CMCI score, indicating that cities with higher levels of e-governance tend to exhibit higher levels of urban competitiveness. This suggests that improvements in digital governance may contribute to a stronger overall competitive standing among highly urbanized cities. While some cities exhibit a clear alignment between their e-governance indices and CMCI scores, inconsistencies among others suggest that additional factors may influence this relationship. These findings underscore

the importance of using more nuanced evaluation methods when leveraging legislative content to assess e-governance performance.

Text Mining: Structural Topic Modeling

Like the first text mining process, the ordinances were subjected to preprocessing steps. This time, the ordinance data was modified to include the computed e-governance index and CMCI scores for each city. Figure 13 shows the Google spreadsheet containing this updated data structure.

The ordinance data—consisting of both ordinance descriptions and full texts—underwent standard preprocessing techniques, including lowercase conversion, stop word removal, elimination of numerical and punctuation characters, stemming, and the removal of words with fewer than three letters. Following this, the vocabulary was reduced by setting a threshold for the minimum number of documents in which a word must appear to be retained. This approach mirrors the vocabulary reduction method used during the computation of the e-governance index. However, it differs from the TF-IDF-based method used in the computation of the e-governance index because instead of relying on term's TF-IDF score, it uses the number of documents in which each word appears across the entire corpus to guide vocabulary reduction. To determine an appropriate threshold, a histogram was generated to visualize how many documents each word appeared across the entire corpus. This visualization guided the selection of a cutoff point that balances vocabulary size reduction with the retention of meaningful terms. The resulting histogram plot is shown in Figure 14.

1	Ordinance Number	Ordinance Description	Full Text	City	E-Governance Index	Overall CMCI Score
2	AS OF 04-11-2024	ANCE ADOPTING THE	equipment, mec	Pasay	0.474025974	56.7826
3	989	SERIES OF 1997 WHIC	a Mite' a	Pasay	0.474025974	56.7826
4	987	RVICES TO THE OFFIC	Administrative Secreta	Pasay	0.474025974	56.7826
5	986	RANGAY 183 ZONE 20	couc OF THE om	Pasay	0.474025974	56.7826

Figure 13. Updated ordinance data in the Google spreadsheet, including columns for the e-governance index and CMCI score for each city.

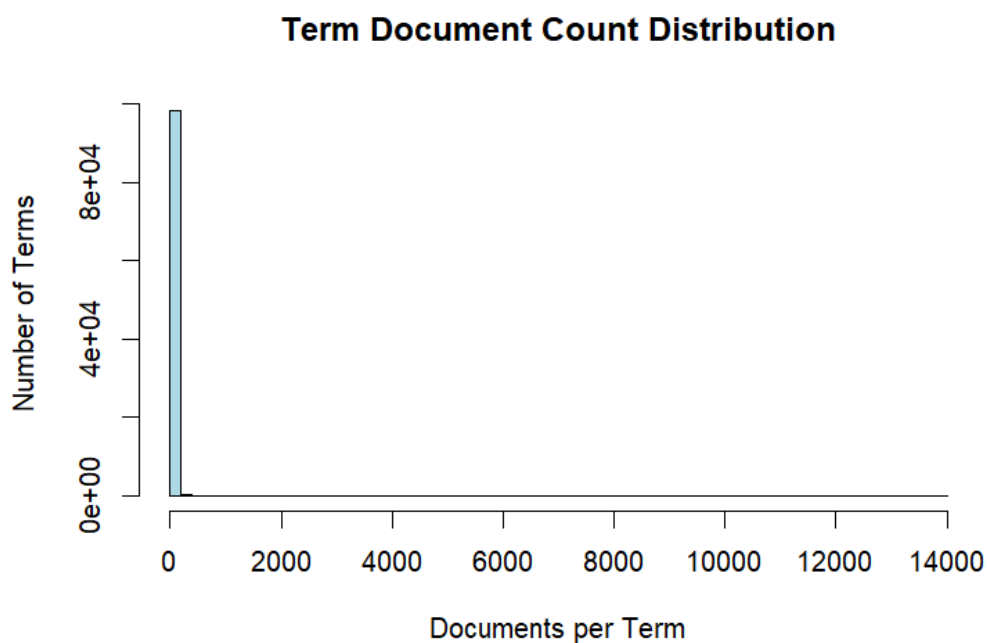


Figure 14. A histogram plot of the number of terms based on how many documents they appear in.

Summary of term document counts:						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
1.000	1.000	1.000	8.903	2.000	13851.000	

Figure 15. Summary statistics of the number of documents each term appears in.

The term-document frequency distribution in the corpus is heavily skewed toward very low frequencies. Based on the summary statistics, the threshold was determined using the first quartile of the number of documents in which each term appears. The summary shows that the minimum, first quartile (Q1), and median counts are all 1, while the mean is much higher (≈ 8.903) and the third quartile (Q3) is only 2. In other words, at least 50% of all unique terms occur in just a single document, and 75% occur in no more than two documents. This extreme gap between the median (1) and the mean (≈ 8.9) reflects a heavily right-skewed (long-tailed) distribution: most terms are very rare, and only a small fraction of terms appear in many documents (with the maximum frequency reaching 13,851). Given this skewed distribution, using the first quartile (Q1) as the cutoff for minimum document frequency is a natural choice to prune the vocabulary. In this case $Q1 = 1$, meaning the threshold is set so that terms must appear in more than one document to be retained. All terms at or below Q1 (those appearing in only one document) are excluded. This criterion is well-justified because Q1 represents the point below which 25% of the terms occur in only a single document, which is similar to the quartile used in the initial vocabulary reduction in the computation of e-governance index. Using Q1 as the threshold ensures we retain terms that appear in at least two documents, which are more likely to be useful information-bearing terms. These remaining terms have some degree of frequency and repetition, meaning they could indicate recurring concepts or topics rather than one-off anomalies. In other words, the Q1 cut-off strikes a balance: it aggressively trims the long tail of infrequent terms to reduce noise, but it does not cut so high as to remove moderately rare terms that might still be important. As a result of setting the threshold to 1, 73,362 words was removed from the original vocabulary of 99,215 terms. This ensured that only

the most meaningful and relevant words were retained for the subsequent processes. After filtering, the vocabulary was reduced to 25,853 words.

After reducing the vocabulary size and conducting initial testing to determine a suitable topic range, several STM models were tested, each with a different number of topics (K). The number of topics ranged from 20 to 145, resulting in six STM models being evaluated. The results of this evaluation are summarized Table 19.

A higher exclusivity score means a topic's words are more unique and less shared across topics. In STM, high exclusivity is desirable for clearer topic distinction and interpretability. Based on the results, a model with $K = 145$ yielded the highest exclusivity score. Similarly, a higher semantic coherence score suggests that the most probable words in a topic frequently co-occur within the same documents, indicating stronger internal consistency. However, semantic coherence tends to be highest when the number of topics is very low, which can oversimplify the thematic structure of the data. In this case, the highest semantic coherence was observed at $K = 20$.

Table 19. Summary of diagnostic metrics for STM models with varying numbers of topics K .

	K	exclus	semcoh	heldout	residual	bound	lbound	em.its
1	20	9.6056	-61.0199	-6.08041	7.908172	-8942892	-8942850	122
2	45	9.7838	-73.5986	-5.88239	5.926704	-8632046	-8631916	156
3	70	9.8441	-79.3879	-5.77448	6.343303	-8478148	-8477918	120
4	95	9.8657	-86.3165	-5.71623	7.943120	-8375448	-8375107	100
5	120	9.8849	-88.4072	-5.70043	11.72088	-8296685	-8296228	75
6	145	9.8916	-86.6457	-5.65507	36.27815	-8208644	-8208064	81

A higher exclusivity score indicates that the words in a topic are more unique to that topic and are not commonly shared across multiple topics. In the context of STM, high

exclusivity is desirable, as it enhances topic distinction and interpretability. Based on the results, a model with $K = 145$ yielded the highest exclusivity score.

Similarly, a higher semantic coherence score suggests that the most probable words in a topic frequently co-occur within the same documents, indicating stronger internal consistency. However, semantic coherence tends to be highest when the number of topics is very low, which can oversimplify the thematic structure of the data. In this case, the highest semantic coherence was observed at $K = 20$.

For the variational lower bound (VLB), a higher value reflects better model convergence. The models with $K = 145$ demonstrated the highest bound and lower bound values, indicating stronger convergence behavior. Regarding residuals, a lower residual value indicates a better model fit. Among the tested topic numbers, the lowest residual was observed at $K = 45$. Held-out likelihood measures a model's ability to predict unseen data. A higher held-out likelihood score implies better generalization. The highest held-out likelihood was again achieved with $K = 145$, reinforcing its strength in predictive performance. The final decision on the optimal number of topics was made by balancing performance across all these diagnostic metrics. While $K = 145$ performed exceptionally well in terms of exclusivity, VLB, and held-out likelihood, exhibited poor semantic coherence, the lowest across all K values tested. This trade-off highlights a common challenge in topic modeling, where optimizing one metric may come at the cost of another. The decision to select the optimal number of topics aligns with the tutorial by Weston et al. (2023), which emphasizes determining the most appropriate number of topics based on the specific goals of the STM model.

After considering all the diagnostic metrics, $K = 95$ was selected as the most appropriate number of topics, offering a balanced performance across metrics. It provided values that were generally moderate or above average in each category, avoiding extremes. It is important to note that in text mining, there is no universally accepted standard for choosing the number of topics (Bai et al., 2021). The optimal K often depends on the specific dataset and the goals of the modeling task. In this study, $K = 95$ yielded the most promising and interpretable results were based on diagnostic metrics.

After determining the appropriate number of topics ($K = 95$), the STM model was constructed using the following parameters listed in Table 20. The selected parameters for the STM model included a topic number of $K = 95$. For the topical prevalence covariates, three variables were considered: city of origin, e-governance index, and CMCI score. These covariates were incorporated so that topic prevalence would be modeled as a function of these city-specific attributes—capturing how much each topic contributes to a document based on its metadata. A key distinction of Structural Topic Modeling (STM) compared to other topic modeling approaches is its ability to integrate metadata directly into the modeling process.

Table 20. The parameters chosen for the STM model.

Parameters	Value
Number of Topics (K)	95
Prevalence Formula (prevalence)	$\text{prevalence} = \alpha_0 + \alpha_1 \text{Overall CMCI Score} + \alpha_2 \text{E-Governance Index} + \alpha_3 \text{City}$
Maximum number of EM iterations (max.em.its)	100

In this study, the e-governance index, CMCI score, and city of origin were included as covariates in the topic prevalence component of the model. However, only the effects of

the e-governance index and CMCI score were analyzed, as they are directly relevant to the objectives of the study. The inclusion of city of origin, while not analyzed in depth, served to enhance the model's completeness by accounting for additional variation that may influence topic distribution. The maximum number of iterations for the STM model was determined based on the highest iteration count observed during yielded from the diagnostic metric of K being equal to 95.

After running the model, 95 topics were generated. Each topic was manually labeled using reference categories from cities such as Pasig and Pasay, which maintain various classifications for their ordinances. Below are the first 10 topics produced by the model:

Table 21. The first 10 topics generated by the STM model with their corresponding labels and words.

Topic Number	Highest Probability	FREX	Lift	Score	Label
1	agricultur, fisheri, fish, farm, engin, water, resourc	fisheri, farm, fish, mcfarmc, aquat, vessel, abe	agnculturist, fishpen, mcfarmci, mollusk, fisher, mcfarmc, agri-fisheri	fisheri, fish, farm, agricultur, mcfarmc, aquat, farmer	FISHERIES
2	construct, instal, project, contractor, infrastructur, site, system	contractor, instal, cabl, infrastructur, construct, telecomm un, site	tunnel, gppb, admnistr, deteriorateddila pid, market-aimtech, metrostor, projectssit	contractor, construct, instal, cabl, infrastructur, telecomm un, bid	INFRASTRUCTURE

Table 21 (cont.).

3	manag, disast, risk, reduct, offic, resourc, system	osm, drmm, align, strategi, aerodrom , airport, reduct	dopt, evaluationof, strateri, cascad, planâ€	drmm-h, drmmc, planâ€	osm, disast, risk, reduct, aerodro m, drmm, strategi	LOCAL DISASTER RISK AND REDUCTION MANAGEMENT
4	insur, liabil, busi, public, accredit, person, requir	iabil, insur, crpu, alegr, lcat- vawc, half-way, lessor	cgl, oversighteth, administratorpropteti, ae«, alessor, clb, commercialbusi		insur, liabil, alegr, ozm, crpu, vista, sub- lessor	BUSINESS AND TRADE REGULATIONS
5	sanggunian g, panlungsod , councilor, mayor, approv, offic, effect	ian, richard, kalakhan g, bayona, pesebr, marlon, maynila	across--board, baangay, officials”, ‘enact, fprovisions’, â€œebi, abakada		richard, bayona, pesebr, advincul a, marlon, ibay, lyn	ORGANIZATION AL STRUCTURE
6	hiv, prevent, aid, diseas, control, offic, sti	hiv, dengu, sti, plhiv, biosecur, surveil, infect	plhiv, -°c, -averag, addrl, addrlraddl, aidssti, ajd		hiv, sti, asf, dengu, biosecur, infect, diseas	HEALTH AND SANITATION
7	panlungsod , sanggunian g, pass, public, market, septemb, mayor	dindo, baribar, israel, novero, renecito, cindi, palermo	-activ, -dz, administrativepersonn el, aldana, angguni, araneta-luzuriaga, attrv		dindo, baribar, israel, none, novero, renecito, cindi	BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT
8	award, search, servic, recognit, committe, incent, tanod	search, award, outstand, winner, recognit, plaqu	huwarang, pcg-prais, tabul, award”, finalist, ilongga, ltia		award, search, plaqu, ilongga, winner, outstand, runner	AWARD AND COMMENDATIO N

Table 21 (cont.).

9	applic, requir, permit, offic, issu, certif, busi	clearanc, applic, issuanc, document, permit, file, secur	“anti-r, sangaunian, institutionorgan, nonbusi, scopecoverag, inter- offic, extort	cdeo-	permit, clearanc, applic, certif, document, licens, transact	BUSINESS AND TRADE REGULATIONS
10	fund, thereof, much, mayor, necessari, subject, procedur	bondal, much, daza, ratif, win, bidder, jejomar	amapola, availablefor, belleza, cactus, carousel, caton	aranda,	bidder, ratif, negoti, win, binay, audit, jejomar	RULES AND REGULATIONS AND DECLARATION

Table 22. Breakdown for each unique topic label.

Unique Topic Label	Topics with this Label	Number of Topics with this Label
FISHERIES	1	1
INFRASTRUCTURE	2, 47, 90	3
LOCAL DISASTER RISK AND REDUCTION MANAGEMENT	3, 93	2
BUSINESS AND TRADE REGULATIONS	4, 9, 18, 19, 23, 28, 49, 58, 70, 73, 78	11
RULES AND REGULATIONS AND DECLARATION	5, 10, 11, 22, 25, 36, 37, 77, 89, 95	10
HEALTH AND SANITATION	6, 39, 69, 72, 88	5
BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT	7, 44, 54, 59	4
AWARD AND COMMENDATION	8	1
BUDGET MANAGEMENT AND ALLOCATION	12, 32, 45, 53, 81, 82	6
TOURISM AND CULTURE	13	1

Table 22 (cont.).

ORGANIZATIONAL STRUCTURE	14, 26, 42, 50, 60, 63, 66, 67, 75	9
IMPOSITION OF PENALTIES	15, 16, 33	3
LAND USE AND ZONING	17, 35, 41, 57, 80, 84, 92	7
ANIMAL WELFARE	20	1
EDUCATION	21, 27	2
EMPLOYEE BENEFITS AND EMPLOYEE WELFARE	24, 31, 68, 85	4
YOUTH PROGRAMS	29, 40	2
RECURRING FESTIVITIES	30	1
ROAD VEHICLE MANAGEMENT	34, 61	2
GENDER AND SEXUALITY	38, 48	2
FOOD SAFETY	43, 62	2
WASTE MANAGEMENT	46, 51, 64	3
AMMENDMENT	52	1
ECOLOGY AND ENVIRONMENTAL PROTECTION	55, 56	2
TRICYLCE PEDICAB FRANCHISING REGULATIONS	65, 91	2
TERRITORIAL BOUNDARIES AND DISPUTES	71	1
SEXUAL HARRASSMENT	74	1
SUBSTANCE REGULATION	76	1
UNCATEGORIZED	79, 86, 94	3
SENIOR CITIZEN	83	1
URBAN DEVELOPMENT	87	1

There are 31 unique topic labels assigned to the 95 topics generated by the model. Table 22 provides a breakdown of which topics fall under each label and how many topics are classified under each one. These topics represent ordinances that address a wide range of issues across different fields. Despite their diversity, they can be meaningfully organized into clusters based on thematic similarities. Below are the identified clusters and their corresponding labels, manually grouped according to their underlying themes.

Table 23. Cluster of topics with their corresponding labels.

Cluster Label	Topic Label
Regulation and Administration	BUSINESS AND TRADE REGULATIONS
	RULES AND REGULATIONS AND DECLARATION
	ORGANIZATIONAL STRUCTURE
	IMPOSITION OF PENALTIES
	LAND USE AND ZONING
	AMMENDMENT
	TERRITORIAL BOUNDARIES AND DISPUTES
	TRICYLCE PEDICAB FRANCHISING REGULATIONS
	ROAD VEHICLE MANAGEMENT
	TERRITORIAL BOUNDARIES AND DISPUTES
	SUBSTANCE REGULATION
Infrastructure and Local Development	BUDGET MANAGEMENT AND ALLOCATION
	INFRASTRUCTURE
	BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT
Sectoral Programs and Community Development	URBAN DEVELOPMENT
	YOUTH PROGRAMS
	EMPLOYEE BENEFITS AND EMPLOYEE WELFARE
	SENIOR CITIZEN
	RECURRING FESTIVITIES

Table 23 (cont.).

	EDUCATION
	TOURISM AND CULTURE
	FISHERIES
Uncategorized	UNCATEGORIZED

Clustering these topics provides a broader understanding of the key issues addressed by the ordinances. For example, the Regulation and Administration cluster includes ordinances related to regulatory and administrative policies in sectors such as business and land use. The Infrastructure and Local Development cluster focuses on ordinances that support urban growth through infrastructure projects and community planning. Finally, the Sectoral Programs and Community Development cluster encompasses ordinances aimed at improving quality of life across various groups—including workers, youth, and even animals—through targeted programs and policies. Together, these clusters demonstrate how the topics derived from the ordinances are interconnected.

Topic prevalence effect and correlation

The regression analysis examining how the covariate variables—e-governance index and CMCI score—affected the topics generated by the model resulted in separate regression tables for each covariate per topic. Below are the first three topics, along with their corresponding intercepts and regression coefficients for both the e-governance index and CMCI score.

Table 24. Regression table of the effect of the CMCI score on the topic proportion of the first 3 topics.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 1	Intercept	8.350e-03	1.869e-03	4.468	7.94e-06 ***
	CMCI Score	-1.199e-04	3.886e-05	-3.085	0.00204 **
Topic 2	Intercept	1.089e-02	3.760e-03	2.897	0.00377 **
	CMCI Score	-2.370e-05	7.776e-05	-0.305	0.76057
Topic 3	Intercept	7.280e-03	4.452e-03	1.635	0.102
	CMCI Score	-5.734e-06	9.279e-05	-0.062	0.951

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 ‘ ‘ 0.001 ‘ ‘ 0.01 ‘ ‘ 0.05 ‘ ‘ 0.1 ‘ ‘ 1.

Table 25. Regression table of the effect of the e-governance index on the topic proportion of the first 3 topics.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 1	Intercept	0.002075	0.001414	1.468	0.142
	E-Governance Index	0.001236	0.003121	0.396	0.692
Topic 2	Intercept	0.011096	0.003027	3.666	0.000247 ***
	E-Governance Index	-0.002850	0.006525	-0.437	0.662253
Topic 3	Intercept	0.011819	0.002921	4.046	5.23e-05 ***
	E-Governance Index	-0.010937	0.006341	-1.725	0.0846 .

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 ‘ ‘ 0.001 ‘ ‘ 0.01 ‘ ‘ 0.05 ‘ ‘ 0.1 ‘ ‘ 1.

The results reveal varying implications regarding how each prevalence covariate influences the proportion of a given topic. To explore this further, the effect of both variables on Topic 28 is examined through the regression tables, along with its corresponding words and assigned label shown below.

Table 26. Topic 28 generated by the STM model with its corresponding labels and words.

Topic Number	Highest Probability	FREX	Lift	Score	Label
28	amus, tax, grant, exempt, oper, franchis, ticket	amus, film, concert, kaabag, flea, game, bingo	“technolog, alixandra, amkor, apostol, based”, expon, iturrald	amus, tax, film, cockpit, franchis, exempt, admiss	BUSINESS AND TRADE REGULATIONS

Table 27. Regression table of the effect of the CMCI score on the topic proportion of the Topic 28.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 28	Intercept	0.0855902	0.0084040	10.184	<2e-16***
	CMCI Score	-0.0014221	0.0001714	-8.297	<2e-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Table 28. Regression table of the effect of the e-governance index on the topic proportion of the Topic 28.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 28	Intercept	-0.021404	0.004851	-4.412	1.03e-05***
	E-Governance Index	0.086796	0.010646	8.153	3.85e-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Topic 28 pertains to business trade and regulations, as indicated by the terms most strongly associated with business operations and infrastructure. Table 15 presents the regression results for the effect of the CMCI score on ordinances related to this topic. When the CMCI score is zero, the expected proportion of ordinances about business trade and regulations is 0.0855902, or approximately 8.5%. The corresponding p-value is less than 0.001 ($p < 2e-16$), indicating that this result is statistically significant. The estimate for the CMCI score is -0.0014221, suggesting that for every one-unit increase in the CMCI score, the proportion of ordinances on this topic decreases by approximately 0.14%. This implies that as a city becomes more competitive, the legislative focus on business trade and regulation declines, possibly due to the reduced need for foundational business-related ordinances in already competitive environments.

The effect of the e-governance index on the topic proportion of Topic 28 indicates that when the e-governance index is zero, the expected topic proportion is -0.021404. While this value is not possible—since topic proportions cannot be negative—it is a statistical artifact of the model and does not carry meaningful interpretation on its own. What is more relevant is the estimate for the e-governance index, which is 0.086796. This means that for every one-unit increase in the e-governance index, the proportion of Topic 28 increases by approximately 8.7%. The p-value for this estimate is less than 0.001 ($3.85e-16 < 0.001$), indicating that the result is statistically significant. This suggests that cities with stronger e-governance initiatives are also more likely to enact ordinances related to business, trade, and regulation.

Looking at the broader picture, the figures below illustrate the effect of the e-governance index and CMCI score on topic proportions across all 95 topics generated by the model.

Effect of E-Governance Index on Topic Prevalence Across Ordinances

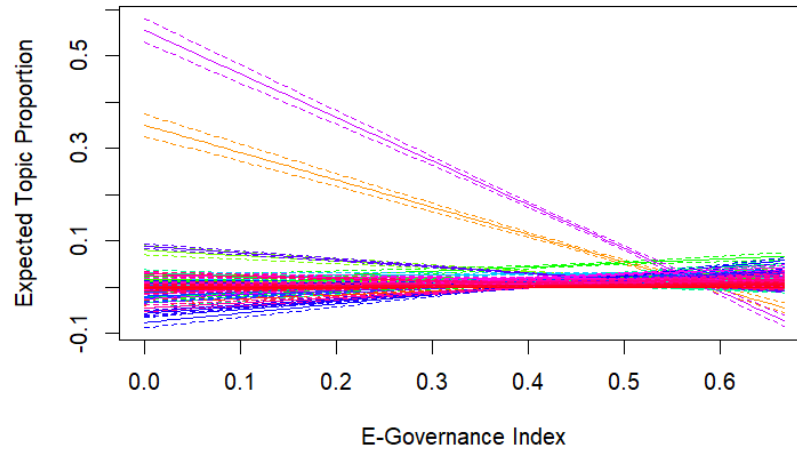


Figure 16. Plot of the effect of the e-governance index to the topic proportion of all 95 generated topics.

Effect of CMCI Score on Topic Prevalence Across Ordinances

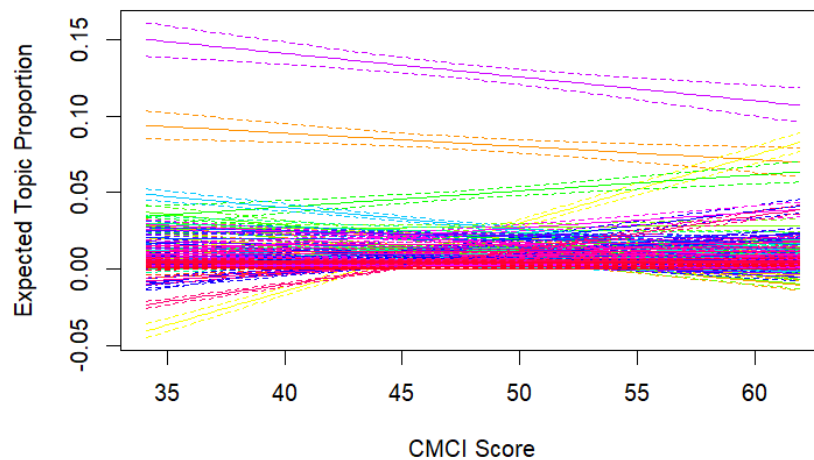


Figure 17. Plot of the effect of the CMCI score to the topic proportion of all 95 generated topics.

Starting with the e-governance index, the plot shows that as the index increases, the topic proportion for most topics also increases. This suggests that, for a given ordinance, the likelihood that it pertains to a specific topic tends to rise with higher e-governance scores. In other words, ordinances from cities with stronger e-governance practices are more likely to address a broader range of issues, reflecting a more dynamic and adaptive legislative environment. However, there are notable outliers—specifically the purple and orange lines, which represent Topic 77 and Topic 10, respectively. Both of these topics are related to rules and regulation declarations. In contrast to the general trend, the topic proportions for these two decrease as the e-governance index increases. This implies that in cities with higher e-governance scores, ordinances related to regulatory declarations become less prominent or less relevant, possibly because such foundational topics are already well-established and no longer a legislative focus. The drop in prevalence of Topics 10 and 77 with higher e-governance scores suggests that declarative or foundational regulatory ordinances become less central as digital governance infrastructure matures.

In contrast to the effect of the e-governance index, the plot illustrating the impact of the CMCI score on topic proportions shows that for most topics, the topic proportion remains relatively stable as the CMCI score increases. This indicates that urban competitiveness, as measured by CMCI, may not directly correlate with a widespread shift in legislative focus for most topic areas. However, there are notable exceptions that deviate from this general trend. Two upward-trending outliers—represented by the green and yellow lines—correspond to Topic 17 and Topic 22, which are associated with land use and zoning and rules and regulations declarations, respectively. The positive slope of these lines suggests that as the CMCI score increases, the prevalence of these topics within

ordinances also increases. This implies that cities with higher urban competitiveness (as indicated by higher CMCI scores) are more likely to produce ordinances related to land management and regulatory declarations. Conversely, a downward trend is observed for Topic 10 and Topic 77, which are also associated with rules and regulations declarations—similar to the trends seen with the e-governance index. In this case, as the CMCI score increases, the contribution of these topics to the overall content of ordinances decreases, suggesting that such topics become less central in the legislative focus of more competitive urban areas.

To sum up, the e-governance index has a positive effect on the topic proportions of most ordinances, meaning that as the e-governance index increases, ordinances across a wide range of categories become more prevalent. This suggests that higher e-governance performance is associated with a broader and more diverse legislative agenda, reflecting various components and initiatives that support digital governance. In particular, this trend indicates that cities with stronger e-governance practices are more likely to pass ordinances that align with modern governance objectives such as transparency, efficiency, infrastructure development, and citizen engagement. However, topics related to rules, regulations, and declarations tend to have lower topic proportions in high e-governance settings. This may be because such ordinances focus on foundational or procedural aspects, which can become less central as governance becomes more digitized and institutional frameworks are already well established. This finding implies that improvements in e-governance are not only about adopting technology but also about expanding the legislative scope to include a wider array of policy areas. It also suggests a shift away from basic regulatory declarations toward more strategic and adaptive governance, enabling cities to

better respond to the evolving needs of digital society. On the other hand, the CMCI score exhibits a more stable effect on the topic proportions of most topics. This suggests that many ordinances are associated with high CMCI scores, indicating that legislation across various categories captures the multi-faceted nature of urban competitiveness. As such, the results from the text mining of ordinances serve as useful indicators of the CMCI score, which, in turn, reflects a city's level of competitiveness. This supports the idea that ordinances can serve as a mirror of a city's overall competitive standing. This reinforces the value of using legislative content—such as ordinances—as a lens to evaluate urban development. Policymakers and planners can leverage ordinance analysis as a complementary approach to CMCI in identifying strategic focus areas or gaps in competitiveness-related governance.

Visualization

A shiny web application was created to effectively showcase the result of the topic modeling. The homepage of the website is shown below wherein users are given a brief description of what the website entails:

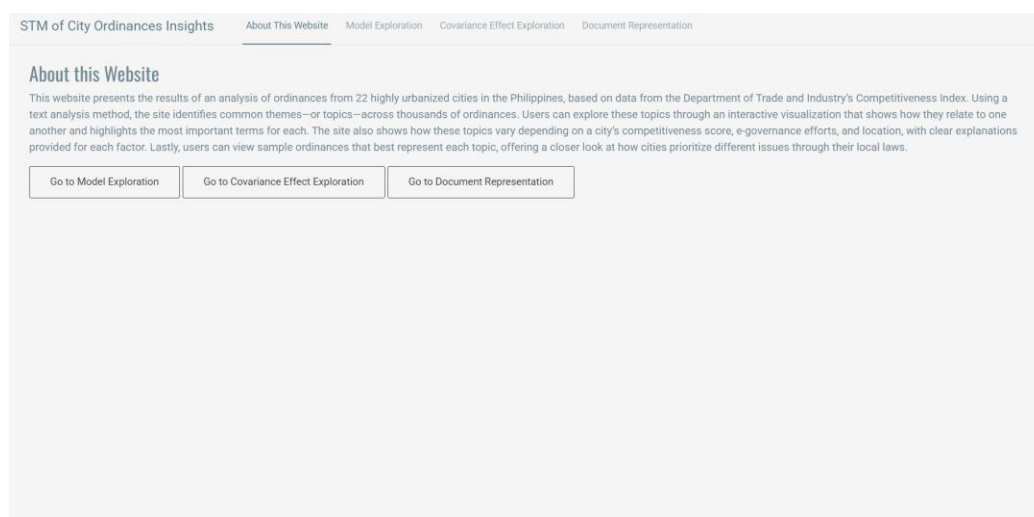


Figure 18. Homepage of the website containing information about what the website entails.

The homepage features an “About the Website” section, along with navigation buttons that lead to different parts of the site: “Model Exploration,” “Covariate Effect Comparison,” and “Document Representation.” These buttons guide users to key results presented on the website, including the discovered topics, how various factors (covariates) influence topic prevalence, and the most representative ordinances for each topic.

Another section of the website is the “Model Exploration” page, which presents the topics using LDAVis—an interactive visualization tool that gives users a global overview of the topics and how they differ from one another. It also allows for a more detailed inspection by letting users explore the specific terms most strongly associated with each individual topic. A preview of this section is shown in Figure 19. The “Covariate Effect” section illustrates how different covariates influence the topic proportions across all topics. This page is divided into three parts, each exploring how one of the covariates used in the prevalence model—e-governance index, CMCI score, and city of origin—affects topic prevalence across the ordinances. Each section includes a corresponding graph and a brief explanation to help users interpret the results. Figure 20, 21, and 22 provide a preview of these three sections. Finally, the last section of the website is “Document Representation,” which displays the top 5 most representative ordinances for each topic. This section gives users a clearer understanding of the types of ordinances that make up a specific topic by showcasing actual examples that best reflect it. A preview of this section is shown in Figure 23.

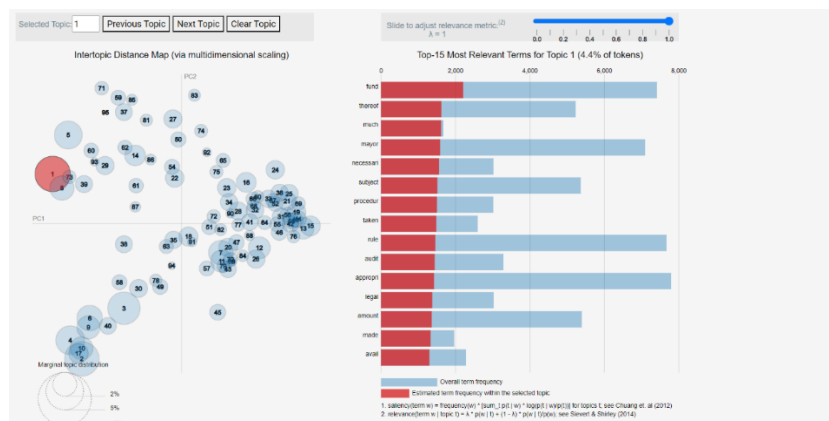


Figure 19. The “Model Exploration” with Topic 1 highlighted and the most relevant terms for Topic 1.



Figure 20. The section for how e-governance index affects the topic proportion of Topic 1.



Figure 21. The section for how CMCI score affects the topic proportion of Topic 1.

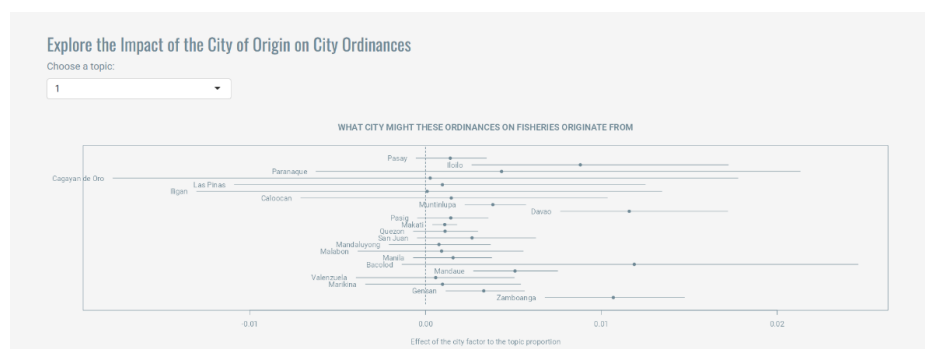


Figure 22. The section for how city of origin affects the topic proportion of Topic 1.

Explore the Top 5 Most Representative Documents

Topic Number	Ordinance Number	Ordinance Description	City
TOPIC 1 (FISHERIES)	ORDN. # 12-2013-798	AN ORDINANCE PROVIDING FOR THE DEVELOPMENT, MANAGEMENT AND CONSERVATION OF THE FISHERIES AND AQUATIC RESOURCES IN THE CITY OF MANDAUE, INCORPORATING ALL LAWS PERTINENT THERETO, AND FOR OTHER PURPOSES.	MANDAUE
	CITY ORDINANCE NO. 981	CITY LEVEL-UP FOR AGRICULTURE AND BIOSYSTEMS ENGINEERING ORDINANCE.	BACOLOD
	ORDINANCE NO. 05-053	REGISTRATION OF FISHING VESSELS THREE GROSS TONNAGE BELOW THAT UTILIZES THE TERRITORIAL WATERS OF MUNTINLUPA.	MUNTINLUPA
	ORDINANCE NO. 5768 S. 2016	AN ORDINANCE DECLARING THE EXTENT OF THE MUNICIPAL WATERS OF THE CITY OF PASAY AND DESIGNATING AREAS OPEN AND SUBJECT TO DEVELOPMENT, IN ACCORDANCE WITH THE SOCIAL AND ECONOMIC PRIORITIES OF THE CITY.	PASAY
	ORDINANCE NO. 8051	AN ORDINANCE ESTABLISHING AND IDENTIFYING BY DELINEATION AND DELIMITATION THE BOUNDARIES OF THE MUNICIPAL WATERS OF THE CITY OF MANILA FOR THE PURPOSE OF IMPLEMENTING, ENFORCING MONITORING AND MANAGING THE MUNICIPAL WATERS.	MANILA

Figure 23. The most representative ordinances for Topic 1 labeled as Fisheries.

The development of the website serves as a crucial platform for making the results of the ordinance analysis accessible, interactive, and user-friendly. By integrating visual tools such as LDAVis, covariate effect graphs, and document representation tables, the site allows users to explore complex topic modeling results in an intuitive manner. Each section of the website—Model Exploration, Covariate Effect Comparison, and Document Representation—was designed to provide meaningful insights into how local ordinances reflect patterns in e-governance and urban competitiveness. Ultimately, the website enhances the interpretability of the findings and promotes greater transparency, enabling

both researchers and policymakers to engage with the data and draw actionable conclusions from it.

C2C Learning and its Practical Application

Returning to the main objective of this study—which is to promote widespread urban competitiveness through city-to-city (C2C) learning enabled by text mining—Topic 28 serves as a compelling example of how this objective is realized. As previously illustrated in the visualization section, C2C learning takes place when cities are compared in terms of their topic proportions and how key variables influence those proportions. Topic 28, identified as relating to business, trade, and regulations, provides a useful case study.

The analysis shows that as a city becomes more urban competitive—as measured by the CMCI score—the proportion of ordinances related to business trade and regulations tends to decrease, likely because foundational economic policies are already in place. In contrast, cities with stronger e-governance initiatives tend to have a higher proportion of ordinances in this area, suggesting that digital governance may support further development in business-related policy.

Beyond these variables, the STM prevalence model also accounts for the city of origin as a covariate influencing topic proportion. For Topic 28, the regression results based on city origin are summarized in the table below:

Table 29. Regression table of the effect of the city of origin on the topic proportion of the Topic 28.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 28	Intercept	0.00526	0.012215	0.431	0.667
	Cagayan de Oro	-0.0054	0.034153	-0.158	0.874
	Caloocan	-0.0021	0.019575	-0.107	0.915
	Davao	0.0141	0.013388	1.050	0.294

Table 29 (cont.).

Gensan	0.00351	0.012602	0.278	0.7809
Iligan	0.01541	0.037417	0.412	0.6805
Iloilo	-0.00425	0.013756	-0.309	0.7576
Las Piñas	0.00415	0.024566	0.169	0.8658
Makati	-0.00080	0.012255	-0.065	0.9479
Malabon	0.01142	0.015782	0.724	0.4692
Mandaluyong	0.02340	0.014418	1.623	0.1046
Mandaue	0.01195	0.012892	0.927	0.3540
Manila	0.02437	0.012786	1.906	0.0567.
Marikina	0.00362	0.014827	0.244	0.8071
Muntinlupa	0.02205	0.012373	1.782	0.0747.
Parañaque	-0.00296	0.019707	-0.150	0.8806
Pasay	0.00609	0.012596	0.483	0.6289
Pasig	-0.00050	0.012628	-0.040	0.9684
Quezon	0.00719	0.012792	0.562	0.5741
San Juan	0.01187	0.013746	0.864	0.3879
Valenzuela	0.013891	0.015422	0.901	0.3678
Zamboanga	0.156609	0.015210	10.296	<2e-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

The regression table shows that among all cities, only Zamboanga exhibits a statistically significant positive result, with an estimate of 0.1566 and a p-value of less than 0.001 ($p < 2e-16$). This indicates that Zamboanga contributes to a 15.66% increase in the topic proportion for Topic 28. In other words, ordinances classified under this topic are most likely to have originated from Zamboanga. This finding is further validated by the most representative documents for this topic, as shown in the table below, which confirms that Topic 28 predominantly consists of ordinances from Zamboanga.

Table 30. The top 5 ordinances contributing to Topic 28, along with the corresponding city each ordinance originates from.

Topic Number	Ordinance Number	Ordinance Description	City
Topic 28 (BUSINESS AND TRADE REGULATIONS)	Ordinance No. 287	An ordinance granting the Zamboanga Puericulture Center No. 144 INC. And rotary club of Zamboanga city exemption from the payment of amusement tax on the proceeds of the tickets sold as sponsor or promoter of the Jed Madela dinner concert dubbed as "September to remember with Jed Madela" to be staged at 7pm on September 23, 2006, at the grand Astoria regency, Zamboanga city.	Zamboanga
	Ordinance No. 286	An ordinance granting exemption on the payment of amusement tax from the proceeds of tickets sold during the mega concert of miss Sharon Cuneta sponsored or promoted by the Universidad de Zamboanga Alumni Association, INC., on October 07, 2006 at the Summit Centre, Zamboanga city.	Zamboanga

Table 30 (cont.).

Ordinance No. 295	An ordinance granting exemption on the payment of 30% amusement tax from the proceeds of the musical concert, featuring Mr. Chris Araneta Cayzer, sponsored by the St. Ignatius de Loyola parish, Tetuan, Zamboanga city, on February 10, 2007 at the summit Centre, Tetuan, Zamboanga city.	Zamboanga
Ordinance No. 294	An ordinance granting exemption on the payment of 30% amusement tax from proceeds of the concert featuring Nina and Nyoy Volaten sponsored or promoted by the shelter, inc., on December 21, 2006, at the Astoria regency convention Centre, Pasonanca, this city.	Zamboanga
Ordinance No. 293	An ordinance granting exemption on the payment of 30% amusement tax from the proceeds of the concert dubbed as "All of me" sponsored by the Simeon Natividad Foundation, INC.,	Zamboanga

Table 30 (cont.).

	on December 18, 2006 at the Astoria regency convention Centre.
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The ordinances representing Topic 28 primarily focus on imposing taxes on various amusement or entertainment activities, such as concerts. This further validates the topic's classification under business, trade, and regulations. Combining these results, it can be observed that as Zamboanga becomes more competitive, ordinances related to business, trade, and regulations become less necessary, reflecting the city's increasing competitiveness. Conversely, as e-governance initiatives in Zamboanga strengthen, the need for such ordinances rises. These findings offer valuable insights for policymakers in Zamboanga, enabling them to prioritize legislative efforts based on whether their focus is on enhancing urban competitiveness, advancing e-governance initiatives, or achieving both objectives. Additionally, policymakers from other cities can use these results to guide the types of ordinances they might enact to promote either stronger e-governance efforts, greater urban competitiveness, or achieving both depending on their development goals.

The practical enactment of C2C learning is primarily achieved through the website created as part of this study. This platform serves as a vital tool for policymakers and citizens to explore and better understand the ordinances enacted within their cities. By presenting the results of the analysis—specifically the effects of the e-governance index and CMCI score on topic prevalence, the city origins of different topics, and the specific ordinances associated with each topic, the website facilitates cross-city comparisons and knowledge sharing. In doing so, it enables users to identify best practices, assess gaps, and learn from the legislative strategies of other cities.

SUMMARY AND CONCLUSION

The application of text mining to ordinances remains a relatively new and niche area of research, partly due to the complexity of interpreting its results. However, studies like this demonstrate how analyzing ordinances can yield valuable insights into how local policies in the Philippines align with broader e-governance initiatives and urban competitiveness.

This study aimed to explore how urban competitiveness can be enhanced and shared across cities through city-to-city (C2C) learning. In pursuit of this goal, the study successfully was able to achieve this by first, developing an e-governance index by mapping ordinances to specific e-governance components—namely online services, telecommunications infrastructure, and human capital—based on classifications inspired by the United Nations' e-government framework. This index quantified each city's level of e-governance initiatives by counting the number of ordinances falling into each category. Second, the integration of e-governance index with the CMCI score through the use of STM. This allowed for the analysis of how these two variables influenced the prevalence of specific policy topics across different cities. Third, the promotion of C2C learning through the creation of an interactive website visualizing the results of the topic modeling analysis. This platform enables policymakers and citizens to explore how ordinances are distributed across topics, how these align with e-governance and competitiveness scores, and which cities contribute most to specific areas of governance.

The results reveal meaningful patterns—ordinances from various categories and cities show a growing alignment with e-governance principles, as reflected by the increasing e-governance index scores. This suggests a rising adoption of ICT tools and

digital processes in local governance. Notably, the regression analysis supports a positive relationship between the e-governance index and CMCI score, highlighting how stronger e-governance initiatives are associated with higher urban competitiveness. This reinforces the role of e-governance as a driver of urban development.

Additionally, cities that scored relatively high across the e-governance components—such as Online Services, Telecommunications Infrastructure, and Human Capital—often exhibit visible government-led digital initiatives. For example, cities like Mandaue and Quezon were found to have large counts of ordinances categorized under these components, which aligns with concrete efforts like fiber-optic infrastructure projects and high literacy rates, respectively.

Conversely, the relatively stable effect of the CMCI score on topic proportions indicates that ordinances already embody the multi-dimensional nature of competitiveness. This means that cities are legislating in ways that support the broader goals of a competitive and progressive nation.

However, while the general trend reflects strong e-governance and competitiveness efforts, it is important to interpret the results at the individual city and topic level. Not all cities perform equally across all categories, and understanding these nuances is essential for enabling C2C learning. For example, the study also demonstrates how C2C learning can be facilitated through the website, as illustrated in the sample case from Topic 28. This case shows how Zamboanga, depending on its developmental goals, can strategically create ordinances to support its objectives. Specifically, ordinances related to business, trade, and regulations become more relevant as e-governance initiatives strengthen, while their

significance diminishes as the city becomes more urban competitive. These insights are also valuable for other cities that share similar developmental goals with Zamboanga.

This establishes how valuable the interactive website is. It provides users with a detailed breakdown of how topics align with e-governance, CMCI scores, and city origins. By examining how a specific topic varies in prevalence, e-governance alignment, and competitiveness across cities, users can begin to identify best practices and opportunities for improvement. The inclusion of e-governance index, CMCI score, and city of origin as covariates was a deliberate choice to capture these dynamics. When viewed together, they offer a more holistic understanding of how ordinances reflect digital governance efforts and urban development.

It's also important to note that limitations—such as the reliance on the UN's e-governance formula for constructing the index—should be addressed in future research. Refining the index to better fit the local context could lead to more accurate evaluations of a city's e-governance maturity. This study illustrates the potential of using text mining to understand legislative patterns and their relationship to e-governance and competitiveness. By combining data-driven analysis with an accessible web platform, it enables stakeholders to explore, compare, and learn from how cities legislate. In doing so, it lays the groundwork for more informed policy-making, cross-city collaboration, and the advancement of inclusive and competitive urban governance in the Philippines.

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APPENDICES

Appendix Table 1. 95 topics generated by the STM model with their corresponding labels and words

Topic Number	Highest Probability	FREX	Lift	Score	Label
1	agricultur, fisheri, fish, farm, engin, water, resourc	fisheri, farm, fish, mcfarmc, aquat, vessel, abe	agnculturist, fishpen, mcfarmci, mollusk, fisher, mcfarmc, agri-fisheri	fisheri, fish, farm, agricultur, mcfarmc, aquat, farmer	FISHERIES
2	construct, instal, project, contractor, infrastructur, site, system	contractor, instal, cabl, infrastructur, construct, telecomm un, site	tunnel, gppb, admnistr, deteriorateddilapi d, market-aimtech, metrostor, projectssit	contractor, construct, instal, cabl, infrastructur, telecomm un, bid	INFRASTRUCTURE
3	manag, disast, risk, reduct, offic, resourc, system	osm, drrm, align, strategi, aerodrom, airport, reduct	dopt, drrm-h, evaluationof, drrmc, strateri, cascadi, planâ€	osm, disast, risk, reduct, aerodrom, drrm, strategi	LOCAL DISASTER RISK AND REDUCTION MANAGEMENT
4	insur, liabil, busi, public, accredit, person, requir	liabil, insur, crpu, alegr, lcat-vawc, half-way, lessor	cgl, oversighteth, administratorprop erti, æ«, alessor, clb, commercialbusi	insur, liabil, alegr, ozm, crpu, vista, sub-lessor	BUSINESS AND TRADE REGULATIONS
5	sanggunia ng, panlungso d, councilor, mayor, approv, offic	ian, richard, kalakhang, bayona, pesebr, marlon, maynila	across--board, baangay, officials”, ‘enact, provisions’, æœbi, abakada	richard, bayona, pesebr, advincula, marlon, ibay, lyn	ORGANIZATIONAL STRUCTURE

Appendix Table 1 (cont.).

6	hiv, prevent, aid, diseas, control, offic, sti	hiv, dengue, sti, plhiv, biosecur, surveil, infect	plhiv, -°c, -averag, addrl, addrlraddl, aidssti, ajd	hiv, sti, asf, dengue, biosecur, infect, diseas	HEALTH AND SANITATION
7	panlung sod, sanggun iang, pass, public, market, septemb , mayor	dindo, baribar, israel, novero, renecito, cindi, palermo	-activ, -dz, administrativepers onnel, aldana, angguni, araneta- luzuriaga, attrv	dindo, baribar, israel, none, novero, renecito, cindi	BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT
8	award, search, servic, recognit, committ e, incent, tanod	search, award, outstand, winner, recognit, plaqu, exemplari	huwarang, pcg- prais, tabul, award”, finalist, ilongga, ltia	award, search, plaqu, ilongga, winner, outstand, runner	AWARD AND COMMENDATIO N
9	applic, requir, permit, offic, issu, certif, busi	clearanc, applic, issuanc, document, permit, file, secur	“anti-r, cdeo- sangaunian, institutionorgan, nonbusi, scopecoverag, inter-offic, extort	permit, clearanc, applic, certif, document, licens, transact	BUSINESS AND TRADE REGULATIONS
10	fund, thereof, much, mayor, necessar i, subject, procedu r	bondal, much, daza, ratif, win, bidder, jejomar	amapola, aranda, availablefor, belleza, cactus, carousel, caton	bidder, ratif, negoti, win, binay, audit, jejomar	RULES AND REGULATIONS AND DECLARATION

Appendix Table 1 (cont.).

11	member, ordn, panlung sod, cort, herebi, offic, presid	twelfth, berced, glenn, mikhail, jeric, demetrio, emmari	iiiai, leos, mgo, oddress, presenttwelfth, -- session, aban	ordn, cort, berced, twelfth, glenn, ordinonc, beethoven	RULES AND REGULATIONS AND DECLARATION
12	offic, php, year, budget, expens, total, member	lorna, ooa, oooo, oao, ecc, appendix, bonus	€ot, €ri, adiust, approvep, bgdget, booo, capitai	bonus, oooo, ooo, oof, surplus, maint, appendix	BUDGET MANAGEMENT AND ALLOCATION
13	tourism, disabl, person, cultur, art, offic, histori	disabl, tourism, pwds, histori, art, cultur, pwd	ourism, disabil, pdao, pcahto	rism, outing, “disabl, cultur, pdao, pwds, histori, bedridden	TOURISM AND CULTURE
14	member, leader, ordn, sanggun iang, herebi, panlung sod, session	fourteenth, nilo, carmelino, malcolm, cinco- remedio, sanchez, cynthia	fourteenth, presidentsk, vvi, cabatingan, carmelino, cortes- zafra, dalli	fourteenth, ordn, nilo, fortuna, cabahug, carmelino, malcolm	ORGANIZATION AL STRUCTURE
15	thereof, provid, penalti, violat, within, jurisdict, public	penalti, thereof, violat, jurisdict, territori, prohibit, within	audiovisu, linecordon, pinbal, shirtless, retract, smoke-belch, bonnet	penalti, violat, prohibit, territori, jurisdict, within, thereof	IMPOSITION OF PENALTIES

Appendix Table 1 (cont.).

16	violat, impleme nt, councilo r, fine, offic, place, public	ovr, covid, asistio, tthhee, compromis , africa, marylou	constr, henson, stroll, aassiiissttiioo	feedmil, nubl, ucc,	ovr, asistio, covid, violat, tthhee, fine, africa	IMPOSITION OF PENALTIES
17	lot, suar, transfer, titl, area, certif, meter	hoa, registri, saleabl, deed, land- --landless, payata, pambansa	hoa, --landless, - landless, aguardient, alcantara, alcav	sp-, lot, suar, saleabl, hoa, meter, registri		LAND USE AND ZONING
18	establish , busi, oper, requir, permit, owner, relat	cemeteri, parlor, establish, videok, funer, karaok, swim	corps, feescharg, sibulao, jardin, mampang, taluksangay, cabaluay	establish, busi, cemeteri, oper, parlor, videok, internet		BUSINESS AND TRADE REGULATIONS
19	camera, cctv, video, busi, instal, system, record	camera, cctv, video, circuit, pawnshop, televis, bus	automak, backrest, bi-annu, cctc, cordanc, criminol, dasmarifia	cctv, camera, circuit, video, televis, pawnshop, surveil		BUSINESS AND TRADE REGULATIONS
20	anim, vaccin, rabi, pet, dog, owner, offic	rabi, vaccin, dog, pet, stray, bite, cat	astray, rabid, volunteer', "anim	prccc, stray, "ali, dog, pet, pound, stray	rabi, anim, vaccin, dog, pet, pound, stray	ANIMAL WELFARE

Appendix Table 1 (cont.).

21	colleg, ord, board, sec, presid, educ, truste	truste, colleg, cockfight, ord, derbi, bot, instructor	athletesplay, cockptt, derbv, itsdiscret, thecolleg, accordancewith, andperson	colleg, truste, ord, cockpit, cockfight, derbi, bot	EDUCATION
22	govern, local, act, impleme nt, republ, known, rule	act, local, republ, govern, implement, pursuant, known	mc-pdao, peopleâ€™™, joi, “anti-viol, act, non-financi, dilg	act, local, implement, govern, republ, otherwis, known	RULES AND REGULATIONS AND DECLARATION
23	properti, real, sale, tax, auction, valu, price	auction, real, pin, redempt, properti, bid, acquir	acq, invencion, maspasan, arrearag, bachoco, badlan, bañe	auction, real, properti, tax, bid, delinqu, sale	BUSINESS AND TRADE REGULATIONS
24	employ, peso, offic, jobseek, properti, owner, ibig	jobseek, ibig, butuanon, pag, vacanc, cabancalan , pilit	droinog, coboncolon, everrust, expropriot, jtridic, nolot, notct	jobseek, ibig, butuanon, pag, employ, vacanc, pilit	EMPLOYEE BENEFITS AND EMPLOYEE WELFARE
25	audit, subject, fund, rule, procedu r, law, amount	audit, typhoon, procedur, mcdrmf, devast, fund- quick, municip	-los, â€œlawinâ€™, â€™œninaâ€™, â€™œompongâ€™, assistancesubsidi, bauan, bradley	audit, typhoon, procedur, budget, municip, taken, fund	RULES AND REGULATIONS AND DECLARATION

Appendix Table 1 (cont.).

26	member, account, offic, php, amount, budget, panlung sod	realign, thirteenth, accotint, non-officegener, flores-cabahug, accotnt, berced	-represent, ‘duli, accoint, accolint, accordinglyport, accordinglyto, accorurt	realign, cort, ordn, account, suico, thirteenth, berced	ORGANIZATION AL STRUCTURE
27	school, student, educ, scholars hip, high, program, , grant	scholarship, , student, school, elementari, scholar, teacher, enrol	-migrat, “ofw, “qualiti, allowance....., banwang, co-curricular, cohort	school, scholarship, , student, elementari, educ, scholar, enrol	EDUCATION
28	amus, tax, grant, exempt, oper, franchis, ticket	amus, film, concert, kaabag, flea, game, bingo	“technolog, alixandra, amkor, apostol, based”, expon, iturrald	amus, tax, film, cockpit, franchis, exempt, admiss	BUSINESS AND TRADE REGULATIONS
29	program, , develop, youth, plan, project, activ, commun iti	youth, program, develop, plan, livelihood, train, kabataan	bequest, malacaã‘ang, mcde, mtplc, panlalawigang, youth-rel, gip	youth, program, develop, plan, kabataan, project, train	YOUTH PROGRAMS
30	week, everi, year, celebr, day, activ, month	bibl, celebr, week, farmers’, coconut, ilonggo, prayer	culinari, heroin, “araw, “ilonggo, pnrc, bounty, brotherhood	celebr, bibl, week, coconut, farmers’, kang, ilonggo	RECURRING FESTIVITIES

Appendix Table 1 (cont.).

31	employ, labor, worker, servic, offic, peso, tourism	tcdo, oversea, employ, bhw, labor, dole, tour	-peso, companiesfirm, conciliation-medi, innapartel, mpc, unmanag, chb	tcdo, oversea, employ, worker, bhw, tourism, labor	EMPLOYEE BENEFITS AND EMPLOYEE WELFARE
32	hundr, peso, amount, thousan d, million, appropri , fund	million, hundr, billion, eight, seven, nine, unappropri	interspeci, pamasko, anfra, ayalaalabang, chess, dcdrrmfip, e-citizen	million, hundr, budget, peso, supplement, lbp, billion	BUDGET MANAGEMENT AND ALLOCATION
33	person, fine, violat, offens, provis, peso, public	fine, offens, imprison, discret, offend, penal, court	clothinggar, cordon, evas, outlook, personindividu, simbang, detainee	fine, offens, violat, imprison, offend, penalti, person	IMPOSITION OF PENALTIES
34	street, park, road, avenu, name, zone, traffic	junction, renam, dadianga, lagao, street, corner, stretch	abelardo, alfaro, apopong, baliwasan, bangoy, chino, culianan	street, avenu, park, renam, corner, road, traffic	ROAD VEHICLE MANAGEMENT
35	lot, along, agricult ur, residenti , other, refer, market	borja, capistrano, yacapin, verez, osmeña, along, hay	abejuela, agricul, borja, camaman, gaerlan, indahag, limketkai	lot, along, borja, agricultur, asf, capistrano, yacapin	LAND USE AND ZONING

Appendix Table 1 (cont.).

36	author, mayor, govern, sign, agreeme nt, enter, bank	agreement, behalf, bank, reput, matina, moa, loan	amigo, calinan, catalunan, marilog, pwede, therma, “sa	agreement, loan, bank, behalf, enter, donat, memorand um	RULES AND REGULATIONS AND DECLARATION
37	coun, councilor, emerg, resolut, health, public, order	yllana, benzon, edwin, coun, olivarez, quizon, merli	bustamante-men, clinker, enric, intecti, jacquelyn, olez, olivar	coun, yllana, parafiaqu, covid-, merli, contagion, olivarez	RULES AND REGULATIONS AND DECLARATION
38	right, women, gender, human, develop, health, sexual	adolesc, reproduct, gad, zod, women, gender, men	“educ, resilience”, zod, kalipunan, lipe, post-nat, ciph	adolesc, gender, sexual, women, zod, reproduct, gad	GENDER AND SEXUALITY
39	health, nutrit, birth, mother, servic, provid, newborn	newborn, nutrit, breastfeed, mnchn, mother, pregnanc, infant	-age, childcar, community-ori, folic, mnchn, nhts-pr, parenter	nutrit, newborn, birth, mother, pregnant, mnchn, matern	HEALTH AND SANITATION
40	children, child, protect, care, parent, right, develop	child, children, childhood, vawc, juvenil, abus, surviv	affront, age-, anteced, batter, center-bas, child-focus, child-mind	child, children, parent, abus, vawc, childhood, care	YOUTH PROGRAMS

Appendix Table 1 (cont.).

41	locat, negoti, squar, meter, land, author, mayor	pt- eusebio, pinagbuhat an, robert, acquisit, palc, preliminari	avida, maybunga, pt-, residences”, - b-b	cadlot, polar, -	expropri, negoti, pt- squar, parcel, meter, eusebio	LAND USE AND ZONING
42	provinci , provinc, member, resolut, offic, panlala wigan, presid	provinci, garin, defensor, panlalawig an, distura, mondejar, christin	distura, cajilig, sullano, germinanda, dorothi	palabrica, rolito,	provinci, provinc, panlalawig an, garin, defensor, governor, “tingting”	ORGANIZATION AL STRUCTURE
43	food, establish , health, requir, water, offic, sanit	food, fortif, handler, rice, bfad, trash, hand	handwash, barbecu, footbath, formica, fortif, fortific, knead		food, fortif, bfad, handler, fortifi, test, sanit	FOOD SAFETY
44	council, creat, function , committ e, repres, member, offic	function, chairman, repres, creat, council, committe, composit	chinatown, chinese-filipino, dapco, legislative- execut, mccw, pccpc, pccpc-twg		function, creat, council, committe, chairman, composit, repres	BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT
45	expens, equip, mainten, fund, offic, oper, appropri	expens, unexpend, equip, mainten, realignmen ttransf, procur, suppli	alapitan, unimpl, kepublika, ldrmfef, ldrmfip	amelori, icrd, ldrmfip	covid- unexpend, expens, amelori, equip, realignmen ttransf, unoblig	BUDGET MANAGEMENT AND ALLOCATION

Appendix Table 1 (cont.).

46	wast, bag, use, water, plastic, solid, tank	septag, bag, septic, tank, wastewat, plastic, sewag	auto-repair, disposaltreat, downspout, interceptor, microb, oxid, scum	bag, septag, septic, tank, wastewat, wast, recycl	WASTE MANAGEMENT
47	hous, project, develop, privat, board, demolit, evict	evict, demolit, psp, ppp, ipg, homeless, propon	siat, specif, non- accept, ppps, çth, activityundertak, beneficiarf	evict, psp, demolit, ppp, hous, ipg, macbosh	INFRASTRUCTU RE
48	person, sexual, gender, public, harass, act, discrimi n	harass, bisexu, wine, sozi, discrimin, sexual, gender-bas	asexu, discrjmin, genitalia, homophobia, iccpr, leer, polysexu	sexual, gender, harass, discrimin, sozi, bisexu, beverag	GENDER AND SEXUALITY
49	sale, use, product, sell, prohibit, good, manufac tur	firecrack, helmet, pyrotechn, crash, accoutr, fake, counterfeit	--", aoil, bawang, chinstrap, firecrack, glow, hiser	function, creat, council, committe, chairman, composit, repres	BUSINESS AND TRADE REGULATIONS
50	member, "yes", mayor, author, eee, sanggun iang, panlung sod	"yes", alemania, abragan, emborong, hynson, capangpan gan, sangcoan	alemania, baslayan, cesarv, mababangloob, marlen, member- author, providencio	"yes", larrazab, alemania, pacana, abragan, marianito, hynson	ORGANIZATION AL STRUCTURE

Appendix Table 1 (cont.).

51	oper, garbag, water, excav, collect, percent, follow	excav, pushcart, dig, tic, overhead, undergrou nd, garbag	“kariton”, callejo, coordinatingmonit or, initialprovision	cafd, carrion, utilityservi c, undergrou nd, dig, overhead	WASTE MANAGEMENT
52	amend, seri, known, otherwis , entitl, code, articl	amend, entitl, otherwis, known, seri, paragraph, section	anti-nuis, dcrc, ownedoper, re-inst, revisingmodifi, roadsstreet	codeâ€, amend, entitl, seri, otherwis, known, revis, articl	AMMENDMENT
53	appropri , budget, fund, hundr, supplem ent, general, amount	expenditur, lee, supplemen t, ira, enterpris, statement, eight	availa, expenditw, ouaiio, supplementa, synthes, xxxiv	supplemen t, budget, expenditur, lee, million, enterpris, hundr	BUDGET MANAGEMENT AND ALLOCATION
54	bar, panlung sod, sanggun iang, examin, offic, oper, servic	suprem, sleep, pldt, creditor, examine, fundnon- offic, misami	creditor, examine, cgdh, joj, ston, gopi, hermag	tow, bar, sleep, ---, suprem, creditor, pldt	BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT
55	nois, sound, environ ment, map, public, pollut, area	nois, e-trik, sound, pollut, amplifi, emiss, map	“nois, logarithm, ambient, analyst, land-land, lithium	evida, nois, e-trik, pollut, sound, tricycl, emiss, map	ECOLOGY AND ENVIRONMENT AL PROTECTION

Appendix Table 1 (cont.).

56	area, mangro v, tree, natur, plant, protect, resourc	mangrov, maricultur, tree, heritag, coastal, forest, landscap	afforest, chabacano, earth-bal, areasspac, clbcs	dock, clbc,	mangrov, coastal, forest, tree, heritag, reforest, maricultur	ECOLOGY AND ENVIRONMENT AL PROTECTION
57	meter, area, zone, per, land, fee, commer ci	fraction, reclassifi, ton, meter, kva, hectar, reclam	alunan, batomelong, commercialmix, csd---d, faucet, grassland, pls--d		meter, zone, fraction, hectar, reclassifi, residenti, kva	LAND USE AND ZONING
58	tax, fee, payment , busi, properti, real, charg	amnesti, surcharg, deadlin, extend, payment, tax, taxpay	“extens, act’, arrear, bounc, dayrit, driverâ€™™, eliaz		tax, surcharg, fee, deadlin, amnesti, taxpay, real	BUSINESS AND TRADE REGULATIONS
59	ang, mga, lungsod, kautusa ng, para, panlung sod, kautusa n	kautusang, kautusan, halagang, isang, libong, inaamyend ahan, traysikel	‘calam, “councilor’, “kautusang, akreditasyon, alituntunin, bahagi, buwan		kautusang, kautusan, halagang, libong, para, bilang, piso	BARANGAY AFFAIRS AND BARANGAY DEVELOPMENT
60	offic, posit, administ r, creat, servic, assist, salari	plantilla, posit, organiz, staf, pattern, abolish, grade	assistont, icomplet, parenthet, aboltshntng, adminisfrot, administrot, aicj		posit, plantilla, grade, salari, organiz, abolish, creat	ORGANIZATION AL STRUCTURE

Appendix Table 1 (cont.).

61	vehicl, traffic, motor, driver, park, use, tow	vehicl, tow, thoroughfa r, motor, signal, intersect, overtak	“park, roundabout, vehicle-, towag	deceler, uvvrp, overtak,	vehicl, tow, traffic, motor, driver, thoroughfa r, overtak	ROAD VEHICLE MANAGEMENT
62	meat, product, anim, inspect, water, poultry, consum pt	meat, slaughter, poultry, carcass, inspector, consumpt, livestock	bwtdt, thenat, veterinarians veterinary-rel,	incontact, tini, meat, chill	meat, anim, poultry, slaughter, inspect, veterinaria n, veterinari	FOOD SAFETY
63	member, floorlea d, ordn, session, offic, herebi, panlung sod	floorlead, borbajo, procopio, rosal, eleventh, gobalani, noeleen	“repair, ainual, approvedsep, attestedcertifi, bleventh	aftd, approc, eleventh, rosal, ordn, fortuna, procopio	floorlead, borbajo, eleventh, rosal, ordn, fortuna, procopio	ORGANIZATION AL STRUCTURE
64	wast, lpg, garbag, public, place, violat, collect	lpg, cylind, jail, contraband , gasolin, wast, petroleum	filth, clutter, dumpinglitteringth row, dumpingthrowingl itt, environmentalsanit ari	beeper, contraband, glitteringth rowingl	lpg, wast, cylind, garbag, contraband , jail, gasolin	WASTE MANAGEMENT
65	amend, tricycl, includ, therein, oper, rout, termin	therein, unload, load, tricycl, rout, furthermor , jeepney	altura, quiricada, smokey, tri-wheel, arlegui	españa, rmcs, tri-wheel,	tricycl, unload, rout, driver, load, toda, passeng	TRICYLCE PEDICAB FRANCHISING REGULATIONS

Appendix Table 1 (cont.).

66	sangguniang, panlung sod, approv, mayor, ordain, cuneta, herebi	trinidad, imelda, antonino, wenceslao, cuneta, rubiano, petallo	honlex, operatepli, prohibition”, re-constitut, re-nam, roso, scourg	cuneta, trinidad, calixto, wenceslao, generoso, antonino, lord	ORGANIZATION AL STRUCTURE
67	councilor, sangguniang, panlung sod, approv, presid, santo, mayor	donnabel, anthoni, aileen, noel, manguerra, alvina, vendivel	estuch, mandarin, rosani, “duay”, c€c, donnabel, honkin	councilor, santo, aileen, alvina, anthoni, calixto, vendivel	ORGANIZATION AL STRUCTURE
68	cooper, assist, financi, member, employe, death, tanod	cooper, ngo’, tanod, confin, bbp, burial, death	paglaya, relocateesbeneficiari, sicknessinjuri, siifc, beneficiaryi, buriat, certificationrecom mend	cooper, loan, tanod, death, ngo’, bbp, relocateesb eneficiari	EMPLOYEE BENEFITS AND EMPLOYEE WELFARE
69	hospit, nurs, offic, medic, iii, servic, district	consign, pcgh, nurs, hospit, apl, ladrido, niel	hemoglobin, ladrido, niel, pototan, reson, federico, sjmch	hospit, nurs, consign, barotac, pcgh, medic, technologi st	HEALTH AND SANITATION
70	declar, exist, ordin, sangguniang, resolut, law, municip	declar, ordin, exist, note, bdp, share-appropri, inconform	inconform, share-appropri, existitng, rembo-appropri, resolutt, confrom, bdp	declar, ordin, exist, bdp, note, municip, inop	RULES AND REGULATIONS AND DECLARATION

Appendix Table 1 (cont.).

71	street, avenue, boundar i, south, baranga y, north, herebi	merg, delin, gil, gallina, tripa, puyat, west	eastsouth, extedsa, gallinaaurora, north-west, origi, roadandrew, sundoy	avenue, street, west, merg, delin, south, north	TERRITORIAL BOUNDARIES AND DISPUTES
72	quaranti n, commun iti, covid- enhanc, impos, diseas, nation	quarantin, mask, covid- mecq, curfew, lift, iatf	“frontliners”, ahernandez, atthediscret, beitordain, bythec, earloop, inthec	quarantin, covid- curfew, iatf, mask, mecq, beverag	HEALTH AND SANITATION
73	tax, properti, real, assess, person, code, dec	dec, levi, tct, assessor, assess, sef, tax	cuenta, distraint, joint-stock, thirty- first, “ad, “depreci, “econom	tax, real, tct, properti, dec, levi, assess	BUSINESS AND TRADE REGULATIONS
74	person, traffick, purpos, servic, offic, act, protect	traffick, spider, anti- traffick, exploit, fraud, prostitut, debt	“anti-traffick, affick, bondag, coscolluela, cybercrim, fromto, jacqu	traffick, spider, bondag, slaveri, anti- traffick, sexual, backho	SEXUAL HARRASSMENT
75	offic, member, session, presid, sanggun ian, may, purok	purok, sanggunian , adjourn, question, relinquish, mc-tipc, quorum	bloc, comrpt, grammat, houserul, inchoat, movant, non-govemment	purok, adjourn, relinquish, mc-tipc, sanggunian , ordn, elect	ORGANIZATION AL STRUCTURE

Appendix Table 1 (cont.).

76	tobacco, smoke, product, advertis, public, place, use	tobacco, vape, smoke, e-cigarette, cigarette, advertise, nicotine	advertisementpro mot, brandsvari, briberi, buildingengin, cartridg, denorm, e-cigarette	tobacco, smoke, advertise, cigarette, fctc, e-cigarette, vape	SUBSTANCE REGULATION
77	law, rule, regul, conform, budget, review, exist	conform, review, bel-air, sub-, urdaneta, --s, dasmariã‘	anti-obstruct, areview, dasmariã‘, gudalup, ha--, ha--- , jmo	conform, sub-, review, budget, supplement, audit, budgetari	RULES AND REGULATIONS AND DECLARATION
78	market, public, stall, fee, applic, event, rental	stall, stallhold, rental, mcma, leas, rent, market	crdinonc, durot, rtw, lwill, rainforest, stallsbooth, accessexit	stall, market, rental, stallhold, leas, mcma, fee	BUSINESS AND TRADE REGULATIONS
79	salari, drug, budget, unit, schedul, personnel, entri	tranch, civilian, madac, border, plmun, cadac, step	anlalawigan, employeespersonnel, madac, marijuana, remand, plmun, tranch	tranch, salari, drug, madac, civilian, border, grade	UNCATEGORIZED
80	valu, road, land, lot, access, r-a, interior	sub-class, r-a, deg, bdri, thenc, r-d, const	adionisio, aflor, alcob, ambulancerescu, ascafio, aveedsa, bdry-pqu	r-a, thenc, deg, sub-class, lot, bdri, r-d	LAND USE AND ZONING

Appendix Table 1 (cont.).

81	peso, php, thousand, assist, financial, fire, amount	sharer, houseown, mcrrdmf, fire, renter, movi, alabang	kabulusan, mefi, basilan, beneficiariesfamili, benselcom, biazon, buhi	sharer, fire, alabang, houseown, mcrrdmf, movi, php	BUDGET MANAGEMENT AND ALLOCATION
82	purpos, fund, general, provid, therefor, appropri, santo	santo, purpos, therefor, general, fund, sarangani, institution	acyep, avawc, blaen, columbari, cpe-ieu, eco-wast, edukar	santo, fund, purpos, institution, general, therefor, provid	BUDGET MANAGEMENT AND ALLOCATION
83	citizen, senior, parent, solo, benefit, provid, social	solo, senior, citizen, parent, osca, privileg, card	“famili, “parent, nowadays, pauperindig, schedule”	solo, senior, parent, citizen, privileg, osca, card	SENIOR CITIZEN
84	beneficiari, lot, civil, hous, structur, owner, offic	registrar, census, band, beneficiari, error, nso, cleric	hierarchi, lcro, lotshous, structuredwel, censussurvey, rent-fre, allocationlot	beneficiari, lot, breq, censussurvey, registrar, census, hectar	LAND USE AND ZONING
85	employe, grant, month, incent, personnel, servic, cash	rtc, cash, judg, incent, pei, trial, pageant	awol, contract--servic, contractappoint, department-assist, metc, non-uniform, pakyaw	incent, rtc, employe, cash, judg, pei, salari	EMPLOYEE BENEFITS AND EMPLOYEE WELFARE

Appendix Table 1 (cont.).

86	justo, cuneta, effect, approv, offic, roberto, day	lorca, teodulo, justo, arabia, uldarico, alvarez, romulo	con-, honroberto, honromulo, honteodulo, honuldarico, lust, lorca	justo, arabia, lorca, alvarez, teodulo, uldarico, alcera	UNCATEGORIZED
87	board, invest, pamantasan, univers, enterpris, , incent, regent	regent, pamantasan, invest, enterpris, univers, investor, faculti	arbit, bmbes, msmedc, statesmanship, studentri, synthesi, leipo	regent, pamantasan, invest, enterpris, incent, faculti, univers	URBAN DEVELOPMENT
88	health, drug, care, medic, test, servic, facil	mental, drug, blood, test, clinic, patient, health	ambulatori, sars-cov, non-hospit, nsd, anti-tuberculosi, creatingestablish, donor?	health, drug, test, mental, patient, blood, medic	HEALTH AND SANITATION
89	sp-, establish, , councilor, public, district, mandat, locat	sp-, wwwcdeo-sanggunian onlin, sora, tandang, maclang, trace, novalich	baesa, download, krus, kyusipass, leandro, locsin, nc-	sp-, novalich, covid-, wwwcdeo-sanggunian onlin, sora, tandang, scanner	RULES AND REGULATIONS AND DECLARATION
90	build, structur, permit, offic, construct, green, offici	obo, drill, green, demolish, oscp, build, berd	decisionsord, non-contract, officer, otlenlion, pursuonf, sub-contractor, backroom	build, green, structur, obo, oscp, berd, permit	INFRASTRUCTURE

Appendix Table 1 (cont.).

91	oper, franchis, pedicab, tricycl, number, fee, driver	pedicab, tfh, franchis, tpfro, tpmo, plate, poda	ajalon, cautious, humilidad, lanetermin, leveriza-, pcpfi, ridership	pedicab, tricycl, franchis, tpfro, tfh, toda, poda	TRICYLCE PEDICAB FRANCHISING REGULATIONS
92	use, zone, area, land, build, develop, lot	clup, deep, gfa, abut, zone, ludmg, manufactur	-fronton, “author, agz, aisar, ambf, authorities”, bgz	zone, gfa, lot, ludmg, manufactur , land, clup	LAND USE AND ZONING
93	disast, energi, climat, manag, develop, plan, inform	climat, energi, evacu, resili, disast, mitig, ict	bles, chemicalga, climate-rel, disaster-affect, disaster-rel, energ, fire-fight	energi, climat, disast, evacu, resili, risk, cbms	LOCAL DISASTER RISK AND REDUCTION MANAGEMENT
94	eee, toward, point, origin, back, see, drive	agapita, mulawin, acal, pelagio, logarta, mantawi, o·f	bop, comejo, “oo, bak, brel, comejo- vital, das	mantawi, agapita, mulawin, logarta, brion, acal, o·f	UNCATEGORIZE D
95	addit, generat, special, way, use, author, public	addit, generat, special, way, use, author, increas	addit, generat, way, special, increas, use, author	addit, generat, way, special, use, increas, author	RULES AND REGULATIONS AND DECLARATION

Appendix Table 2. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 1.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 1	Intercept	0.008325	0.0018727	4.446	8.84E-06***
	CMCI Score	-0.0001194	0.000039	-3.061	0.00221**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 3. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 2.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 2	Intercept	1.09E-02	3.77E-03	2.89	0.00386**
	CMCI Score	-2.36E-05	7.79E-05	-0.302	0.76228

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 4. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 3.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 3	Intercept	7.31E-03	4.50E-03	1.625	0.104
	CMCI Score	-6.41E-06	9.40E-05	-0.068	0.946

Appendix Table 5. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 4.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 4	Intercept	3.54E-03	2.47E-03	1.433	0.152
	CMCI Score	-1.39E-05	5.06E-05	-0.275	0.783

Appendix Table 6. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 5.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 5	Intercept	-3.03E-02	3.59E-03	-8.444	<2E-16***
	CMCI Score	7.14E-04	7.87E-05	9.077	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 7. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 6.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 6	Intercept	1.59E-02	3.89E-03	4.104	4.09E-05***
	CMCI Score	-2.42E-04	7.92E-05	-3.05	0.00229**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 8. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 7.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 7	Intercept	1.22E-02	1.99E-03	6.113	1.00E-09***
	CMCI Score	-2.16E-04	4.10E-05	-5.274	1.35E-07***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 9. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 8.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 8	Intercept	4.23E-03	3.04E-03	1.392	0.164
	CMCI Score	-5.08E-06	6.34E-05	-0.08	0.936

Appendix Table 10. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 9.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 9	Intercept	8.47E-03	3.66E-03	2.314	0.0207*
	CMCI Score	-7.51E-06	7.71E-05	-0.097	0.9224

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 11. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 10.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 10	Intercept	0.1236042	0.0147015	8.408	<2E-16***
	CMCI Score	-0.0008665	0.0003059	-2.833	0.00462**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 12. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 11.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 11	Intercept	0.0769957	0.0060352	12.76	<2E-16***
	CMCI Score	-0.0014149	0.0001208	-11.72	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 13. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 12.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 12	Intercept	1.17E-02	2.12E-03	5.548	2.94E-08***
	CMCI Score	-2.08E-04	4.28E-05	-4.862	1.17E-06***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 14. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 13.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 13	Intercept	-1.09E-03	3.43E-03	-0.317	0.751
	CMCI Score	1.36E-04	7.20E-05	1.89	0.0588.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 15. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 14.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 14	Intercept	3.27E-02	3.89E-03	8.405	<2E-16***
	CMCI Score	-5.92E-04	7.79E-05	-7.601	3.12E-14***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 16. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 15.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 15	Intercept	0.0447454	0.0055865	8.01	1.24E-15***
	CMCI Score	-0.0004982	0.000114	-4.369	1.26E-05***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 17. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 16.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 16	Intercept	3.46E-03	2.03E-03	1.707	0.0879.
	CMCI Score	-3.18E-05	4.18E-05	-0.761	0.4468

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 18. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 17.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 17	Intercept	-0.1930772	0.0085151	-22.68	<2E-16***
	CMCI Score	0.0044627	0.0001849	24.14	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 19. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 18.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 18	Intercept	4.45E-02	4.37E-03	10.183	<2E-16***
	CMCI Score	-7.08E-04	8.80E-05	-8.053	8.70E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 20. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 19.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 19	Intercept	3.56E-03	3.07E-03	1.159	0.247
	CMCI Score	6.55E-06	6.41E-05	0.102	0.919

Appendix Table 21. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 20.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 20	Intercept	9.28E-03	3.22E-03	2.885	0.00392**
	CMCI Score	-1.07E-04	6.66E-05	-1.612	0.10694

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 22. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 21.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 21	Intercept	1.87E-02	2.77E-03	6.769	1.35E-11***
	CMCI Score	-3.29E-04	5.53E-05	-5.943	2.87E-09***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 23. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 22.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 22	Intercept	-0.0080273	0.0046467	-1.728	0.0841.
	CMCI Score	0.0004099	0.0000969	4.23	2.35E-05***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 24. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 23.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 23	Intercept	-8.66E-03	4.63E-03	-1.869	0.061642.
	CMCI Score	3.29E-04	9.79E-05	3.359	0.000784***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 25. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 24.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 24	Intercept	3.47E-03	1.32E-03	2.625	0.00868**
	CMCI Score	-4.61E-05	2.73E-05	-1.686	0.0918.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 26. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 25.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 25	Intercept	2.33E-02	6.28E-03	3.714	0.000205***
	CMCI Score	8.38E-05	1.30E-04	0.644	0.519569

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 27. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 26.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 26	Intercept	0.074659	0.005874	12.71	<2E-16***
	CMCI Score	-0.001365	0.0001192	-11.45	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 28. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 27.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 27	Intercept	1.27E-02	5.58E-03	2.279	0.0227*
	CMCI Score	2.33E-05	1.15E-04	0.203	0.8388

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 29. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 28.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 28	Intercept	0.0855536	0.0084484	10.127	<2E-16***
	CMCI Score	-0.0014216	0.0001724	-8.247	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 30. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 29.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 29	Intercept	7.01E-03	3.24E-03	2.161	0.0307*
	CMCI Score	1.59E-04	6.75E-05	2.347	0.019*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 31. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 30.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 30	Intercept	-0.0036059	0.0042363	-0.851	0.39468
	CMCI Score	0.0002772	0.0000893	3.104	0.00191**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 32. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 31.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 31	Intercept	-5.23E-04	3.32E-03	-0.157	0.875
	CMCI Score	1.24E-04	7.08E-05	1.754	0.0794.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 33. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 32.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 32	Intercept	-0.001306	0.010265	-0.127	0.899
	CMCI Score	0.001049	0.000215	4.879	1.08E-06***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 34. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 33.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 33	Intercept	-7.01E-03	4.07E-03	-1.723	0.08492.
	CMCI Score	2.93E-04	8.53E-05	3.437	0.00059***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 35. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 34.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 34	Intercept	0.0488914	0.0080626	6.064	1.36E-09***
	CMCI Score	-0.0005867	0.0001659	-3.535	0.000408***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 36. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 35.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 35	Intercept	4.21E-03	1.67E-03	2.522	0.0117*
	CMCI Score	-5.46E-05	3.43E-05	-1.593	0.1111

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 37. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 36.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 36	Intercept	8.14E-03	4.44E-03	1.835	0.0666.
	CMCI Score	8.53E-05	9.40E-05	0.907	0.3644

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 38. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 37.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 37	Intercept	1.61E-02	2.81E-03	5.715	1.12E-08***
	CMCI Score	-2.79E-04	5.74E-05	-4.854	1.22E-06***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 39. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 38.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 38	Intercept	9.65E-03	3.14E-03	3.073	0.00212**
	CMCI Score	-1.12E-04	6.54E-05	-1.711	0.08701.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 40. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 39.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 39	Intercept	6.64E-03	3.27E-03	2.034	0.042*
	CMCI Score	-4.39E-05	6.90E-05	-0.636	0.525

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 41. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 40.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 40	Intercept	-5.01E-04	4.03E-03	-0.124	0.9012
	CMCI Score	1.43E-04	8.47E-05	1.689	0.0913.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 42. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 41.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 41	Intercept	0.0436609	0.00832	5.248	1.56E-07***
	CMCI Score	-0.000402	0.0001726	-2.329	0.0199*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 43. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 42.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 42	Intercept	-1.14E-02	2.97E-03	-3.849	0.000119***
	CMCI Score	3.22E-04	6.27E-05	5.142	2.76E-07***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 44. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 43.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 43	Intercept	2.13E-03	3.04E-03	0.699	0.484
	CMCI Score	3.80E-05	6.33E-05	0.601	0.548

Appendix Table 45. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 44.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 44	Intercept	1.50E-02	3.92E-03	3.825	0.000131***
	CMCI Score	-3.99E-05	8.21E-05	-0.486	0.627181

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 46. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 45.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 45	Intercept	-4.21E-03	4.54E-03	-0.927	0.3541
	CMCI Score	2.87E-04	9.80E-05	2.93	0.0034**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 47. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 46.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 46	Intercept	7.48E-03	4.04E-03	1.852	0.0641.
	CMCI Score	-4.07E-05	8.47E-05	-0.48	0.6309

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 48. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 47.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 47	Intercept	1.53E-02	3.48E-03	4.39	1.14E-05***
	CMCI Score	-2.20E-04	7.19E-05	-3.065	0.00218**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 49. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 48.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 48	Intercept	6.33E-03	3.08E-03	2.057	0.0397*
	CMCI Score	-4.98E-05	6.35E-05	-0.785	0.4325

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 50. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 49.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 49	Intercept	0.0265533	0.0048734	5.449	5.16E-08***
	CMCI Score	-0.0004065	0.0001006	-4.04	5.37E-05***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 51. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 50.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 50	Intercept	1.06E-02	2.00E-03	5.264	1.43E-07***
	CMCI Score	-1.96E-04	4.14E-05	-4.727	2.30E-06***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 52. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 51.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 51	Intercept	7.26E-03	2.63E-03	2.762	0.00575**
	CMCI Score	-7.41E-05	5.47E-05	-1.355	0.1755

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 53. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 52.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 52	Intercept	0.0992025	0.0061902	16.03	<2E-16***
	CMCI Score	-0.0014736	0.0001288	-11.44	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 54. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 53.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 53	Intercept	-2.86E-03	3.13E-03	-0.915	0.3602
	CMCI Score	1.64E-04	6.58E-05	2.487	0.0129*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 55. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 54.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 54	Intercept	2.55E-03	1.86E-03	1.372	0.17
	CMCI Score	-2.03E-05	3.82E-05	-0.533	0.594

Appendix Table 56. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 55.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 55	Intercept	7.70E-03	2.98E-03	2.586	0.00971**
	CMCI Score	-8.50E-05	6.09E-05	-1.396	0.16273

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 57. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 56.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 56	Intercept	1.62E-02	3.21E-03	5.065	4.14E-07***
	CMCI Score	-2.50E-04	6.53E-05	-3.828	0.00013***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 58. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 57.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 57	Intercept	2.92E-02	4.12E-03	7.089	1.42E-12***
	CMCI Score	-4.40E-04	8.47E-05	-5.199	2.03E-07***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 59. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 58.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 58	Intercept	0.0410546	0.0070297	5.84	5.33E-09***
	CMCI Score	-0.0004054	0.0001474	-2.75	0.00597**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 60. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 59.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 59	Intercept	-0.0047189	0.0057711	-0.818	0.41356
	CMCI Score	0.0003617	0.0001213	2.982	0.00287**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 61. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 60.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 60	Intercept	0.0226606	0.0073039	3.103	0.00192**
	CMCI Score	-0.0000709	0.0001519	-0.467	0.64065

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 62. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 61.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 61	Intercept	7.73E-03	3.98E-03	1.942	0.0522.
	CMCI Score	8.93E-06	8.28E-05	0.108	0.9141

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 63. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 62.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 62	Intercept	1.04E-02	3.15E-03	3.301	0.000967***
	CMCI Score	-1.29E-04	6.50E-05	-1.981	0.047659*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 64. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 63.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 63	Intercept	4.53E-02	4.76E-03	9.527	<2E-16***
	CMCI Score	-8.19E-04	9.57E-05	-8.553	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 65. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 64.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 64	Intercept	1.55E-02	3.90E-03	3.97	7.23E-05***
	CMCI Score	-2.02E-04	8.03E-05	-2.514	0.0119*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 66. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 65.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 65	Intercept	-0.0730952	0.0069199	-10.56	<2E-16***
	CMCI Score	0.001848	0.0001479	12.5	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 67. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 66.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 66	Intercept	-5.27E-02	4.25E-03	-12.41	<2E-16***
	CMCI Score	1.24E-03	9.25E-05	13.36	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 68. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 67.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 67	Intercept	-4.76E-02	4.49E-03	-10.59	<2E-16***
	CMCI Score	1.14E-03	9.81E-05	11.66	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 69. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 68.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 68	Intercept	3.62E-03	2.59E-03	1.396	0.163
	CMCI Score	1.79E-06	5.38E-05	0.033	0.973

Appendix Table 70. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 69.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 69	Intercept	-1.35E-03	2.87E-03	-0.472	0.637
	CMCI Score	1.25E-04	6.03E-05	2.073	0.0382*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 71. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 70.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 70	Intercept	3.18E-02	3.52E-03	9.015	<2E-16***
	CMCI Score	-1.91E-04	7.31E-05	-2.616	0.00891**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 72. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 71.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 71	Intercept	-4.11E-03	2.87E-03	-1.433	0.15184
	CMCI Score	1.78E-04	6.09E-05	2.922	0.00348**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 73. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 72.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 72	Intercept	0.0297177	0.0073412	4.048	5.19E-05***
	CMCI Score	-0.0003422	0.0001535	-2.23	0.0258*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 74. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 73.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 73	Intercept	2.44E-02	3.89E-03	6.267	3.79E-10***
	CMCI Score	-4.42E-04	7.89E-05	-5.599	2.20E-08***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 75. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 74.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 74	Intercept	1.73E-03	1.49E-03	1.158	0.247
	CMCI Score	-7.10E-06	3.16E-05	-0.224	0.822

Appendix Table 76. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 75.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 75	Intercept	7.55E-03	1.97E-03	3.835	0.000126***
	CMCI Score	-1.11E-04	4.09E-05	-2.703	0.006874**

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 77. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 76.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 76	Intercept	9.89E-03	3.32E-03	2.975	0.00294**
	CMCI Score	-1.08E-04	6.76E-05	-1.593	0.1112

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 78. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 77.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 77	Intercept	0.2041739	0.0173727	11.753	<2E-16***
	CMCI Score	-0.0015696	0.0003615	-4.341	1.43E-05***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 79. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 78.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 78	Intercept	1.32E-02	3.56E-03	3.703	0.000214***
	CMCI Score	-1.63E-04	7.43E-05	-2.196	0.028132*

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 80. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 79.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 79	Intercept	8.18E-03	2.62E-03	3.128	0.00176**
	CMCI Score	-1.04E-04	5.41E-05	-1.915	0.05549.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 81. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 80.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 80	Intercept	1.56E-03	1.82E-03	0.858	0.391
	CMCI Score	1.01E-05	3.83E-05	0.264	0.791

Appendix Table 82. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 81.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 81	Intercept	-0.0239593	0.0075803	-3.161	0.00158**
	CMCI Score	0.0010182	0.0001591	6.4	1.60E-10***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 83. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 82.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 82	Intercept	5.73E-02	4.82E-03	11.89	<2E-16***
	CMCI Score	-8.18E-04	9.87E-05	-8.288	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 84. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 83.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 83	Intercept	1.14E-02	5.25E-03	2.166	0.0304*
	CMCI Score	-3.88E-05	1.09E-04	-0.354	0.7232

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 85. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 84.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 84	Intercept	4.36E-03	2.46E-03	1.769	0.0769.
	CMCI Score	-1.47E-05	5.15E-05	-0.285	0.7753

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 86. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 85.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 85	Intercept	1.27E-02	5.96E-03	2.126	0.0336*
	CMCI Score	7.47E-05	1.24E-04	0.603	0.5464

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 87. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 86.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 86	Intercept	-4.06E-02	4.36E-03	-9.32	<2E-16***
	CMCI Score	9.58E-04	9.54E-05	10.04	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 88. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 87.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 87	Intercept	7.96E-04	3.72E-03	0.214	0.831
	CMCI Score	1.09E-04	7.67E-05	1.415	0.157

Appendix Table 89. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 88.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 88	Intercept	1.19E-02	4.93E-03	2.41	0.016*
	CMCI Score	-4.46E-06	1.04E-04	-0.043	0.966

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 90. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 89.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 89	Intercept	-0.1003506	0.0038944	-25.77	<2E-16***
	CMCI Score	0.0022509	0.0000848	26.55	<2E-16***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 91. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 90.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 90	Intercept	2.21E-03	2.98E-03	0.742	0.458
	CMCI Score	5.41E-05	6.22E-05	0.87	0.384

Appendix Table 92. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 91.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 91	Intercept	5.94E-03	3.73E-03	1.59	0.112
	CMCI Score	-2.20E-05	7.85E-05	-0.279	0.78

Appendix Table 93. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 92.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 92	Intercept	4.33E-03	2.59E-03	1.668	0.0953.
	CMCI Score	-1.17E-05	5.46E-05	-0.215	0.8301

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 94. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 93.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 93	Intercept	3.17E-03	3.02E-03	1.048	0.295
	CMCI Score	3.16E-05	6.42E-05	0.493	0.622

Appendix Table 95. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 94.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 94	Intercept	-1.85E-03	1.70E-03	-1.092	0.2748
	CMCI Score	6.61E-05	3.67E-05	1.802	0.0716.

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 96. Regression Table of the Effect of the CMCI Score on the Topic Proportion of Topic 95.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 95	Intercept	5.56E-04	3.47E-05	16.009	<2E-16***
	CMCI Score	-4.62E-06	7.22E-07	-6.407	1.53E-10***

*Significance codes for the intercepts and CMCI score estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 97. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 1.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 1	Intercept	0.002076	0.00141	1.473	0.141
	E-Governance Index	0.001233	0.003103	0.397	0.691

Appendix Table 98. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 2.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 2	Intercept	0.011033	0.003013	3.662	0.000251***
	E-Governance Index	-0.002708	0.006481	-0.418	0.676069

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 99. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 3.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 3	Intercept	0.011802	0.002925	4.035	5.48E-05***
	E-Governance Index	-0.010923	0.006346	-1.721	0.0852.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 100. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 4.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 3	Intercept	0.0025495	0.0018448	1.382	0.167
	E-Governance Index	0.0008689	0.0039331	0.221	0.825

Appendix Table 101. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 5.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 5	Intercept	-0.032987	0.003192	-10.33	<2E-16***
	E-Governance Index	0.0808	0.007417	10.89	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 102. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 6.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 6	Intercept	0.001126	0.002174	0.518	0.605
	E-Governance Index	0.007464	0.004856	1.537	0.124

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 103. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 7.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 7	Intercept	0.004094	0.001555	2.633	0.00847**
	E-Governance Index	-0.00479	0.003347	-1.431	0.1525

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 104. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 8.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 8	Intercept	0.005966	0.00215	2.776	0.00552**
	E-Governance Index	-0.004347	0.004604	-0.944	0.34516

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 105. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 9.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 9	Intercept	0.003	0.00272	1.103	0.27
	E-Governance Index	0.011176	0.006022	1.856	0.0635.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 106. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 10.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 10	Intercept	0.35042	0.01159	30.24	<2E-16***
	E-Governance Index	-0.59156	0.02428	-24.37	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 107. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 11.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 11	Intercept	0.015458	0.003651	4.235	2.30E-05***
	E-Governance Index	-0.012851	0.007794	-1.649	0.0992.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 108. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 12.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 12	Intercept	0.0008399	0.0015572	0.539	0.59
	E-Governance Index	0.0022901	0.0033795	0.678	0.498

Appendix Table 109. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 13.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 13	Intercept	0.001626	0.002273	0.715	0.474
	E-Governance Index	0.008349	0.00497	1.68	0.093.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 110. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 14.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 14	Intercept	0.007972	0.002532	3.148	0.00164**
	E-Governance Index	-0.007574	0.005494	-1.379	0.168

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 111. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 15.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 15	Intercept	0.004955	0.004168	1.189	0.23456
	E-Governance Index	0.035394	0.009169	3.86	0.000114***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 112. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 16.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 16	Intercept	0.006574	0.001669	3.94	8.19E-05***
	E-Governance Index	-0.010221	0.003509	-2.912	0.00359**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 113. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 17.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 17	Intercept	0.008392	0.0058	1.447	0.1479
	E-Governance Index	0.02351	0.01264	1.86	0.0629.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 114. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 18.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 18	Intercept	0.0007605	0.0029848	0.255	0.798884
	E-Governance Index	0.0220829	0.0067006	3.296	0.000984***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 115. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 19.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 19	Intercept	0.0036936	0.0023105	1.599	0.11
	E-Governance Index	0.0007336	0.0050879	0.144	0.885

Appendix Table 116. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 20.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 20	Intercept	0.005343	0.002412	2.215	0.0268*
	E-Governance Index	-0.00244	0.00522	-0.467	0.6402

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 117. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 21.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 21	Intercept	-0.0005047	0.0018884	-0.267	0.7893
	E-Governance Index	0.0077278	0.0041697	1.853	0.0639.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 118. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 22.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 22	Intercept	0.008653	0.002998	2.886	0.0039**
	E-Governance Index	0.006074	0.006549	0.928	0.3537

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 119. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 23.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 23	Intercept	0.005712	0.003565	1.602	0.109
	E-Governance Index	0.002943	0.007765	0.379	0.705

Appendix Table 120. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 24.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 24	Intercept	0.0016444	0.0010094	1.629	0.103
	E-Governance Index	-0.0007696	0.0021574	-0.357	0.721

Appendix Table 121. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 25.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 25	Intercept	0.080022	0.005122	15.62	<2E-16***
	E-Governance Index	-0.116271	0.010814	-10.75	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 122. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 26.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 26	Intercept	0.015029	0.003967	3.788	0.000152***
	E-Governance Index	-0.011782	0.008492	-1.387	0.165341

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 123. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 27.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 27	Intercept	0.009171	0.003645	2.516	0.0119*
	E-Governance Index	0.009888	0.007909	1.25	0.2113

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 124. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 28.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 28	Intercept	-0.021372	0.004897	-4.364	1.29E-05***
	E-Governance Index	0.086716	0.010722	8.088	6.59E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 125. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 29.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 29	Intercept	0.028888	0.002389	12.092	<2E-16***
	E-Governance Index	-0.031641	0.005002	-6.326	2.59E-10***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 126. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 30.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 30	Intercept	0.016935	0.003422	4.949	7.56E-07***
	E-Governance Index	-0.015932	0.007371	-2.162	0.0307*

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 127. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 31.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 31	Intercept	-0.001658	0.002358	-0.703	0.48213
	E-Governance Index	0.015435	0.00521	2.963	0.00306**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 128. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 32.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 32	Intercept	0.008695	0.007909	1.099	0.272
	E-Governance Index	0.087316	0.017307	5.045	4.59E-07***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 129. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 33.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 33	Intercept	-0.020072	0.002856	-7.029	2.18E-12***
	E-Governance Index	0.0595	0.006367	9.345	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 130. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 34.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 34	Intercept	0.012611	0.005085	2.48	0.0131*
	E-Governance Index	0.018598	0.01088	1.709	0.0874.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 131. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 35.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 35	Intercept	0.003258	0.001165	2.795	0.0052**
	E-Governance Index	-0.003617	0.002561	-1.413	0.1578

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 132. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 36.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 36	Intercept	0.025689	0.003357	7.653	2.10E-14***
	E-Governance Index	-0.029972	0.007072	-4.238	2.27E-05***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 133. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 37.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 37	Intercept	-0.008391	0.002058	-4.077	4.58E-05***
	E-Governance Index	0.024761	0.00464	5.336	9.65E-08***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 134. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 38.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 38	Intercept	0.0039133	0.0021918	1.785	0.0742.
	E-Governance Index	0.0009487	0.0046885	0.202	0.8396

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 135. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 39.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 39	Intercept	0.00169	0.002674	0.632	0.527
	E-Governance Index	0.006269	0.005857	1.07	0.284

Appendix Table 136. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 40.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 40	Intercept	-0.0006895	0.0025725	-0.268	0.78868
	E-Governance Index	0.0154545	0.0055827	2.768	0.00564**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 137. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 41.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 41	Intercept	-0.034326	0.005824	-5.894	3.85E-09***
	E-Governance Index	0.129411	0.012934	10.005	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 138. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 42.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 42	Intercept	0.031031	0.003569	8.695	<2E-16***
	E-Governance Index	-0.059846	0.007267	-8.235	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 139. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 43.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 43	Intercept	-4.35E-05	1.93E-03	-0.023	0.982
	E-Governance Index	8.54E-03	4.14E-03	2.061	0.0393*

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 140. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 44.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 44	Intercept	0.001029	0.003227	0.319	0.749786
	E-Governance Index	0.026534	0.007407	3.582	0.000342***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 141. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 45.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 45	Intercept	0.002942	0.003543	0.83	0.406
	E-Governance Index	0.014573	0.007686	1.896	0.058.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 142. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 46.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 46	Intercept	0.003452	0.002996	1.152	0.249
	E-Governance Index	0.004805	0.006545	0.734	0.463

Appendix Table 143. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 47.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 47	Intercept	-0.003736	0.002518	-1.484	0.137875
	E-Governance Index	0.019118	0.00555	3.445	0.000573***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 144. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 48.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 48	Intercept	-0.001957	0.002387	-0.82	0.4123
	E-Governance Index	0.012794	0.005384	2.376	0.0175*

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 145. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 49.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 49	Intercept	0.009584	0.003081	3.111	0.00187**
	E-Governance Index	-0.005052	0.006613	-0.764	0.4449

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 146. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 50.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 50	Intercept	-0.004555	0.001388	-3.283	0.00103**
	E-Governance Index	0.012866	0.003135	4.104	4.09E-05***

Appendix Table 147. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 51.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 51	Intercept	-1.27E-05	1.92E-03	-0.007	0.9947
	E-Governance Index	8.11E-03	4.21E-03	1.924	0.0544.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 148. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 52.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 52	Intercept	0.023148	0.004619	5.012	5.46E-07***
	E-Governance Index	0.013368	0.010134	1.319	0.187

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 149. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 53.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 53	Intercept	0.001445	0.002542	0.568	0.57
	E-Governance Index	0.007852	0.00554	1.417	0.156

Appendix Table 150. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 54.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 54	Intercept	-0.0008872	0.0013192	-0.673	0.501
	E-Governance Index	0.0054278	0.002897	1.874	0.061.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 151. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 55.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 55	Intercept	-0.0004423	0.0022255	-0.199	0.8425
	E-Governance Index	0.0092109	0.0049817	1.849	0.0645.

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 152. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 56.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 56	Intercept	0.00616	0.002339	2.634	0.00844**
	E-Governance Index	-0.003855	0.004985	-0.773	0.43931

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 153. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 57.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 57	Intercept	0.005538	0.002677	2.069	0.0386*
	E-Governance Index	0.005788	0.005692	1.017	0.3093

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 154. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 58.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 58	Intercept	-0.01052	0.00538	-1.956	0.0505.
	E-Governance Index	0.07207	0.01194	6.035	1.63E-09***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 155. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 59.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 59	Intercept	-0.023325	0.004267	-5.467	4.66E-08***
	E-Governance Index	0.078771	0.009598	8.207	2.47E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 156. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 60.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 60	Intercept	-0.019444	0.005424	-3.585	0.000338***
	E-Governance Index	0.085523	0.011808	7.243	4.62E-13***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 157. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 61.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 61	Intercept	0.008468	0.003195	2.65	0.00805**
	E-Governance Index	-0.001	0.007087	-0.141	0.88774

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 158. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 62.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 62	Intercept	0.001144	0.002329	0.491	0.623
	E-Governance Index	0.007087	0.005128	1.382	0.167

Appendix Table 159. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 63.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 63	Intercept	0.009589	0.003012	3.184	0.00146**
	E-Governance Index	-0.007044	0.006534	-1.078	0.28105

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 160. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 64.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 64	Intercept	0.006193	0.003052	2.029	0.0425*
	E-Governance Index	-0.000653	0.006572	-0.099	0.9209

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 161. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 65.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 65	Intercept	-0.075933	0.005711	-13.29	<2E-16***
	E-Governance Index	0.200292	0.012842	15.6	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 162. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 66.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 66	Intercept	-0.056116	0.003528	-15.91	<2E-16***
	E-Governance Index	0.13716	0.008142	16.85	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 163. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 67.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 67	Intercept	-0.052212	0.004126	-12.65	<2E-16***
	E-Governance Index	0.130133	0.009546	13.63	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 164. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 68.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 68	Intercept	-0.001425	0.001693	-0.842	0.39988
	E-Governance Index	0.011432	0.003718	3.075	0.00211**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 165. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 69.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 69	Intercept	-0.001475	0.002265	-0.651	0.51495
	E-Governance Index	0.013652	0.005038	2.71	0.00674**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 166. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 70.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 70	Intercept	0.088744	0.002619	33.88	<2E-16***
	E-Governance Index	-0.146038	0.005577	-26.18	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 167. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 71.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 71	Intercept	-0.003074	0.002301	-1.336	0.18158
	E-Governance Index	0.016237	0.005211	3.116	0.00184**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 168. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 72.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 72	Intercept	-0.024518	0.004604	-5.325	1.03E-07***
	E-Governance Index	0.083854	0.01009	8.311	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 169. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 73.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 73	Intercept	-0.00493	0.002079	-2.371	0.0177*
	E-Governance Index	0.018343	0.004621	3.969	7.24E-05***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 170. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 74.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 74	Intercept	0.002746	0.001048	2.621	0.00878**
	E-Governance Index	-0.003147	0.002205	-1.427	0.15353

Appendix Table 171. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 75.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 75	Intercept	0.003277	0.00151	2.17	0.03*
	E-Governance Index	-0.002144	0.003292	-0.651	0.515

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 172. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 76.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 76	Intercept	0.002652	0.002446	1.084	0.278
	E-Governance Index	0.004965	0.005368	0.925	0.355

Appendix Table 173. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 77.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 77	Intercept	0.55599	0.01273	43.67	<2E-16***
	E-Governance Index	-0.94202	0.0273	-34.5	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 174. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 78.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 78	Intercept	-0.00215	0.002553	-0.842	0.3997
	E-Governance Index	0.017004	0.005622	3.024	0.0025**

Appendix Table 175. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 79.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 79	Intercept	-0.00311	0.001936	-1.607	0.10814
	E-Governance Index	0.014146	0.004365	3.241	0.00119**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 176. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 80.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 80	Intercept	-0.002345	0.001539	-1.523	0.12774
	E-Governance Index	0.00979	0.003447	2.84	0.00452**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 177. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 81.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 81	Intercept	-0.01401	0.00548	-2.556	0.0106*
	E-Governance Index	0.08498	0.01206	7.049	1.89E-12***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 178. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 82.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 82	Intercept	0.025245	0.003525	7.161	8.43E-13***
	E-Governance Index	-0.014854	0.007478	-1.986	0.047*

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 179. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 83.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 83	Intercept	-0.008606	0.003407	-2.526	0.0115*
	E-Governance Index	0.040552	0.007604	5.333	9.83E-08***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 180. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 84.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 84	Intercept	0.006486	0.001981	3.274	0.00106**
	E-Governance Index	-0.006052	0.004198	-1.442	0.1494

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 181. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 85.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 85	Intercept	0.001813	0.004524	0.401	0.68861
	E-Governance Index	0.031474	0.009854	3.194	0.00141**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 182. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 86.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 86	Intercept	-0.045148	0.003682	-12.26	<2E-16***
	E-Governance Index	0.110761	0.008521	13	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 183. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 87.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 87	Intercept	-0.007872	0.002998	-2.626	0.00865**
	E-Governance Index	0.030895	0.006649	4.647	3.41E-06***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 184. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 88.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 88	Intercept	-0.0002951	0.0038191	-0.077	0.93842
	E-Governance Index	0.0264723	0.0085448	3.098	0.00195**

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 185. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 89.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 89	Intercept	0.029007	0.002377	12.21	<2E-16***
	E-Governance Index	-0.049257	0.004904	-10.04	<2E-16***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 186. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 90.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 90	Intercept	0.0004625	0.0020185	0.229	0.8188
	E-Governance Index	0.0093849	0.0045609	2.058	0.0396*

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 187. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 91.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 91	Intercept	-0.007681	0.002652	-2.896	0.00379**
	E-Governance Index	0.02751	0.005897	4.665	3.11E-06***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '****' 0.001 '**', 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 188. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 92.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 92	Intercept	0.002752	0.002007	1.371	0.17
	E-Governance Index	0.0022	0.004367	0.504	0.614

Appendix Table 189. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 93.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 93	Intercept	0.0008055	0.0023326	0.345	0.73
	E-Governance Index	0.0083778	0.005252	1.595	0.111

Appendix Table 190. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 94.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 94	Intercept	-0.003948	0.001416	-2.788	0.00532**
	E-Governance Index	0.01164	0.003203	3.634	0.00028***

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 191. Regression Table of the Effect of the E-Governance Index on the Topic Proportion of Topic 95.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 95	Intercept	3.64E-04	2.44E-05	14.918	<2E-16***
	E-Governance Index	-6.05E-05	5.15E-05	-1.176	0.24

*Significance codes for the intercepts and e-governance index estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 192. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 1.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 1	Intercept	0.0121895	0.0069034	1.766	0.0775.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 193. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 2.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 2	Intercept	0.0053778	0.0079656	0.675	0.4996
	Davao	0.0201046	0.0088892	2.262	0.0237*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 194. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 3.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 3	Intercept	0.0035126	0.0081002	0.434	0.665

Appendix Table 195. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 4.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 4	Intercept	0.01052	0.008226	1.279	0.201

Appendix Table 196. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 5.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 5	Intercept	1.49E-04	5.11E-03	0.029	0.977
	Pasay	5.19E-02	6.84E-03	7.581	3.65E-14***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 197. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 6.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 6	Intercept	0.02818	0.01185	2.379	0.0174*
	Caloocan	-0.02606	0.01323	-1.969	0.049*
	Davao	-0.02535	0.0119	-2.13	0.0332*
	Iligan	-0.02791	0.01554	-1.796	0.0725.
	Las Piñas	-0.02615	0.01477	-1.77	0.0767.
	Makati	-0.0262	0.01192	-2.198	0.0279*
	Mandaue	-0.02549	0.01199	-2.126	0.0336*
	Manila	-0.02378	0.01201	-1.981	0.0476*
	Muntinlupa	-0.02506	0.01178	-2.127	0.0335*
	Pasay	-0.02206	0.0122	-1.808	0.0706.
	Pasig	-0.0249	0.01203	-2.071	0.0384*
	Quezon	-0.02492	0.01209	-2.06	0.0394*
	Valenzuela	-0.02153	0.012	-1.795	0.0727.
	Zamboanga	-0.02222	0.01205	-1.843	0.0653.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 198. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 7.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 7	Intercept	0.31225	0.02841	10.99	<2E-16***
	Cagayan de Oro	-0.31227	0.02945	-10.6	<2E-16***
	Caloocan	-0.31217	0.02862	-10.91	<2E-16***
	Davao	-0.31165	0.02841	-10.97	<2E-16***
	Gensa	-0.31165	0.02841	-10.97	<2E-16***
	Iligan	-0.31199	0.02887	-10.81	<2E-16***
	Iloilo	-0.31223	0.02845	-10.97	<2E-16***
	Las Piñas	-0.28395	0.02672	-10.63	<2E-16***
	Makati	-0.31159	0.02839	-10.98	<2E-16***
	Malabon	-0.31185	0.02846	-10.96	<2E-16***
	Mandaluyong	-0.31109	0.02845	-10.93	<2E-16***
	Mandaue	-0.31215	0.02842	-10.98	<2E-16***
	Manila	-0.31151	0.02839	-10.97	<2E-16***
	Marikina	-0.31141	0.02855	-10.91	<2E-16***
	Muntinlupa	-0.31126	0.02839	-10.96	<2E-16***
	Paranaque	-0.31223	0.02865	-10.9	<2E-16***
	Pasay	-0.31218	0.02843	-10.98	<2E-16***
	Pasig	-0.31112	0.02842	-10.95	<2E-16***
	Quezon	-0.31159	0.02844	-10.96	<2E-16***
	San Juan	-0.31138	0.02847	-10.94	<2E-16***
	Valenzuela	-0.31175	0.02848	-10.95	<2E-16***
	Zamboanga	-0.31139	0.02844	-10.95	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 199. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 8.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 8	Intercept	0.010683	0.008612	1.24	0.215

Appendix Table 200. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 9.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 9	Intercept	0.021025	0.009408	2.235	0.02544*
	Cagayan de Oro	0.164914	0.059471	2.773	0.00556**
	Marikina	-0.02036	0.009957	-2.045	0.04089*
	Muntinlupa	-0.018476	0.009483	-1.948	0.05139.
	Pasig	-0.016304	0.00966	-1.688	0.09147.
	Quezon	-0.015955	0.009559	-1.669	0.09511.
	Valenzuela	-0.017586	0.010098	-1.742	0.08161.
	Zamboanga	-0.017402	0.009764	-1.782	0.07473.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 201. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 10.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 10	Intercept	0.0003279	0.0275791	0.012	0.991
	Makati	0.2035873	0.027782	7.328	2.46E-13***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 202. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 11.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 11	Intercept	9.55E-05	8.49E-03	0.011	0.991
	Mandaue	1.74E-01	1.12E-02	15.571	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 203. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 12.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 12	Intercept	6.47E-05	3.45E-03	0.019	0.985
	Malabon	1.21E-02	6.36E-03	1.897	0.0578.
	Mandaue	2.06E-02	4.38E-03	4.698	2.65E-06***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 204. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 13.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 13	Intercept	0.0100873	0.0079018	1.277	0.2018
	Iloilo	0.0193765	0.010407	1.862	0.0626.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 205. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 14.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 14	Intercept	2.20E-04	6.09E-03	0.036	0.971
	Mandaue	7.15E-02	7.78E-03	9.199	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 206. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 15.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 15	Intercept	0.0052739	0.0086371	0.611	0.54147
	Davao	0.028052	0.0097714	2.871	0.0041**
	Gensan	0.0254315	0.009239	2.753	0.00592**
	Malabon	0.030775	0.0124386	2.474	0.01337*
	Mandaue	0.0710073	0.0100182	7.088	1.43E-12***
	Marikina	0.0400845	0.0127352	3.148	0.00165**
	Pasig	0.0277556	0.0092096	3.014	0.00259**
	Quezon	0.037764	0.0092532	4.081	4.51E-05***
	San Juan	0.0161665	0.0098233	1.646	0.09984
	Valenzuela	0.1886853	0.0170298	11.08	<2E-16***
	Zamboanga	0.049378	0.0099117	4.982	6.38E-07
	Valenzuela	-0.31175	0.02848	-10.95	<2E-16***
	Zamboanga	-0.31139	0.02844	-10.95	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 207. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 16.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 16	Intercept	8.33E-05	3.40E-03	0.024	0.98
	Caloocan	2.06E-01	5.00E-02	4.115	3.90E-05***
	Iloilo	3.81E-02	8.54E-03	4.463	8.13E-06***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 208. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 17.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 17	Intercept	0.0098621	0.0144958	0.68	0.496
	Manila	0.0696059	0.0155091	4.488	7.24E-06***
	Quezon	0.1572048	0.016822	9.345	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 209. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 18.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 18	Intercept	0.0030674	0.0066162	0.464	0.64293
	Davao	0.0320977	0.0075362	4.259	2.07E-05***
	Gensan	0.0204316	0.0068243	2.994	0.00276**
	Mandaluyong	0.0376976	0.0085761	4.396	1.11E-05***
	Pasig	0.0115793	0.006809	1.701	0.08905.
	Valenzuela	0.03365	0.0103218	3.26	0.00112**
	Zamboanga	0.0435442	0.0071448	6.095	1.13E-09***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 210. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 19.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 19	Intercept	0.0006369	0.0049182	0.129	0.897
	Mandaluyong	0.0117853	0.0057246	2.059	0.0395*
	Paranaque	0.0216041	0.01205	1.793	0.073.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 211. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 20.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 20	Intercept	0.0008086	0.0047351	0.171	0.8644
	Malabon	0.0140152	0.0074143	1.89	0.0587.
	Mandaluyong	0.0137561	0.0063113	2.18	0.0293*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 212. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 21.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 21	Intercept	0.0003035	0.0035547	0.085	0.931969
	Davao	0.0085579	0.0040613	2.107	0.03512*
	Malabon	0.0145367	0.0060918	2.386	0.017034*
	Mandaue	0.0067344	0.0039742	1.695	0.090183.
	Marikina	0.0296379	0.0066365	4.466	8.04E-06***
	Zamboanga	0.0149081	0.004299	3.468	0.000526***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 213. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 22.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 22	Intercept	0.027283	0.010468	2.606	0.00916**
	Makati	-0.023561	0.010551	-2.233	0.02557*
	Muntinlupa	-0.020004	0.010436	-1.917	0.05529.
	Valenzuela	-0.022411	0.011504	-1.948	0.05143.
	Zamboanga	-0.017803	0.010656	-1.671	0.09482.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 213. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 23.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 23	Intercept	0.0075601	0.0085859	0.881	0.3786
	Mandaluyong	0.0180359	0.0102811	1.754	0.0794.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 214. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 24.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 24	Intercept	1.89E-04	2.49E-03	0.076	0.9395
	Mandaue	6.98E-03	2.96E-03	2.359	0.0184*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 215. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 25.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 25	Intercept	1.13E-03	1.25E-02	0.09	0.92811
	Makati	5.07E-02	1.26E-02	4.023	5.78E-05***
	Muntinlupa	3.52E-02	1.28E-02	2.75	0.00596**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 216. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 26.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 26	Intercept	3.08E-04	9.59E-03	0.032	0.974
	Mandaue	1.71E-01	1.13E-02	15.085	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 217. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 27.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 27	Intercept	0.0130647	0.0104023	1.256	0.20916
	Zamboanga	0.0348976	0.0113175	3.084	0.00205**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 218. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 28.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 28	Intercept	0.0033327	0.0122185	0.273	0.785
	Mandaluyong	0.0244762	0.0135902	1.801	0.0717.
	Manila	0.0269067	0.0129049	2.085	0.0371*
	Muntinlupa	0.0250638	0.0124234	2.017	0.0437*
	Zamboanga	0.1562817	0.0147827	10.572	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 219. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 29.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 29	Intercept	0.0147159	0.0078495	1.875	0.0608.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 220. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 30.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 30	Intercept	0.008972	0.0104291	0.86	0.3896
	Gensan	0.0200575	0.0103277	1.942	0.0521.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 221. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 31.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 31	Intercept	0.018398	0.01013	1.816	0.0694.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 222. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 32.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 32	Intercept	0.003508	0.0181974	0.193	0.84714
	Makati	0.0503139	0.0182612	2.755	0.00587**
	Malabon	0.1405351	0.026143	5.376	7.76E-08***
	Manila	0.0919766	0.0199977	4.599	4.28E-06***
	Muntinlupa	0.0869611	0.0186304	4.668	3.07E-06***
	Pasig	0.0877062	0.019386	4.524	6.11E-06***
	Quezon	0.0347675	0.0190321	1.827	0.06775.
	San Juan	0.046385	0.0206535	2.246	0.02473*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 223. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 33.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 33	Intercept	0.0165051	0.0097824	1.687	0.09158.
	Pasay	0.0288245	0.0105659	2.728	0.00638**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 224. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 34.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 34	Intercept	0.0077517	0.0118911	0.652	0.51448
	Davao	0.0221482	0.0128968	1.717	0.08594.
	Gensan	0.0377513	0.0126868	2.976	0.00293**
	Las Piñas	0.0765603	0.0370956	2.064	0.03905*
	Manila	0.0333105	0.0132217	2.519	0.01177*
	Quezon	0.0230471	0.0122182	1.886	0.05928.
	Zamboanga	0.0977532	0.012936	7.557	4.39E-14***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 225. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 35.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 35	Intercept	0.012131	0.008654	1.402	0.161
	Cagayan de Oro	0.265114	0.026928	9.845	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 226. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 36.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 36	Intercept	0.0067912	0.0090527	0.75	0.453
	Davao	0.1119029	0.0127151	8.801	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 227. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 37.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 37	Intercept	0.006532	0.006632	0.985	0.3246
	Paranaque	0.42934	0.060812	7.06	1.74E-12***
	Valenzuela	0.014272	0.007395	1.93	0.0536.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 228. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 38.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 38	Intercept	0.0065969	0.0067611	0.976	0.3292
	Iloilo	0.0202868	0.0093983	2.159	0.0309*
	San Juan	0.0158265	0.0069758	2.269	0.0233*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 '.' 1.

Appendix Table 229. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 39.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 39	Intercept	0.0048085	0.0063769	0.754	0.4508
	Malabon	0.0188225	0.0093998	2.002	0.0453*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 230. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 40.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 40	Intercept	0.0060732	0.0072791	0.834	0.404

Appendix Table 231. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 41.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 41	Intercept	0.0054801	0.0147788	0.371	0.710788
	Muntinlupa	0.0806196	0.015348	5.253	1.52E-07***
	Pasig	0.1526526	0.0168203	9.075	<2E-16***
	San Juan	0.0695436	0.018518	3.755	0.000174***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 232. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 42.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 42	Intercept	4.67E-05	4.33E-03	0.011	0.991
	Iloilo	2.63E-01	2.21E-02	11.889	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 233. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 43.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 43	Intercept	0.0015122	0.0048982	0.309	0.75754
	Davao	0.0165476	0.0062268	2.657	0.00788
	Marikina	0.0204315	0.0073039	2.797	0.00516

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 234. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 44.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 44	Intercept	0.0152742	0.0094294	1.62	0.105

Appendix Table 235. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 45.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 45	Intercept	0.0005276	0.0066469	0.079	0.936737
	Caloocan	0.1058099	0.0200769	5.27	1.38E-07
	Makati	0.0125944	0.0066869	1.883	0.059661
	Manila	0.0290884	0.0075337	3.861	0.000113

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 236. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 46.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 46	Intercept	0.0173508	0.0106005	1.637	0.102

Appendix Table 237. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 47.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 47	Intercept	0.0079141	0.0077325	1.023	0.3061
	Malabon	0.0324983	0.0130426	2.492	0.0127*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 238. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 48.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 48	Intercept	1.35E-03	4.93E-03	0.274	0.784
	Davao	1.10E-02	5.87E-03	1.872	0.0612.
	Iloilo	1.36E-02	8.02E-03	1.689	0.0912.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 239. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 49.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 49	Intercept	0.0004911	0.0070097	0.07	0.9441
	Davao	0.0193353	0.008286	2.333	0.0196*
	Gensan	0.0189145	0.0077516	2.44	0.0147*
	Mandaluyong	0.0220053	0.0084108	2.616	0.0089**
	Marikina	0.0512023	0.0130043	3.937	8.28E-05***
	San Juan	0.0208937	0.0088371	2.364	0.0181*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 240. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 50.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 50	Intercept	8.45E-05	2.21E-03	0.038	0.9695
	Iligan	6.15E-01	8.29E-02	7.423	1.21E-13***
	San Juan	6.49E-03	2.98E-03	2.176	0.0296*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 241. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 51.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 51	Intercept	1.48E-03	4.27E-03	0.348	0.728
	Davao	8.37E-03	4.88E-03	1.713	0.0866.
	Marikina	1.33E-02	6.86E-03	1.944	0.0519.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 242. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 52.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 52	Intercept	0.017408	0.012356	1.409	0.158873
	Davao	0.063449	0.014823	4.281	1.88E-05***
	Gensan	0.040684	0.013102	3.105	0.001905**
	Malabon	0.025628	0.015571	1.646	0.099818.
	Mandaluyong	0.04095	0.013801	2.967	0.00301**
	Marikina	0.062662	0.018291	3.426	0.000615***
	Pasig	0.046993	0.012482	3.765	0.000167***
	San Juan	0.065338	0.013307	4.91	9.20E-07***
	Valenzuela	0.015824	2.345	0.019032	0.015824*
	Zamboanga	0.012936	3.619	0.000296	0.012936***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 243. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 53.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 53	Intercept	6.87E-04	5.88E-03	0.117	0.9069
	Iloilo	7.40E-02	1.25E-02	5.942	2.89E-09***
	Manila	1.53E-02	6.49E-03	2.349	0.0188*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 244. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 54.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 54	Intercept	0.007504	0.006346	1.182	0.237

Appendix Table 245. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 55.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 55	Intercept	0.04331	0.01362	3.18	0.00148**
	Cagayan de Oro	-0.0432	0.01835	-2.354	0.01859*
	Davao	-0.04012	0.01361	-2.947	0.00322**
	Gensan	-0.04216	0.0136	-3.101	0.00193**
	Iligan	-0.04239	0.01662	-2.55	0.01078*
	Iloilo	-0.04004	0.014	-2.861	0.00423**
	Las Piñas	-0.03758	0.0175	-2.148	0.03176*
	Makati	-0.04044	0.01368	-2.957	0.00311**
	Malabon	-0.02842	0.01297	-2.192	0.02842*
	Mandaluyong	-0.02895	0.01401	-2.066	0.03887*
	Mandaue	-0.04163	0.01359	-3.063	0.0022**
	Manila	-0.03896	0.01398	-2.786	0.00535**
	Marikina	-0.04218	0.01372	-3.074	0.00212**
	Muntinlupa	-0.04126	0.01371	-3.009	0.00263**
	Paranaque	-0.04307	0.01482	-2.907	0.00366**
	Pasay	-0.03616	0.01415	-2.555	0.01063*
	Pasig	-0.03599	0.0134	-2.687	0.00722**
	Quezon	-0.04129	0.01376	-3.001	0.00269**
	San Juan	-0.03544	0.01413	-2.509	0.01213*
	Valenzuela	-0.03717	0.01461	-2.543	0.01099*
	Zamboanga	-0.04205	0.01361	-3.089	0.00201**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 246. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 56.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 56	Intercept	0.06723	0.01391	4.831	1.37E-06***
	Cagayan de Oro	-0.06674	0.01918	-3.48	0.000503***
	Caloocan	-0.06716	0.0153	-4.388	1.15E-05***
	Davao	-0.05288	0.01416	-3.734	0.000189***
	Gensan	-0.05312	0.01403	-3.785	0.000154***
	Iligan	-0.06665	0.01682	-3.963	7.45E-05***
	Iloilo	-0.05412	0.01414	-3.826	0.000131***
	Makati	-0.06553	0.01389	-4.716	2.43E-06***
	Malabon	-0.06301	0.01371	-4.596	4.35E-06***
	Mandaluyong	-0.065	0.01397	-4.651	3.33E-06***
	Mandaue	-0.06197	0.01426	-4.346	1.40E-05***
	Manila	-0.06307	0.01382	-4.565	5.05E-06***
	Marikina	-0.06062	0.01499	-4.043	5.30E-05***
	Muntinlupa	-0.06368	0.01388	-4.587	4.54E-06***
	Paranaque	-0.06727	0.01537	-4.377	1.21E-05***
	Pasay	-0.06517	0.01382	-4.716	2.42E-06***
	Pasig	-0.06491	0.01404	-4.622	3.84E-06***
	Quezon	-0.0649	0.01382	-4.694	2.70E-06***
	San Juan	-0.06232	0.01425	-4.374	1.23E-05***
	Valenzuela	-0.06634	0.01436	-4.618	3.90E-06***
	Zamboanga	-0.05315	0.01469	-3.618	0.000298***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 247. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 57.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 57	Intercept	0.0053321	0.0077094	0.692	0.489
	Iloilo	0.0470347	0.0093055	5.055	4.37E-07***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 248. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 58.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 58	Intercept	0.0035838	0.0123971	0.289	0.772519
	Mandaluyong	0.0529579	0.0140112	3.78	0.000158
	Marikina	0.2390651	0.0225947	10.581	<2E-16***
	Muntinlupa	0.0244746	0.0127552	1.919	0.055031.
	Paranaque	0.0550744	0.0311958	1.765	0.077511.
	Pasig	0.0274256	0.0131536	2.085	0.03708*
	Quezon	0.0334641	0.0130359	2.567	0.010267*
	San Juan	0.0402406	0.0140215	2.87	0.004112**
	Valenzuela	0.0607358	0.0180177	3.371	0.000751***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 249. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 59.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 59	Intercept	1.27E-04	1.05E-02	0.012	0.99
	Muntinlupa	8.61E-02	1.16E-02	7.401	1.43E-13***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 250. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 60.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 60	Intercept	0.03196	0.018549	1.723	0.0849
	Malabon	0.049644	0.027707	1.792	0.0732
	Pasig	0.036114	0.019748	1.829	0.0675

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 251. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 61.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 61	Intercept	8.57E-03	9.23E-03	0.929	0.353

Appendix Table 252. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 62.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 62	Intercept	0.001333	0.0053858	0.248	0.805

Appendix Table 253. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 63.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 63	Intercept	-1.16E-05	7.76E-03	-0.001	0.999
	Mandaue	1.02E-01	9.57E-03	10.672	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 254. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 64.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 64	Intercept	0.0016825	0.006605	0.255	0.79893
	Mandaluyong	0.0227022	0.0080342	2.826	0.00472**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 255. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 65.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 65	Intercept	0.0033304	0.0110477	0.301	0.76307
	Las Piñas	0.1437087	0.0502554	2.86	0.00425**
	Manila	0.1644626	0.0121545	13.531	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 256. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 66.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 66	Intercept	1.78E-04	5.98E-03	0.03	0.976
	Pasay	9.36E-02	9.00E-03	10.403	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 257. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 67.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 67	Intercept	4.96E-04	6.82E-03	0.073	0.942
	Pasay	8.70E-02	8.44E-03	10.313	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 258. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 68.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 68	Intercept	0.008257	0.0067584	1.222	0.222
	Pasay	8.70E-02	8.44E-03	10.313	<2E-16***

Appendix Table 259. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 69.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 69	Intercept	4.99E-04	4.80E-03	0.104	0.91713
	Gensan	1.36E-02	5.19E-03	2.625	0.00867**
	Iloilo	2.12E-02	7.51E-03	2.817	0.00485**
	Manila	1.30E-02	5.26E-03	2.479	0.0132*
	Pasig	1.11E-02	5.20E-03	2.127	0.03342*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 260. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 70.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 70	Intercept	0.0074645	0.0066932	1.115	0.265
	Gensan	0.043957	0.0067058	6.555	5.76E-11***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 261. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 71.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 71	Intercept	7.60E-04	4.60E-03	0.165	0.8687
	Pasay	1.19E-02	5.08E-03	2.338	0.0194*

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 262. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 72.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 72	Intercept	0.00612	0.011432	0.535	0.59244
	Caloocan	0.162977	0.031966	5.098	3.47E-07***
	Mandaluyong	0.042035	0.013515	3.11	0.00187**
	Marikina	0.052461	0.017213	3.048	0.00231**
	Paranaque	0.074059	0.030887	2.398	0.01651*
	San Juan	0.042502	0.013445	3.161	0.00157**
	Valenzuela	0.256225	0.029067	8.815	<2E-16***
	Zamboanga	-0.001593	0.01177	-0.135	0.89237***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 263. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 73.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 73	Intercept	2.88E-04	4.64E-03	0.062	0.9505
	Caloocan	2.75E-02	1.41E-02	1.961	0.0499*
	Mandaluyong	6.35E-02	9.29E-03	6.829	8.93E-12***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 264. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 74.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 74	Intercept	0.0004334	0.0025813	0.168	0.86667
	Caloocan	0.0155403	0.0052395	2.966	0.00302**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 265. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 75.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 75	Intercept	0.0040844	0.0049081	0.832	0.405

Appendix Table 266. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 76.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 76	Intercept	0.024315	0.008961	2.713	0.00667**
	Davao	-0.017409	0.009404	-1.851	0.06415.
	Iloilo	-0.016581	0.009644	-1.719	0.08558.
	Makati	-0.021654	0.009045	-2.394	0.01668*
	Mandaue	-0.021034	0.00914	-2.301	0.02139*
	Manila	-0.019706	0.009272	-2.125	0.03358*
	Marikina	-0.017602	0.010365	-1.698	0.08949.
	Muntinlupa	-0.021126	0.008956	-2.359	0.01834*
	Paranaque	-0.023883	0.011285	-2.116	0.03433*
	Pasig	-0.019812	0.009445	-2.098	0.03596*
	Quezon	-0.020569	0.009096	-2.261	0.02376*
	San Juan	-0.021747	0.009163	-2.373	0.01764*
	Valenzuela	-0.022408	0.009677	-2.316	0.0206*
	Zamboanga	-0.015794	0.009188	-1.719	0.08563.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 267. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 77.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 77	Intercept	0.0030496	0.0290666	0.105	0.916
	Davao	0.3152817	0.0293952	10.726	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 268. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 78.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 78	Intercept	0.0011628	0.0055412	0.21	0.83379
	Davao	0.0133601	0.0063974	2.088	0.03678*
	Marikina	0.0398125	0.0091312	4.36	1.31E-05***
	Pasig	0.018144	0.0061476	2.951	0.00317**
	Valenzuela	0.0147724	0.0080664	1.831	0.06707.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 269. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 79.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 79	Intercept	3.96E-04	4.23E-03	0.093	0.926
	Davao	2.50E-02	5.74E-03	4.357	1.33E-05***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 270. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 80.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 80	Intercept	8.27E-04	3.64E-03	0.227	0.8202
	Marikina	9.94E-03	5.10E-03	1.948	0.0515.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 271. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 81.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 81	Intercept	0.0012433	0.0137855	0.09	0.928
	Muntinlupa	0.1472074	0.0146869	10.023	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 272 Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 82.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 82	Intercept	0.024315	0.008961	2.713	0.00667**
		0.0031434	0.0078551	0.4	0.689
	Davao	-0.017409	0.009404	-1.851	0.06415.
	Malabon	0.0249857	0.0123653	2.021	0.0433*
	Iloilo	-0.016581	0.009644	-1.719	0.08558.
	Mandaluyong	0.0201941	0.0098167	2.057	0.0397*
	Makati	-0.021654	0.009045	-2.394	0.01668*
	Manila	0.0158219	0.0086896	1.821	0.0687.
	Mandaue	-0.021034	0.00914	-2.301	0.02139*
	Marikina	0.022168	0.0116066	1.91	0.0562.
	Manila	-0.019706	0.009272	-2.125	0.03358*
	Pasig	0.0156502	0.0081566	1.919	0.055.
	Marikina	-0.017602	0.010365	-1.698	0.08949
	San Juan	0.0165126	0.0089843		0.0661.
	Muntinlupa	-0.021126	0.008956	-2.359	0.01834*

Appendix Table 272 (cont.).

Paranaque	-0.023883	0.011285	-2.116	0.03433*
Pasig	-0.019812	0.009445	-2.098	0.03596*
Quezon	-0.020569	0.009096	-2.261	0.02376*
San Juan	-0.021747	0.009163	-2.373	0.01764*
Valenzuela	-0.022408	0.009677	-2.316	0.0206*
Zamboanga	-0.015794	0.009188	-1.719	0.08563.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 273. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 83.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 83 Intercept	0.0031434	0.0078551	0.4	0.689
Malabon	0.0249857	0.0123653	2.021	0.0433*
Mandaluyong	0.0201941	0.0098167	2.057	0.0397*
Manila	0.0158219	0.0086896	1.821	0.0687.
Marikina	0.022168	0.0116066	1.91	0.0562.
Pasig	0.0156502	0.0081566	1.919	0.055.
San Juan	0.0165126	0.0089843	1.838	0.0661.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 274. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 84.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 84 Intercept	0.0004803	0.0047172	0.102	0.9189
Davao	0.0100074	0.0055153	1.814	0.0696.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 275. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 85.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 85 Intercept	0.0004803	0.0047172	0.102	0.9189
Malabon	0.0288086	0.0134824	2.137	0.03264*
Mandaluyong	0.0204782	0.0115361	1.775	0.0759.
Mandaue	0.0212619	0.0109095	1.949	0.05132.
Manila	0.0284288	0.0110407	2.575	0.01004*
Muntinlupa	0.0233866	0.0105669	2.213	0.0269*
Pasig	0.020202	0.0109648	1.842	0.06543.

Appendix Table 275 (cont.).

San Juan	0.0399496	0.0124319	3.213	0.00131**
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*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 276. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 86.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 86 Intercept	1.64E-04	6.44E-03	0.025	0.98
Pasay	7.37E-02	8.35E-03	8.83	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 277. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 87.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 87 Intercept	0.0008907	0.006155	0.145	0.88494
Davao	0.0153264	0.0072105	2.126	0.03356*
Pasay	0.0185825	0.0067674	2.746	0.00604**
Pasig	0.0123421	0.0067708	1.823	0.06835.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 278. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 88.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 88 Intercept	0.00922	0.009705	0.95	0.342121
Malabon	0.022352	0.012112	1.845	0.065003.
Valenzuela	0.047211	0.013528	3.49	0.000485***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 279. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 89.

Coefficients	Estimate	Std. Error	t value	Pr(> t)
Topic 89 Intercept	5.39E-05	4.52E-03	0.012	0.99
Cagayan de Oro	7.80E-02	4.74E-02	1.645	0.1.
Quezon	1.01E-01	5.53E-03	18.32	<2E-16***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 280. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 90.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 90	Intercept	2.69E-03	4.98E-03	0.54	0.589

Appendix Table 281. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 91.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 91	Intercept	0.0043623	0.0067702	0.644	0.5194
	Pasay	0.0128356	0.007521	1.707	0.0879.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 282. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 92.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 92	Intercept	0.018402	0.010577	1.74	0.081909.
	Cagayan de Oro	0.129832	0.039355	3.299	0.000973***
	Mandaue	-0.017588	0.010543	-1.668	0.095292.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 283. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 93.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 92	Intercept	0.018402	0.010577	1.74	0.081909.
	Cagayan de Oro	0.129832	0.039355	3.299	0.000973***
	Mandaue	-0.017588	0.010543	-1.668	0.095292.

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1.

Appendix Table 284. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 94.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 94	Intercept	1.41E-04	2.72E-03	0.052	0.95851
	Pasay	8.68E-03	3.16E-03	2.744	0.00607**

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '****' 0.001 '***' 0.01 '**' 0.05 '.' 0.1 ' ' 1

Appendix Table 285. Regression Table of the Effect of City of Origin on the Topic Proportion of Topic 95.

Coefficients		Estimate	Std. Error	t value	Pr(> t)
Topic 95	Intercept	9.10E-05	5.32E-05	1.713	0.08681.
	Davao	3.82E-04	5.83E-05	6.563	5.46E-11***
	Gensan	3.78E-04	5.54E-05	6.83	8.83E-12***
	Las	4.60E-04	1.59E-04	2.904	0.00369**
	Makati	2.02E-04	5.36E-05	3.761	0.00017***
	Malabon	3.09E-04	7.12E-05	4.336	1.46E-05***
	Mandaluyong	4.91E-04	6.41E-05	7.652	2.11E-14***
	Manila	2.63E-04	5.54E-05	4.739	2.17E-06***
	Marikina	2.96E-04	6.90E-05	4.298	1.73E-05***
	Muntinlupa	3.34E-04	5.52E-05	6.049	1.50E-09***
	Pasay	9.31E-05	5.58E-05	1.67	0.09503.
	Pasig	3.27E-04	5.80E-05	5.644	1.70E-08***
	Quezon	2.70E-04	5.62E-05	4.811	1.52E-06***
	San Juan	4.99E-04	6.31E-05	7.904	2.90E-15***
	Valenzuela	1.60E-04	6.71E-05	2.384	0.01712*
	Zamboanga	3.49E-04	5.98E-05	5.838	5.40E-09***

*Significance codes for the intercepts and city of origin estimates are as follows: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Appendix Table 286. Rank of the Highly Urbanized Cities (HUCs) from the Department of Trade and Industry's (DTI) Cities and Municipalities Competitive Index (CMCI).

City	CMCI Score
Quezon	61.8475
Pasay	56.7826
Manila	56.2923
Iloilo	51.0439
Caloocan	49.5927
Muntinlupa	49.0816
Cagayan de Oro	48.612
Makati	47.1135
Pasig	44.5875
Davao	44.5875
Valenzuela	44.2105
Gensan	42.1879
Iligan	42.0494
Mandaue	41.6661

Appendix Table 286 (cont.).

Bacolod	41.1841
Mandaluyong	37.7354
Zamboanga	37.4007
Marikina	36.6775
Paranaque	36.3001
Las Pinas	35.1899
Malabon	34.8983
San Juan	34.1455

Appendix Table 287: Cities by their total number of collected ordinances and their corresponding year range of publication.

City	Number of Ordinances	Publication Year Range
Makati	5522	1988-2022
Muntinlupa	1607	1997-2024
Quezon	845	2010-2024
Pasay	774	1992-2024
Pasig	735	2005-2024
Mandaue	726	2008-2024
Gensan	711	2007-2018
Manila	640	1997-2022
Zamboanga	571	2019-2024
Davao	429	1956-2018
San Juan	295	2020-2023
Mandaluyong	292	2020-2024
Iloilo	189	2016-2024
Marikina	132	2001-2024
Valenzuela	122	2020
Malabon	106	2013-2019
Bacolod	53	2021-2022
Caloocan	34	2004-2020
Parañaque	31	2014-2020
Las Piñas	19	2010-2023
Iligan	13	2022
Cagayan de Oro	7	2006-2021

Appendix Figure 1. QR Code for Access of the Website Created.

